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Rates Vol Lab

Forward steepener and vol divergence

CITI'S TAKE

Even after the recent cheapening, we prefer to sell vol on bounce rather than buy on dip. Long-dated forward curves have diverged from vol and look modestly flat. Instead of selling rates options outright, we see better risk/reward in forward steepeners and shorting Blues convexity adjustment as short vol proxy. Note: Futures trading involves substantial risk of loss.

Vol Surface — In the US, 30y tails continue to look to be the most undervalued based on our macro-vol fair value model. In EUR, top-left drove the cheapening in implied vols over the last month, while the 1y2y vol ratio at 3m expiry is hovering around the highest in four years.

Realized vol — In the US, hedging in the morning continues to result in higher realized vol than hedging in the afternoon.

Skew — In both the US and EUR, short-dated payer skews have extended their recent rebound. We expect the US payer skews outperformance to be short-lived.

SOFR convexity adjustments (USD) — Blues and Golds convexity adjustments have finally started to decline, and we believe there is more room to go.

CMS curve vol — Implied curve correlation on US 5s30s remains low, which reflects the persistent demand for curve caps. In EUR, we flag the richness of 6m implied vols on 2s5s, both outright and against 5s30s.

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See Appendix A-1 for Analyst Certification, Important Disclosures and Research Analyst Affiliations.

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Weekly Focus: Forward steepener and vol divergence

Mike Chang

Even after the recent cheapening, we prefer to sell vol on bounce rather than buy on dip. Long-dated forward curves have diverged from vol and look modestly flat. Instead of selling rates options outright, we see better risk/reward in forward steepeners and shorting Blues convexity adjustment as short vol proxy.

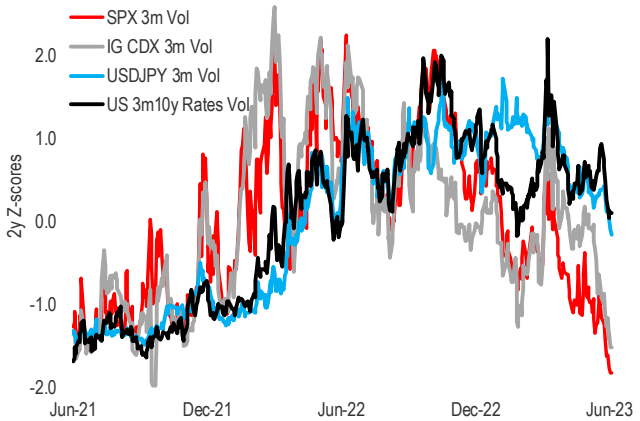
Sell vol on bounce rather than buy on dip

Last month, we noted that a resolution on the debt ceiling would likely be followed by a cheapening in US short-dated rates vol, which at the time looked elevated relative to EUR rates vol in outright levels and in terms of the inversion of the vol slopes (US Rates Weekly - The ceiling can't hold us). Since then, not only has US rates vol cheapened as we have expected, but implied vols for almost all asset classes have collapsed (Figure 1). US 3m10y swaption vol has fallen close to its 12-month low while the decline in risky asset vols (SPX and IG CDX) has been even more notable, reaching their multi-year lows.

Given that the pace of the cheapening over the past two weeks has been one of the most extreme historically (Figure 2), we would not be surprised to see some stabilization in the near term. However, we still would avoid being outright long vol here and prefer to sell on bounce. In the scenario where the Fed delivers a hawkish skip and "kicks the can" to the July meeting, we believe that rates could continue to trade within a range, which would keep implied vols anchored. Range-bound rates would also entice systematic vol sellers to increase their activity. Even if the Fed delivers a 25bp hike at the upcoming meeting, they could temper the hawkishness by noting the possibility of a July skip. In either scenario, we see rates vol facing structural headwind for the course of this year as we get closer to the last hike. As we have previously shown (US Rates Weekly - The trend can be your friend), historically rates vol tends to decline in the months immediately following the last rate hike, including the late 70s/early 80s (where the analysis is based on the subsequent realized vol).

The risk to our view would be a surprisingly strong CPI print followed by a hawkish 25bp hike with upward revisions in the Fed's forecasts, which would suggest that the last hike is not within sight. Under this tail risk scenario, implied vols should rebound sharply higher from here to reflect a much wider right side of the rate distribution going forward.

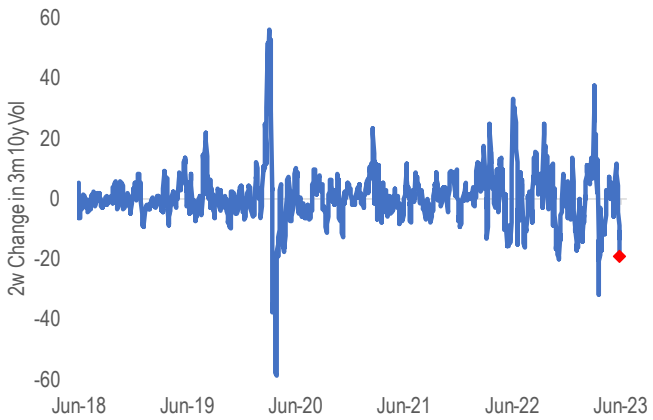
Figure 1. Implied vols for almost all asset classes have collapsed with risky asset vols falling to their multi-year lows



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Source: Citi Research

Figure 2. The pace of the recent cheapening in 3m10y vol has been one of the most extreme, suggesting possible near-term stabilization



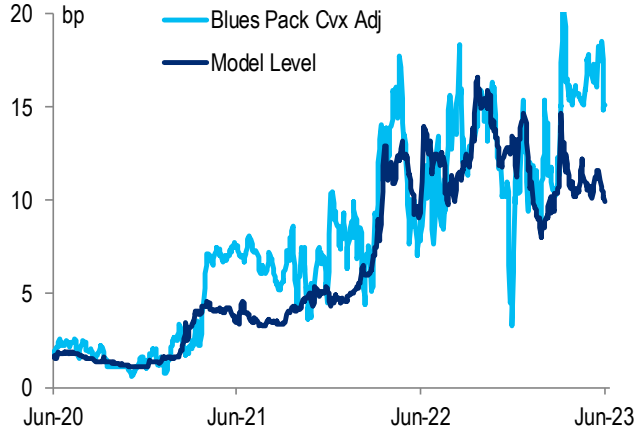
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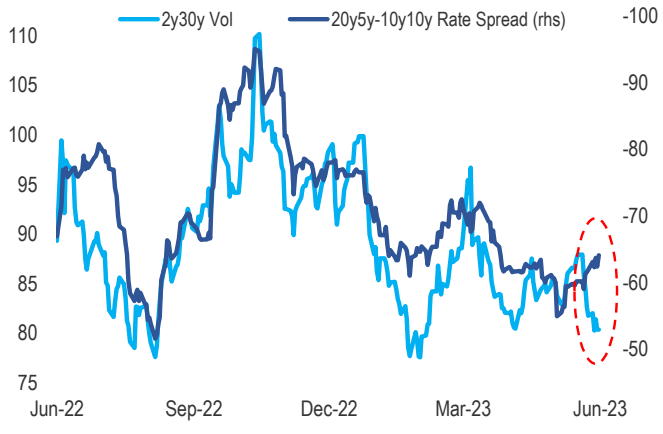
Rather than selling rates options outright with short-dated implied vols close to their YTD lows, we prefer to express our structural bearish vol bias through other structures with relatively more attractive entry levels. We maintain our previously recommended idea of shorting Blues convexity adjustment via long SOFR futures against pay in swaps as a short vol proxy (Rates Vol Lab - Short Blues convexity via futures/swaps). Blues and Golds convexity adjustments have been stubbornly high outright and relative to our model, which is highly dependent on implied vol, but they have finally started to normalize in the past week (Figure 3). If implied vols remain unchanged and continue to trade around their YTD lows, we believe that the Blues convexity adjustment could continue to compress towards our model fair value, which is currently about 5bps lower. Note: Futures trading involves substantial risk of loss.

Figure 3. Blues convexity adjustment has finally started to normalize and can continue to compress towards our model (5bps lower), which is highly dependent on the implied vol



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Figure 4. Vol and long-dated forward curve have moved inversely as expected until the recent divergence where the curve bear flattened slightly despite the sharp collapse in vol



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Long-dated forward steepener and vol divergence

We have previously highlighted the inverse relationship between long-dated forward curves and implied vol due to the net short/(long) convexity exposure that is embedded in long-dated forward curve steepeners/(flatteners). The inverse relationship has been broadly consistent over the past year until very recently with the long-dated forward curve bear flattening slightly despite the sharp collapse in vol that is mentioned above (Figure 4). While the divergence is still relatively modest at the moment, it is worth watching given that a hawkish skip at the FOMC meeting with upward revision in the median Fed dots could cause vol to remain muted with knee-jerk bear flattening in the curve. Given that the hawkish skip is the likely outcome, we prefer to wait until after the FOMC meeting to consider possibly establishing a long-dated steepener as a positive carry / short vol trade.

Because long-dated forward steepeners have net negative convexity, we can back out implied yield breakevens from the carry and compare them to the recent realized yield moves. Having a breakeven/realized vol ratio allows us to compare the positive carry of various long-dated forward steepeners in a standardized fashion (Figure 5).

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Figure 5. We backed out the implied yield breakeven from the steepener's carry and negative convexity and compare it to the recent realized moves. Having a breakeven/realized vol ratio allows us to compare the various forward steepeners in a standardized fashion

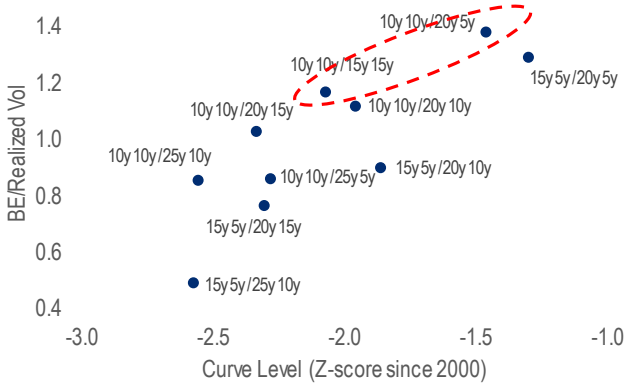
Curve Close of 06/08/23	10y10y/ 15y15y	10y10y/ 20y5y	10y10y/ 20y10y	10y10y/ 20y15y	10y10y/ 25y5y	10y10y/ 25y10y	15y5y/ 20y5y	15y5y/ 20y10y	15y5y/ 20y15y	15y5y/ 25y10y
USD										
curve ,bp	-51.5	-62.4	-76.1	-89.4	-91.5	-105.6	-50.9	-64.5	-77.9	-94.1
3y ZS	-0.91	-0.91	-0.96	-1.00	-1.01	-1.04	-0.98	-1.03	-1.06	-1.11
ZS since 2000	-2.07	-1.47	-1.96	-2.34	-2.28	-2.56	-1.30	-1.87	-2.31	-2.58
1y carry, bp	6.89	10.31	8.31	7.71	5.95	6.04	5.93	3.93	3.33	1.66
daily BE, bp	6.18	7.22	5.71	4.99	4.29	3.97	6.77	4.59	3.71	2.29
1y realized vol, bp	5.30	5.24	5.11	4.85	4.98	4.66	5.24	5.11	4.85	4.66
BE/realized vol	1.17	1.38	1.12	1.03	0.86	0.85	1.29	0.90	0.77	0.49

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Source: Citi Research

Not surprisingly, there is generally a trade-off between carry and entry level, where the steepeners with the most attractive carry usually have the worst entry point (Figure 6). Currently, the 10y10y/15y15y and 20y5y-10y10y steepeners lie on the efficient frontier of this carry vs entry trade-off and have breakeven vs realized vol ratios that are meaningfully above 1. Between the two, 20y5y-10y10y steepener looks slightly more attractive due to a more favorable carry vs entry trade-off when using a more recent 3y Z-score. 25y10y-10y10y and 20y15y-10y10y steepeners also lie on the efficient frontier but their breakeven vs realized vol ratios are not high enough. Time series of a simple estimation of the 1y carry of the 10y10y/20y5y steepener shows that even though the positive carry has pulled back from its peak, it remains elevated relative to the long-run history (Figure 7).

Figure 6. Generally, there is a trade-off between carry and entry level for forward steepeners, and 10y10y/15y15y is on the efficient frontier



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Source: Citi Research

Figure 7. Even though the estimated 1y carry of the 10y10y/20y5y steepener has declined from the peak, it is still elevated



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USD at-the-money swaption vols

USD relative value on the swaption surface

We report levels and z-scores of at-the-money vols, as well as the relative richness/cheapness of vols to rates (based on principal components of the curve).

- Short-dated vols have underperformed intermediates amid the overall vol cheapening following the debt ceiling resolution (Figure 9). US vol slopes have steepened and converged towards EUR vol slopes as we have expected (US Rates Weekly - The ceiling can't hold us).

- Our macro-vol model shows that implied vols are generally cheap with upper- and intermediate vols being the exception (Figure 10). Hence instead of selling rates options outright, we see better risk/reward to be short vol through structures such as long-dated forward steepeners and shorting Blues convexity adjustment.

Figure 8. Implied vols, normals (3m z-score)

exp\tails	1y	2y	3y	5y	7y	10y	30y
1m	139.5 (-1.2)	164.4 (-0.9)	150.6 (-1.0)	136.4 (-1.0)	118.0 (-1.2)	103.0 (-1.3)	79.9 (-1.2)
3m	143.2 (-1.2)	156.4 (-1.2)	148.1 (-1.2)	135.3 (-1.1)	119.6 (-1.3)	106.7 (-1.3)	83.9 (-1.1)
6m	153.1 (-1.1)	155.2 (-1.1)	146.4 (-1.1)	131.0 (-1.1)	118.0 (-1.3)	107.4 (-1.4)	85.3 (-1.2)
1y	155.2 (-1.0)	143.0 (-1.3)	135.3 (-1.2)	121.8 (-1.1)	111.7 (-1.3)	103.8 (-1.4)	83.2 (-1.5)
2y	134.2 (-1.1)	126.2 (-1.3)	120.9 (-1.3)	112.6 (-1.2)	105.5 (-1.3)	98.4 (-1.4)	80.0 (-1.5)
3y	120.7 (-1.2)	115.4 (-1.2)	111.6 (-1.2)	106.2 (-1.1)	100.1 (-1.3)	93.8 (-1.4)	76.9 (-1.6)
5y	107.5 (-0.7)	102.8 (-0.7)	100.2 (-0.8)	95.8 (-1.1)	91.7 (-1.2)	86.7 (-1.3)	72.1 (-1.3)
10y	85.3 (-0.8)	83.2 (-0.8)	81.3 (-0.9)	78.2 (-1.0)	75.6 (-1.1)	72.5 (-1.1)	61.6 (-1.0)

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Close 6/9/23. The table shows ATM implied vols and 3m z-scores in parentheses. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

Figure 9. Change in implied vols since May 12th, normals

exp\tails	1y	2y	3y	5y	7y	10y	30y
1m	-17.0	-0.4	-0.6	-1.5	-5.0	-6.7	-6.5
3m	-24.5	-13.1	-11.5	-7.5	-8.0	-7.5	-8.0
6m	-16.1	-9.8	-9.0	-7.4	-7.3	-6.1	-6.9
1y	-9.4	-8.1	-7.1	-5.0	-4.3	-3.1	-5.3
2y	-3.2	-3.7	-3.6	-3.4	-3.1	-2.4	-3.8
3y	-0.5	-0.6	-1.4	-2.3	-2.1	-1.8	-3.3
5y	0.8	0.7	0.1	-1.0	-1.4	-1.5	-1.3
10y	-0.8	-0.5	-0.6	-0.7	-0.7	-0.5	-0.2

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Note: The table shows changes in ATM implied vols from 5/12/23 to 6/9/23. Source: Citi Research

Figure 10. Macro-vol (rich/cheap to rates), normals (z-score)

exp\tails	1y	2y	3y	5y	7y	10y	30y
1m	-30.6 (-1.3)	-18.0 (-0.9)	-7.7 (-0.5)	-9.8 (-0.6)	-20.5 (-1.1)	-32.3 (-1.5)	-41.3 (-1.7)
3m	-22.3 (-1.0)	-4.9 (-0.3)	2.1 (0.2)	-3.8 (-0.3)	-13.2 (-0.8)	-22.8 (-1.2)	-32.0 (-1.6)
6m	-19.1 (-1.0)	-1.0 (-0.1)	3.1 (0.3)	-5.0 (-0.4)	-10.8 (-0.8)	-18.0 (-1.2)	-25.2 (-1.5)
1y	-6.5 (-0.4)	1.5 (0.1)	1.0 (0.1)	-6.2 (-0.6)	-10.5 (-1.0)	-14.7 (-1.2)	-19.2 (-1.4)
2y	-2.0 (-0.2)	-2.5 (-0.3)	-4.8 (-0.6)	-8.6 (-1.0)	-11.4 (-1.2)	-14.0 (-1.4)	-16.4 (-1.5)
3y	-5.3 (-0.7)	-7.2 (-1.0)	-8.5 (-1.2)	-10.3 (-1.4)	-11.2 (-1.4)	-12.5 (-1.4)	-13.3 (-1.4)
5y	-7.7 (-1.3)	-9.3 (-1.5)	-9.6 (-1.5)	-9.3 (-1.4)	-9.8 (-1.4)	-10.2 (-1.4)	-9.6 (-1.2)
10y	-6.8 (-1.2)	-7.1 (-1.3)	-7.1 (-1.3)	-6.8 (-1.2)	-7.1 (-1.3)	-7.3 (-1.3)	-6.0 (-1.1)

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Close 6/9/23. We regress each ATM point on 3 principal components of the swap and TIPS breakeven curves. We report rich/cheap for each vol as a residual from this regression together with z-scores in parentheses. We calibrate the model to the 2/1/2021-2/28/2022 sample history and test the model out-of-sample using data since 3/1/2022. Source: Citi Research

Figure 11. Rich/cheap to PCA of the vol surface, normals (z-score)

exp\tls	3m	1y	2y	3y	5y	7y	10y	30y
1m	8.1 (0.5)	-6.0 (-0.7)	3.4 (0.5)	1.5 (0.3)	4.3 (0.8)	0.2 (0.1)	-0.7 (-0.2)	0.2 (0.1)
3m	1.2 (0.1)	-5.8 (-1.1)	-3.1 (-0.7)	-1.9 (-0.5)	3.1 (0.9)	0.9 (0.3)	0.2 (0.1)	-0.4 (-0.2)
6m	-1.7 (-0.2)	-1.7 (-0.5)	-0.5 (-0.1)	-0.1 (-0.0)	3.3 (1.1)	1.6 (0.7)	1.6 (0.7)	0.0 (0.0)
1y	-1.7 (-0.3)	-0.7 (-0.1)	-2.3 (-0.5)	-1.7 (-0.5)	1.4 (0.6)	0.7 (0.3)	1.4 (0.8)	-0.8 (-0.5)
2y	-0.3 (-0.0)	-2.1 (-0.5)	-2.4 (-0.7)	-2.0 (-0.7)	0.2 (0.1)	0.1 (0.0)	0.4 (0.3)	-0.9 (-0.8)
3y	-1.3 (-0.3)	-2.0 (-0.6)	-2.0 (-0.7)	-1.6 (-0.7)	0.1 (0.0)	-0.0 (-0.0)	0.0 (0.0)	-1.0 (-0.9)
5y	1.5 (0.8)	0.4 (0.2)	0.2 (0.1)	-0.2 (-0.2)	-0.4 (-0.5)	-0.4 (-0.7)	-0.2 (-0.4)	-0.3 (-0.3)
10y	0.9 (0.7)	0.5 (0.6)	0.5 (0.6)	0.4 (0.5)	0.1 (0.2)	0.1 (0.2)	0.3 (0.3)	0.2 (0.2)

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Close 6/9/23. We calculate rich/cheap and z-scores in parentheses from PCA on the vol surface based on 3 principal components using 2y of data. By design, this framework only yields relative value. Source: Citi Research

USD implied vs realized volatilities

We calculate the relative richness of short-dated implied vols to 1m and 3m realized vols. We calculate realized vols using both intraday (hourly) and close-to-close sampling.

- Intraday realized vols are trading relatively in-line with close-to-close realized vols (Figure 12), with 30y tails being the only exception with slightly higher intraday vol.

- Even after the recent sharp cheapening, implied vols across all tails are still meaningfully above the recent realized vols on both intraday and close-to-close measures (Figure 16).

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Figure 12. 1m intraday/close-to-close ratio of realized vols (1y z-score)

exp\tails	1y	2y	3y	5y	7y	10y	30y
1m	1.19 (0.8)	1.08 (0.3)	1.04 (0.0)	1.01 (-0.3)	1.00 (-0.3)	0.99 (-0.4)	1.13 (-0.1)
3m	1.15 (0.7)	1.08 (0.3)	1.04 (0.0)	1.01 (-0.2)	1.00 (-0.3)	1.00 (-0.3)	1.14 (-0.1)
6m	1.08 (0.3)	1.05 (0.1)	1.02 (-0.1)	1.00 (-0.3)	0.99 (-0.4)	0.99 (-0.4)	1.14 (-0.1)
1y	1.03 (0.0)	1.00 (-0.2)	0.99 (-0.4)	0.98 (-0.5)	0.97 (-0.5)	0.98 (-0.5)	1.13 (-0.1)

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Note: Close 6/9/23. The 30m sampling between 7am and 4pm NY time is used. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

Figure 13. 3m intraday/close-to-close ratio of realized vols (1y z-score)

exp\tails	1y	2y	3y	5y	7y	10y	30y
1m	0.95 (-0.8)	0.93 (-0.9)	0.95 (-0.9)	0.99 (-0.7)	1.00 (-0.7)	1.02 (-0.6)	1.16 (0.1)
3m	0.87 (-1.1)	0.91 (-1.0)	0.94 (-1.0)	0.99 (-0.8)	1.00 (-0.8)	1.01 (-0.6)	1.16 (0.1)
6m	0.89 (-1.1)	0.93 (-0.9)	0.95 (-0.9)	1.00 (-0.7)	1.01 (-0.7)	1.02 (-0.5)	1.16 (0.1)
1y	0.95 (-0.7)	0.98 (-0.6)	1.00 (-0.6)	1.03 (-0.5)	1.03 (-0.4)	1.04 (-0.3)	1.16 (0.1)

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Note: Close 6/9/23. The 30m sampling between 7am and 4pm NY time is used. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

Figure 14. Intraday realized: implied/1m realized ratio (1y z-score)

exp\tails	1y	2y	3y	5y	7y	10y	30y
1m	1.17 (0.7)	1.17 (1.0)	1.12 (1.0)	1.14 (1.1)	1.11 (1.0)	1.09 (0.8)	1.06 (1.1)
3m	1.00 (0.6)	1.05 (0.9)	1.06 (1.0)	1.11 (1.3)	1.11 (1.3)	1.12 (1.3)	1.11 (1.5)
6m	0.94 (0.8)	1.01 (1.0)	1.04 (1.2)	1.08 (1.4)	1.10 (1.5)	1.13 (1.5)	1.13 (1.8)
1y	0.91 (0.9)	0.96 (1.0)	1.01 (1.2)	1.06 (1.5)	1.09 (1.6)	1.13 (1.7)	1.11 (1.9)

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Note: Close 6/9/23. The 30m sampling between 7am and 4pm NY time is used. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

Figure 15. Intraday realized: implied/3m realized ratio (1y z-score)

exp\tails	1y	2y	3y	5y	7y	10y	30y
1m	0.57 (-1.8)	0.73 (-1.5)	0.76 (-1.3)	0.82 (-1.0)	0.81 (-1.0)	0.82 (-1.0)	0.81 (-0.8)
3m	0.56 (-1.8)	0.69 (-1.7)	0.75 (-1.4)	0.82 (-0.9)	0.83 (-0.7)	0.85 (-0.6)	0.85 (-0.1)
6m	0.56 (-1.9)	0.68 (-1.7)	0.74 (-1.3)	0.80 (-0.7)	0.83 (-0.3)	0.86 (0.0)	0.87 (0.5)
1y	0.65 (-1.6)	0.72 (-1.4)	0.77 (-0.9)	0.81 (-0.2)	0.85 (0.3)	0.88 (0.8)	0.87 (1.1)

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Note: Close 6/9/23. The 30m sampling between 7am and 4pm NY time is used. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

Figure 16. Close-to-close realized: implied/1m realized ratio (1y z-score)

exp\tails	1y	2y	3y	5y	7y	10y	30y
1m	1.39 (1.3)	1.26 (1.1)	1.16 (0.8)	1.15 (0.8)	1.11 (0.7)	1.08 (0.5)	1.20 (1.1)
3m	1.16 (1.0)	1.13 (0.9)	1.10 (0.8)	1.13 (1.0)	1.11 (1.0)	1.12 (1.0)	1.26 (1.7)
6m	1.01 (0.8)	1.05 (0.8)	1.06 (0.8)	1.08 (1.0)	1.09 (1.2)	1.12 (1.3)	1.28 (2.0)
1y	0.94 (0.6)	0.96 (0.6)	0.99 (0.7)	1.03 (1.0)	1.06 (1.3)	1.11 (1.6)	1.26 (2.1)

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Note: Close 6/9/23. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

Figure 17. Close-to-close realized: implied/3m realized ratio (1y z-score)

exp\tails	1y	2y	3y	5y	7y	10y	30y
1m	0.54 (-2.1)	0.68 (-1.8)	0.72 (-1.6)	0.81 (-1.2)	0.82 (-1.2)	0.83 (-1.1)	0.94 (-0.5)
3m	0.49 (-2.0)	0.63 (-1.8)	0.70 (-1.7)	0.80 (-1.2)	0.83 (-1.0)	0.86 (-0.8)	0.98 (-0.1)
6m	0.50 (-1.8)	0.63 (-1.6)	0.71 (-1.5)	0.80 (-1.0)	0.83 (-0.7)	0.88 (-0.3)	1.01 (0.4)
1y	0.62 (-1.4)	0.70 (-1.3)	0.76 (-1.0)	0.83 (-0.4)	0.87 (0.0)	0.92 (0.6)	1.00 (0.9)

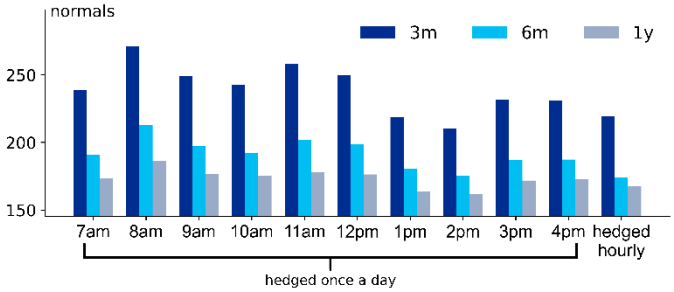
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Note: Close 6/9/23. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

USD day-to-day realized volatilities for a given time snapshot

We calculate day-to-day realized volatilities measured at a given hour within a day for the last 3m, 6m and 1y (EST time)

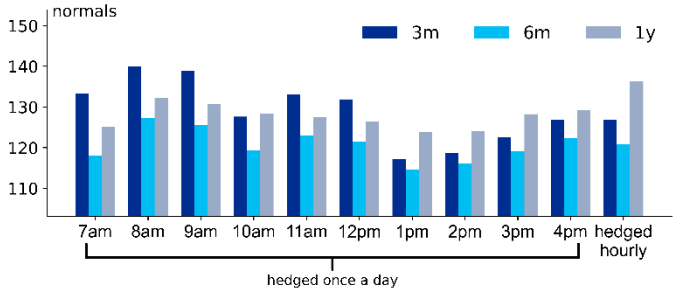
- For both the 2y and 10y rates, hedging in the morning has resulted in higher realized vol, while hedging in the afternoon has delivered lower realized vol. This suggests possible rates whipsaw trading behavior around the economic data releases in the morning.

Figure 18. Realized vol of the 2y rate



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Note: Close 6/9/23, time is NY. Source: Citi Research

Figure 19. Realized vol of the 10y rate



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Note: Close 6/9/23, time is NY. Source: Citi Research

USD expiry and tail switches

We analyze ATM vol switches across expiries and tails.

- Short expiries have cheapened relative to longer expiries, especially on the 10y tails, in recent weeks with the resolution of the debt ceiling. For example, the 6m10y/1y10y vol ratio has reached -1.2 sigmas (Figure 20).

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- Both 5y and 10y tails have outperformed 30y tails. For example, the 1y5y/1y30y and 1y10y/1y30y vol ratios are at +1.3 sigmas and +1.5 sigmas, respectively (Figure 21). This is consistent with our recent view that 10y tails look rich to 30y tails and has scope to mean revert (US Rates Weekly - Oops... We Did It Again).

Figure 20. Short/long expiry ratio (3m z-score)

exp\tails	1y	2y	3y	5y	7y	10y	30y
1m/3m	0.97 (-0.9)	1.05 (-0.2)	1.02 (-0.5)	1.01 (-0.8)	0.99 (-1.0)	0.97 (-1.2)	0.95 (-1.1)
3m/6m	0.94 (-1.6)	1.01 (-1.2)	1.01 (-1.2)	1.03 (-1.0)	1.01 (-1.1)	0.99 (-1.2)	0.98 (-0.9)
6m/1y	0.99 (-1.1)	1.09 (-0.6)	1.08 (-0.7)	1.08 (-0.7)	1.06 (-1.0)	1.03 (-1.2)	1.03 (-0.4)
1y/3y	1.29 (-0.3)	1.24 (-0.6)	1.21 (-0.9)	1.15 (-0.9)	1.12 (-1.3)	1.11 (-0.8)	1.08 (-0.9)
3y/5y	1.12 (-1.9)	1.12 (-2.3)	1.11 (-2.1)	1.11 (-0.6)	1.09 (-0.8)	1.08 (-1.2)	1.07 (-1.9)
5y/10y	1.26 (-0.4)	1.24 (-0.3)	1.23 (-0.5)	1.23 (-1.2)	1.21 (-1.5)	1.20 (-1.6)	1.17 (-1.4)

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Note: Close 6/9/23. The table reports shorter expiry/longer expiry ratios with 3m z-scores in parentheses. Deep red (deep blue) indicates largest (smallest) z-scores.
Source: Citi Research

Figure 21. Short/long tail ratio (3m z-score)

exp\tails	1y/2y	1y/3y	2y/5y	2y/10y	5y/10y	5y/30y	10y/30y
1m	0.85 (-1.7)	0.93 (-1.5)	1.20 (-0.2)	1.60 (0.7)	1.32 (2.1)	1.71 (0.8)	1.29 (-0.6)
3m	0.92 (-1.3)	0.97 (-1.3)	1.16 (-1.1)	1.47 (-0.5)	1.27 (1.3)	1.61 (0.7)	1.27 (-0.1)
1y	1.09 (0.3)	1.15 (0.4)	1.17 (-0.8)	1.38 (-0.4)	1.17 (0.2)	1.46 (1.3)	1.25 (1.5)
2y	1.06 (0.9)	1.11 (0.7)	1.12 (-1.0)	1.28 (-0.5)	1.14 (0.4)	1.41 (1.6)	1.23 (1.3)
5y	1.05 (0.3)	1.07 (0.8)	1.07 (1.3)	1.19 (1.3)	1.10 (0.6)	1.33 (0.9)	1.20 (0.3)
10y	1.02 (0.4)	1.05 (0.6)	1.06 (0.8)	1.15 (0.7)	1.08 (0.4)	1.27 (0.6)	1.18 (0.5)

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Note: Close 6/9/23. The table reports shorter tail/longer tail ratios with 3m z-scores in parentheses. Deep red (deep blue) indicates largest (smallest) z-scores.
Source: Citi Research

The efficient frontier of USD forward vol trades

We analyze all executable forward vol trades, expressed using gamma-neutral calendar spreads and triangles, and report 20 trades on the efficient frontier based on ex-ante Sharpe (measured by the vol-adjusted roll) and value (measured by 1y-zcore).

- Intermediate left-side forward vols, such as 1y2y1y fwd vol (long 3y1y vs short 1y2y1y), are generally the most attractive based on the trade-off between low entry level and attractive vol-adjusted positive carry on the efficient frontier. This is due to the sharper inversion in left-side vol slopes.

Figure 22. Forward vol trades on the efficient frontier (Sharpe vs z-score)

short expry	long expry	fwd vol	gamma-ntrl wghts	short vol, nv	long vol, nv	fwd vol, nv	3m roll, nv	Shrp	1y ZS	3m ZS	short expry	long expry	fwd vol	gamma-ntrl wghts	short vol, nv	long vol, nv	fwd vol, nv	3m roll, nv	Shrp	1y ZS	3m ZS
3m3y2y	9m7y	3m6m7y	254.2/ 100	142.3	114.5	103.9	14.08	0.69	-0.7	-2.0	3m25y	1y25y	3m9m25y	50.7/ 100	87.6	85.5	84.9	1.93	0.24	-1.2	-1.6
1y2y1y	3y1y	1y2y1y	69.8/ 100	145.5	120.7	106.1	7.02	0.54	-1.0	-0.5	6m25y	18m25y	6m1y25y	59.4/ 100	88.1	83.5	82.2	1.65	0.23	-1.2	-1.7
6m3y2y	3y3y	6m30m3y	76.9/ 100	139.3	111.6	105.9	5.16	0.51	-1.0	-1.9	1y25y	18m25y	1y6m25y	82.4/ 100	85.5	83.5	81.8	1.57	0.23	-1.2	-1.6
6m3y1y	3y2y	6m30m2y	99.6/ 100	140.5	115.4	110.7	5.04	0.50	-1.0	-1.8	9m30y	18m30y	9m9m30y	71.6/ 100	84.3	81.6	80.1	1.38	0.21	-1.2	-1.8
1y2y2y	3y2y	1y2y2y	70.0/ 100	139.6	115.4	101.2	6.23	0.48	-1.0	-0.7	6m30y	2y30y	6m18m30y	51.3/ 100	85.3	80.0	79.1	1.25	0.20	-1.2	-1.7
18m5y	3y2y	18m18m2y	28.7/ 100	116.8	115.4	114.1	3.44	0.34	-1.1	-1.2	6m30y	18m30y	6m1y30y	58.9/ 100	85.3	81.6	80.5	1.36	0.19	-1.2	-1.8
2y5y	3y3y	2y1y3y	49.4/ 100	112.6	111.6	109.5	3.22	0.33	-1.2	-1.2	6m20y	18m20y	6m1y20y	59.1/ 100	91.3	86.8	85.9	1.41	0.19	-1.2	-1.7
6m25y	1y25y	6m6m25y	72.0/ 100	88.1	85.5	83.4	2.34	0.29	-1.2	-1.7	1y30y	18m30y	1y6m30y	82.1/ 100	83.2	81.6	80.2	1.27	0.19	-1.2	-1.6
9m25y	18m25y	9m9m25y	72.0/ 100	86.8	83.5	81.8	1.67	0.25	-1.2	-1.8	3m25y	9m25y	3m6m25y	58.0/ 100	87.6	86.8	86.4	1.67	0.19	-1.2	-1.5
6m30y	1y30y	6m6m30y	71.7/ 100	85.3	83.2	81.5	1.90	0.24	-1.2	-1.7	1y20y	18m20y	1y6m20y	82.2/ 100	88.7	86.8	86.0	1.32	0.19	-1.2	-1.6

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Note: Close 6/9/23. The table reports 20 best forward vol trades on the efficient frontier based on ex-ante Sharpe ratio and 1y z-score. We analyze both calendar spreads (underlying forwards with matching mid-points) and triangles. The ex-ante Sharpe is calculated as vol-adjusted roll on the surface. Source: Citi Research

USD forward volatilities

We calculate forward 1y2y, 1y5y, 1y10y and 1y30y volatilities implied from the spot swaption surface. Forward volatilities reflect market expectations about future short-dated vols and may highlight dislocations on the spot vol surface.

- Short-dated forward vols have cheapened recently while long-dated forward vols have richened, causing the forward vol term structures to steepen (Figure 23).

- 1y2y vol is no longer expected to converge and dip below 1y5y vol as it was the case 3 months ago (Figure 28).

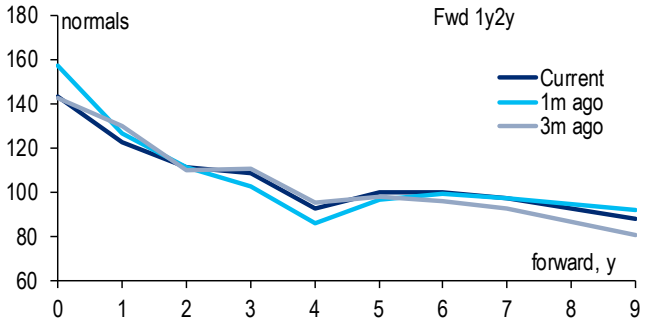
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Figure 23. Forward volatilities

	1y2y							1y5y							1y10y							1y30y						
Yrs		1w	1m	vs 1y		vs 10y			1w	1m	vs 1y		vs 10y			1w	1m	vs 1y		vs 10y			1w	1m	vs 1y		vs 10y	
Fwd	Vol	chg	chg	avg	z-scr	spt	z-scr	Vol	chg	chg	avg	z-scr	spt avg	z-scr	Vol	chg	chg	avg	z-scr	spt avg	z-scr	Vol	chg	chg	avg	z-scr	spt	z-scr
spt	143.0	(7.0)	(13.8)	(1.9)	(0.2)	70.4	2.3	121.8	(4.3)	(7.5)	(7.0)	(0.7)	41.9	1.9	103.8	(3.1)	(5.2)	(9.4)	(1.0)	24.2	1.5	83.2	(2.1)	(7.1)	(11.5)	(1.5)	8.9	0.8
1	122.6	(3.8)	(3.8)	(6.2)	(0.6)	50.1	1.6	112.1	(1.3)	(5.8)	(4.2)	(0.5)	32.2	1.5	97.7	(2.5)	(4.0)	(3.9)	(0.6)	18.1	1.1	77.5	(1.6)	(4.1)	(5.0)	(1.0)	3.2	0.3
2	111.4	(3.6)	0.2	(3.9)	(0.5)	38.8	1.3	106.7	(1.8)	(4.1)	(1.7)	(0.3)	26.8	1.2	92.7	(1.2)	(3.1)	(1.4)	(0.3)	13.0	0.8	71.9	(0.4)	(4.3)	(0.8)	(0.2)	(2.4)	(0.2)
3	108.3	0.2	6.0	(4.2)	(0.4)	35.7	1.2	101.3	(0.8)	(0.2)	0.0	0.0	21.4	1.0	88.7	(0.7)	(0.4)	(1.9)	(0.5)	9.0	0.6	68.3	(0.4)	1.5	1.9	0.6	(6.0)	(0.5)
4	92.9	3.2	6.7	(6.0)	(0.7)	20.3	0.7	89.4	(0.2)	1.6	(5.0)	(1.0)	9.5	0.4	83.7	(0.3)	(0.7)	(2.0)	(0.5)	4.1	0.3	65.8	(0.6)	3.1	1.7	0.6	(8.4)	(0.7)
5	99.7	0.8	2.9	(1.3)	(0.2)	27.1	0.9	91.0	0.5	1.2	(2.2)	(0.4)	11.1	0.5	81.6	0.1	0.0	(1.5)	(0.4)	1.9	0.1	62.8	(0.7)	(0.3)	1.1	0.5	(11.5)	(1.0)
6	99.6	1.6	0.4	1.7	0.3	27.0	0.9	89.8	1.4	1.0	0.7	0.1	9.9	0.4	80.7	0.5	0.3	0.8	0.2	1.0	0.1	59.4	(0.6)	(1.2)	1.2	0.6	(14.8)	(1.3)
7	97.0	5.3	(0.3)	3.0	0.5	24.4	0.8	84.9	2.3	1.4	2.2	0.6	5.0	0.2	76.5	1.2	0.1	1.4	0.5	(3.1)	(0.2)	57.8	0.1	1.8	3.3	1.7	(16.4)	(1.4)
8	92.6	5.4	(1.9)	4.0	0.6	20.0	0.6	78.0	2.3	0.8	3.3	0.9	(1.9)	(0.1)	71.9	1.5	(0.4)	2.3	1.0	(7.7)	(0.5)	53.6	0.4	3.2	3.1	1.4	(20.7)	(1.8)
9	87.7	3.4	(4.2)	1.6	0.2	15.2	0.5	77.6	1.3	(0.8)	3.2	0.8	(2.3)	(0.1)	70.6	0.8	(1.4)	1.4	0.6	(9.0)	(0.6)	53.4	0.1	1.3	2.5	1.4	(20.9)	(1.8)

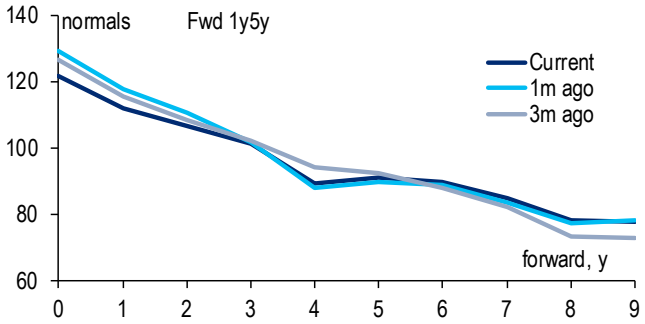
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Note: Close 6/9/23. Forward vols are computed using the standard triangular approximation from spot vols and using 6m realized correlations as a proxy for implied correlations. In addition to levels of forward vols, we calculate 1w and 1m changes, the difference from 1y average (with corresponding z-scores), and the difference from the 10y average of the corresponding 1y expiry spot vol (together with z-scores). Source: Citi Research

Figure 24. Forward vol of 1y2y



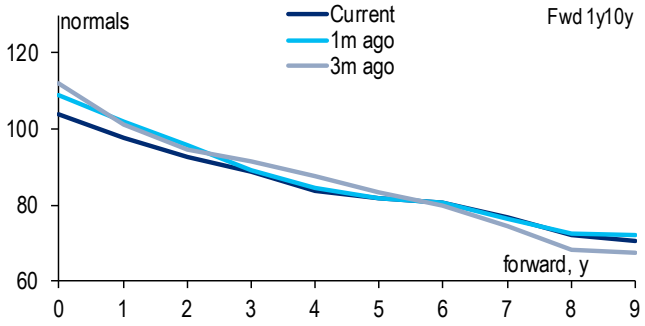
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Note: Close 6/9/23. Source: Citi Research

Figure 25. Forward vol of 1y5y



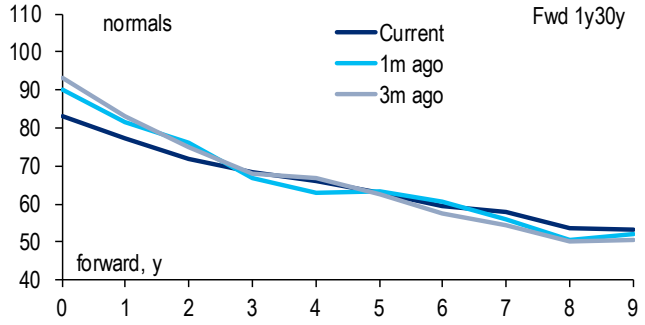
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Figure 26. Forward vol of 1y10y



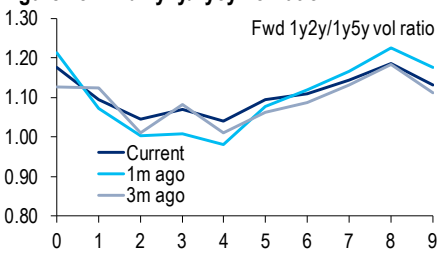
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Figure 27. Forward vol of 1y30y



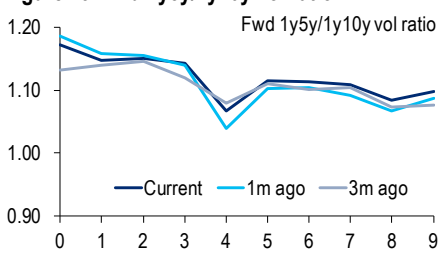
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Note: Close 6/9/23. Source: Citi Research

Figure 28. Fwd 1y2y/1y5y vol ratio



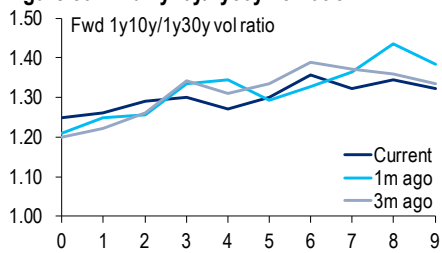
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Figure 29. Fwd 1y5y/1y10y vol ratio



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Figure 30. Fwd 1y10y/1y30y vol ratio



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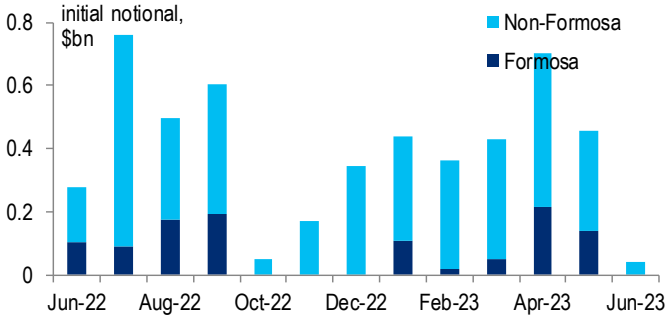
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Long-dated callable USD supply

We report supply and redemptions data for long-dated callable bonds.

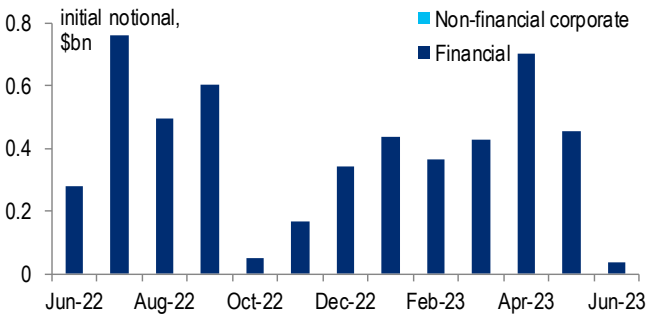
- After peaking in April, long-maturity callable gross issuance has trended lower in May and will likely again in June. The pace of issuance this month is well behind the pace for last June. Overall gross issuance should remain low for the rest of the year given the lack of foreign investors' appetite in the near term.

Figure 31. Gross callable supply: Formosa vs Non-Formosa



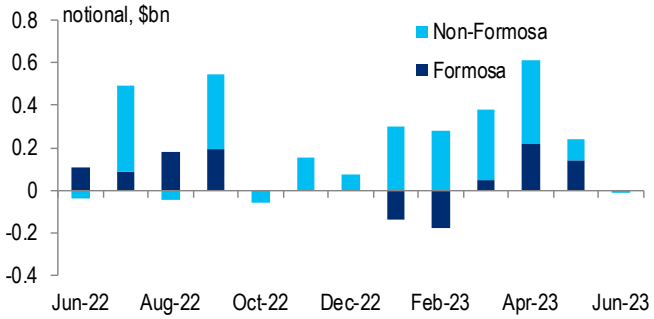
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Note: We include all callable Formosa (zero-coupon and fixed-rate) and zero-coupon non-Formosa bonds. Source: Citi Research

Figure 32. Gross callable supply: Financial vs Non-Financial



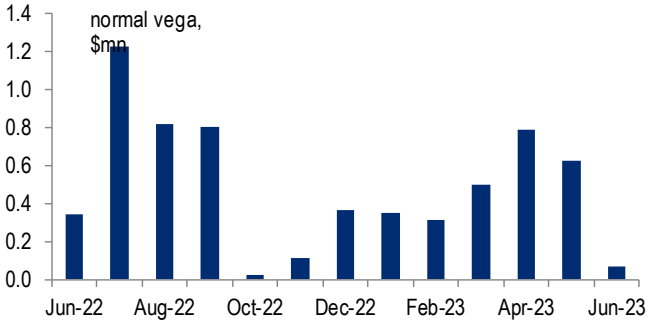
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Note: We include all callable Formosa (zero-coupon and fixed-rate) and zero-coupon non-Formosa bonds. Source: Citi Research

Figure 33. Net callable notional supply



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Source: Citi Research

Figure 34. Vega supply (normal parallel vega)



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Note: We assume (simplistically) that all financial and government bonds are swapped, while all non-financial bonds are not swapped. Source: Citi Research

Figure 35. Recent callable supply estimates

	gross issued, initial notional, $bn					redeemed, $bn					net issued, $bn					Vega, $mn
	Formosa	Non-Formosa	Financial	Non-Financial	All	Formosa	Non-Formosa	Financial	Non-Financial	All	Formosa	Non-Formosa	Financial	Non-Financial	All	
Jun-23 MTD	0.00	0.04	0.04	0.00	0.04	0.00	0.05	0.05	0.00	0.05	0.00	-0.01	-0.01	0.00	-0.01	0.1
May-23 (prev month)	0.14	0.32	0.46	0.00	0.46	0.00	0.22	0.22	0.00	0.22	0.14	0.10	0.24	0.00	0.24	0.6
Jun-22 (prev year)	0.11	0.18	0.28	0.00	0.28	0.00	0.22	0.22	0.00	0.22	0.11	-0.04	0.06	0.00	0.06	0.3
2023 YTD	0.54	1.90	2.44	0.00	2.44	0.45	0.51	0.95	0.00	0.95	0.09	1.39	1.48	0.00	1.48	2.6
2022 YTD	11.08	3.23	13.37	0.93	14.30	9.10	1.81	9.44	1.47	10.91	1.97	1.42	3.93	-0.54	3.39	36.0
2022 (all year)	11.54	5.19	15.80	0.93	16.73	9.10	2.89	10.53	1.47	12.00	2.44	2.30	5.28	-0.54	4.74	39.3

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Note: To estimate vega, we assume (simplistically) that all financial and government bonds are swapped, while all non-financial bonds are not swapped. Source: Citi Research

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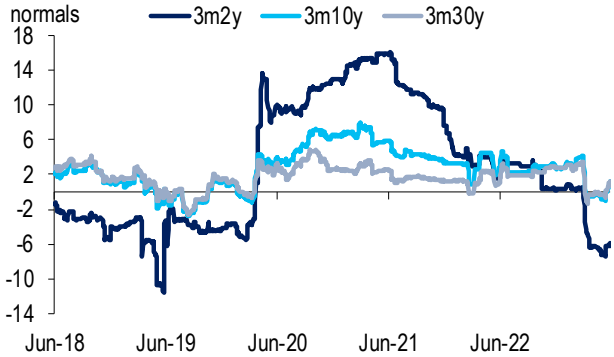
USD swaption skew

USD skew history and z-scores

We calculate z-scores of payer and receiver swaption skews across the surface.

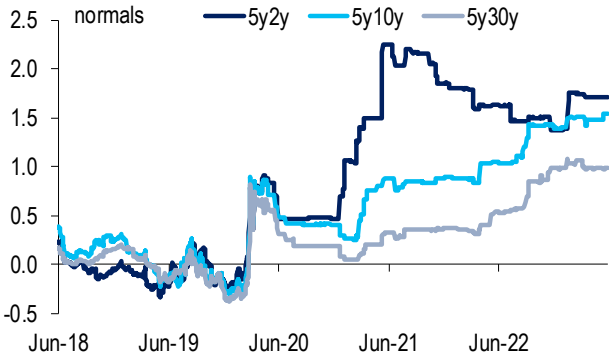
- Payer skews have rebounded recently on the concerns of a breakout of the range in yields on the high side. However, we expect the recent moves in payer skews to be short-lived and see scope for short-dated receiver skews to outperform (US Rates Weekly - Oops... We Did It Again).

Figure 36. 3m expiry payer skew



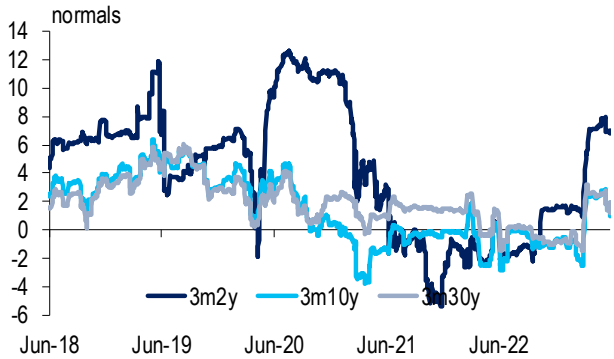
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Note: 25bp-out payer skews for 3m expiry payers. Source: Citi Research

Figure 37. 5y expiry payer skew



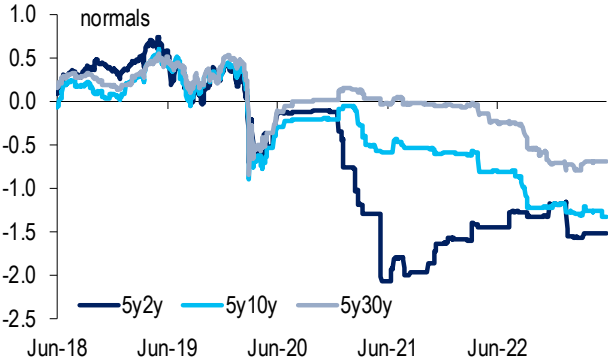
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Note: 25bp-out payer skews for 5y expiry payers. Source: Citi Research

Figure 38. 3m expiry receiver skew



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Note: 25bp-out receiver skews for 3m expiry receivers. Source: Citi Research

Figure 39. 5y expiry receiver skew



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Note: 25bp-out receiver skews for 5y expiry receivers. Source: Citi Research

Figure 40. Skews (1y z-scores) of risk reversals, OTM payer and OTM receivers

	Risk Reversals						Payer skew						Receiver skew					
expr\tail	1y	2y	5y	7y	10y	30y	1y	2y	5y	7y	10y	30y	1y	2y	5y	7y	10y	30y
3m	-17.7 (-1.5)	-12.7 (-1.6)	-2.9 (-1.0)	-1.5 (-0.9)	0.2 (-0.7)	0.0 (-0.6)	-8.6 (-1.6)	-5.9 (-1.6)	-0.7 (-1.0)	0.2 (-0.9)	1.2 (-0.6)	1.2 (-0.5)	9.1 (1.5)	6.8 (1.6)	2.3 (1.0)	1.7 (0.9)	1.0 (0.8)	1.2 (0.8)
6m	-11.5 (-1.6)	-7.6 (-1.7)	-1.1 (-1.1)	0.0 (-1.2)	1.2 (-1.2)	0.4 (-1.1)	-5.6 (-1.6)	-3.5 (-1.7)	-0.1 (-1.1)	0.5 (-1.1)	1.2 (-1.1)	0.9 (-1.0)	6.0 (1.6)	4.1 (1.7)	1.0 (1.1)	0.6 (1.2)	0.0 (1.2)	0.6 (1.2)
1y	-4.7 (-1.4)	-1.3 (-1.1)	0.4 (-1.4)	1.5 (-1.6)	2.6 (-1.7)	2.0 (-0.5)	-2.2 (-1.5)	-0.4 (-1.1)	0.5 (-1.4)	1.1 (-1.6)	1.7 (-1.6)	1.5 (-0.3)	2.5 (1.4)	0.9 (1.1)	0.1 (1.4)	-0.4 (1.6)	-0.9 (1.8)	-0.5 (0.7)
2y	3.3 (0.2)	3.3 (0.3)	2.9 (-1.0)	3.0 (-1.3)	3.2 (-1.4)	1.8 (-0.2)	1.8 (0.2)	1.8 (0.3)	1.6 (-1.0)	1.7 (-1.2)	1.8 (-1.3)	1.2 (0.0)	-1.5 (-0.2)	-1.5 (-0.3)	-1.3 (1.0)	-1.3 (1.3)	-1.4 (1.5)	-0.6 (0.4)
3y	3.7 (0.6)	3.8 (1.4)	3.3 (0.4)	3.2 (0.8)	3.1 (0.8)	2.1 (1.0)	2.0 (0.6)	2.0 (1.4)	1.7 (0.5)	1.7 (0.9)	1.7 (0.9)	1.2 (1.0)	-1.8 (-0.5)	-1.8 (-1.3)	-1.5 (-0.4)	-1.5 (-0.7)	-1.4 (-0.6)	-0.9 (-0.9)
5y	3.3 (0.8)	3.2 (1.0)	2.7 (0.1)	2.7 (0.6)	2.9 (1.0)	1.7 (0.7)	1.7 (0.9)	1.7 (1.0)	1.5 (0.2)	1.5 (0.7)	1.5 (1.1)	1.0 (0.7)	-1.5 (-0.8)	-1.5 (-1.0)	-1.3 (0.0)	-1.2 (-0.6)	-1.3 (-1.0)	-0.7 (-0.6)
10y	2.0 (2.8)	2.0 (2.8)	2.0 (1.1)	2.2 (1.2)	2.4 (1.3)	1.5 (1.3)	1.1 (2.8)	1.1 (2.8)	1.1 (1.1)	1.2 (1.2)	1.3 (1.3)	0.9 (1.3)	-0.9 (-2.7)	-0.9 (-2.8)	-0.9 (-1.1)	-1.0 (-1.2)	-1.1 (-1.3)	-0.6 (-1.3)

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Note: Close 6/9/23. The table reports skews and 1y z-scores in parentheses for risk reversals, payer and receiver skews, defined in normal vol terms. We use 25bp OTM. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

Prepared for Christopher Lee

USD implied vs realized skew

We compare implied swaption skews with realized skews. We calculate realized skews as the model value of OTM options from the empirical SABR model calibrated to the recent (3m) realized dynamics of ATM volatilities and rates.

- Our model is showing that payer skews are overvalued and receiver skews are undervalued across the entire swaption surface. For example, the spread to the model for 3m10y risk reversal is currently at +2.9 sigmas (Figure 41). This is consistent with our view mentioned above that there is room for risk reversals to retrace/decline.

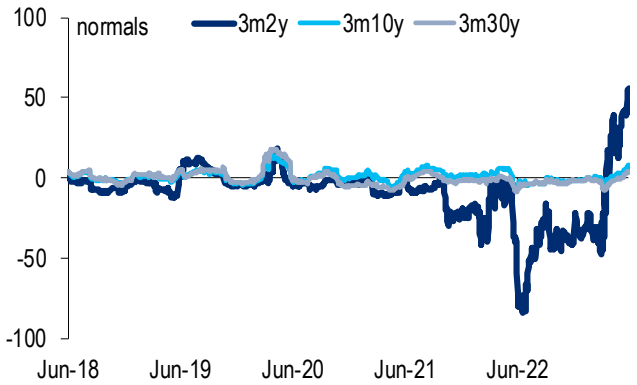
Figure 41. Implied vs realized skews (spread to the model for risk reversals, price/model ratio for payers and receivers)

	Risk Reversals (1Y z-score)					OTM payers (1Y z-score)					OTM receivers (1Y z-score)				
expr\tail	1y	2y	5y	10y	30y	1y	2y	5y	10y	30y	1y	2y	5y	10y	30y
3m	80.5 (1.9)	40.9 (2.1)	5.5 (2.7)	6.0 (2.9)	3.0 (2.5)	1.00 (1.3)	1.01 (1.6)	1.03 (2.9)	1.04 (3.2)	1.03 (2.9)	0.65 (-1.6)	0.80 (-1.8)	0.99 (-2.5)	0.98 (-2.8)	0.99 (-2.6)
6m	49.3 (2.0)	22.8 (2.1)	5.6 (2.4)	5.5 (2.2)	2.2 (2.1)	1.01 (1.7)	1.02 (2.0)	1.03 (2.5)	1.03 (2.4)	1.02 (2.3)	0.77 (-1.7)	0.89 (-1.9)	0.98 (-2.3)	0.98 (-2.3)	0.99 (-2.2)
1y	3.8 (2.2)	0.3 (2.4)	4.6 (2.2)	4.9 (2.0)	2.4 (1.9)	1.02 (2.0)	1.03 (2.4)	1.02 (2.3)	1.03 (2.1)	1.02 (2.0)	1.00 (-2.0)	1.02 (-2.1)	0.98 (-2.2)	0.98 (-2.0)	0.99 (-2.0)
2y	-21.7 (2.4)	-18.9 (2.5)	4.9 (2.4)	4.1 (2.2)	1.4 (2.0)	1.03 (2.7)	1.03 (2.6)	1.02 (2.6)	1.02 (2.3)	1.01 (2.1)	1.24 (-2.3)	1.22 (-2.4)	0.98 (-2.5)	0.98 (-2.3)	0.99 (-2.1)
3y	-24.8 (2.3)	-22.0 (2.3)	4.2 (2.3)	3.4 (2.3)	1.5 (2.5)	1.03 (2.5)	1.03 (2.3)	1.02 (2.5)	1.02 (2.5)	1.01 (2.6)	1.31 (-2.3)	1.29 (-2.2)	0.98 (-2.4)	0.98 (-2.4)	0.99 (-2.6)
5y	3.4 (1.9)	3.5 (1.7)	2.6 (1.5)	2.5 (2.0)	1.2 (1.9)	1.02 (2.0)	1.02 (1.7)	1.01 (1.7)	1.02 (2.1)	1.01 (2.0)	0.99 (-2.0)	0.98 (-1.7)	0.99 (-1.6)	0.99 (-2.1)	0.99 (-2.0)
10y	1.8 (1.7)	1.9 (1.7)	1.9 (1.8)	1.8 (1.8)	0.8 (1.5)	1.01 (1.7)	1.01 (1.7)	1.01 (1.9)	1.01 (1.9)	1.01 (1.5)	0.99 (-1.7)	0.99 (-1.7)	0.99 (-1.9)	0.99 (-1.9)	0.99 (-1.6)

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Note: Close 6/9/23. We calibrate an empirical SABR model to the recent observed dynamics of ATM vols and rates. We use a 2y rolling window to estimate a backbone parameter, and a 3m window to estimate correlation and vol-of-vol parameters. The empirical model matches ATM implied vols exactly, so it's different from the market only in skew parameters. We calculate "realized skew" as the theoretical values of OTM payers, receivers and risk reversals from this empirical model. The table shows market-"realized" spreads of risk reversals, and market/"realized" ratios of OTM payers and receivers together with 1y z-scores of these spreads and ratios in parentheses. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

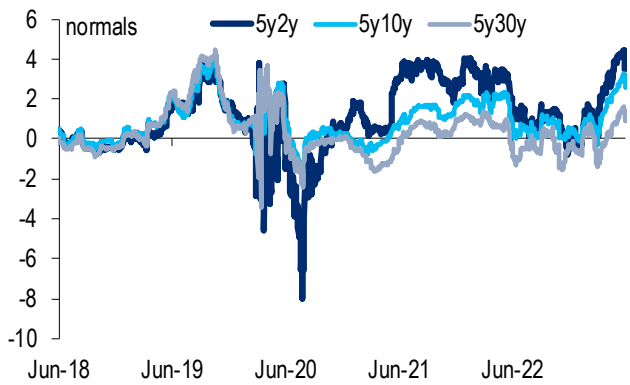
Figure 42. 3m expiry risk reversals/model spread



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Note: The chart shows model-"realized" spreads for 25bp-out risk reversals in 3m expiries using the same methodology as in Figure 41. Source: Citi Research.

Figure 43. 5y expiry risk reversals/model spread



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Note: The chart shows model-"realized" spreads for 100bp-out risk reversals in 5y expiries using the same methodology as in Figure 41. Source: Citi Research.

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USD Board/swaption relative value

We evaluate the relative pricing of Board and swaption volatilities.

- Board vols look to be in-line with their corresponding CTD-matched swaption vols for all maturities. Both board and swaption vols have been trading slightly above their realized vols recently (Figure 44).
- Board puts are trading above swaption payers while board calls are trading below swaption receivers (Figure 46 to Figure 48).
- *Note: Futures trading involves substantial risk of loss.*

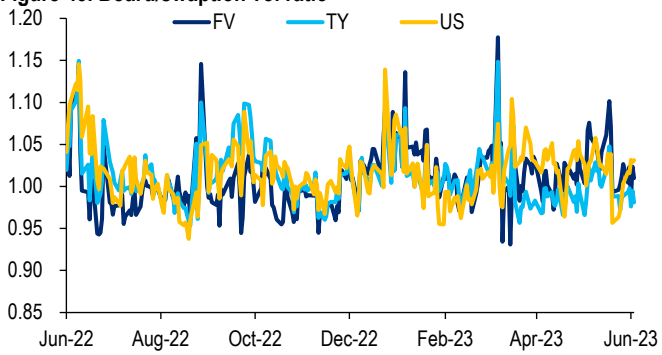
Figure 44. Board vols vs swaptions

option	FVQ3	FVU3	TYQ3	TYU3	USQ3	USU3
Board vol	149.0	148.8	125.5	126.8	99.4	100.0
1w chg	-6.5	-5.1	-3.8	-4.2	1.5	1.2
Board/ 1m RV	1.22	1.21	1.16	1.17	1.17	1.17
Board/ 3m RV	0.77	0.77	0.79	0.80	0.93	0.93
CTD-matched swpn vol	147.5	144.8	128.0	127.3	96.3	98.6
1w chg	-3.9	-4.5	-2.8	-1.9	-0.9	-0.5
swpn/ 1m RV	1.15	1.13	1.16	1.16	1.23	1.26
swpn/ 3m RV	0.77	0.75	0.82	0.82	0.97	0.99
Board/swpn	1.01	1.03	0.98	1.00	1.03	1.01
1w ago	1.03	1.03	0.99	1.01	1.01	1.00
ftrs 1m RV/swap 1m RV	0.96	0.96	0.98	0.98	1.08	1.08
ftrs 3m RV/swap 3m RV	1.00	1.00	1.02	1.02	1.08	1.08

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Note: Close 6/9/23. The table shows implied ATM (interpolated) Board vols vs CTD-matched swaption vols with corresponding realized volatilities. Source: Citi Research

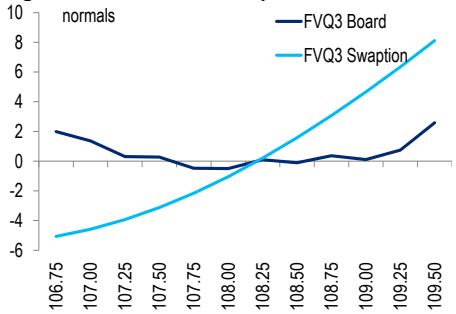
Figure 45. Board/swaption vol ratio



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Note: The chart shows the ratio of ATM Board vol to CTD-matched swaption vols. The second available expiry (rolled on the expiration of front options) is used for the chart. Source: Citi Research

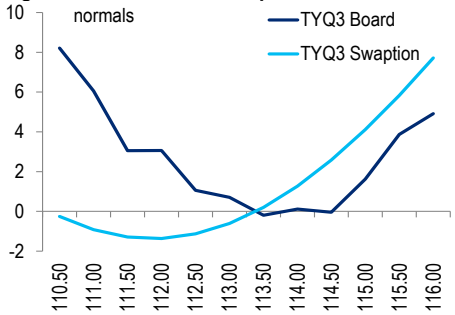
Figure 46. FV Board vs swaption skew



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Note: Close 6/9/23. Swaption strike= Board strike yield+invoice spread. Source: Citi Research

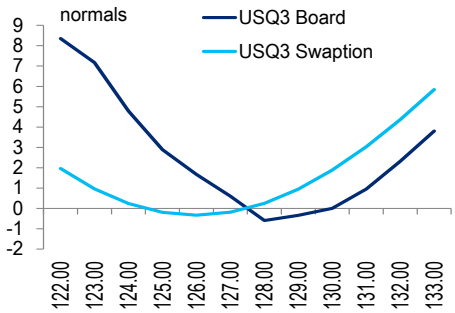
Figure 47. TY Board vs swaption skew



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Note: Close 6/9/23. Swaption strike= Board strike yield+invoice spread. Source: Citi Research

Figure 48. US Board vs swaption skew



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Note: Close 6/9/23. Swaption strike= Board strike yield+invoice spread. Source: Citi Research

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Figure 49. Conditional spread trades

							puts/payers								calls/receivers							
optn	expiry	futures price	fut DV01	swap DV01	swpn ntl, $mn	invc sprd, bp	put strike	strike yield, %	costless payer strike,	strike spread, bp	bp pick-up	put vol	payer vol	vol ratio	call strike	strike yield, %	costless receiver strike, %	strike spread, bp	bp pick-up	call vol	rcvr vol	vol ratio
FVM3	05/26/23	109-7.25	4.22	3.84	109.8	-25.0	109.00	3.76	3.56	-19.6	-5.4	142.8	142.5	1.00	109.25	3.70	3.47	-22.4	-2.7	143.7	144.1	1.00
FVM3	05/26/23	109-7.25	4.22	3.84	109.8	-25.0	108.75	3.82	3.62	-19.5	-5.5	140.9	140.6	1.00	109.50	3.64	3.41	-22.4	-2.6	145.0	145.8	0.99
FVM3	05/26/23	109-7.25	4.22	3.84	109.8	-25.0	108.50	3.88	3.68	-19.9	-5.2	141.0	139.5	1.01	109.75	3.58	3.36	-22.4	-2.6	147.0	147.5	1.00
FVM3	05/26/23	109-7.25	4.22	3.84	109.8	-25.0	108.25	3.94	3.74	-19.9	-5.1	140.2	138.5	1.01	110.00	3.52	3.30	-22.1	-2.9	149.7	149.2	1.00
FVM3	05/26/23	109-7.25	4.22	3.84	109.8	-25.0	108.00	4.00	3.79	-20.2	-4.8	140.0	137.6	1.02	110.25	3.46	3.24	-21.9	-3.1	152.0	151.1	1.01
FVM3	05/26/23	109-7.25	4.22	3.84	109.8	-25.0	107.75	4.06	3.85	-20.2	-4.8	139.1	136.9	1.02	110.50	3.40	3.18	-22.0	-3.0	153.8	153.1	1.00
FVN3	06/23/23	109-25.25	4.19	3.74	111.8	-21.5	109.75	3.55	3.48	-7.0	-14.5	151.2	145.1	1.04	110.00	3.49	3.38	-11.1	-10.4	152.8	147.1	1.04
FVN3	06/23/23	109-25.25	4.19	3.74	111.8	-21.5	109.50	3.61	3.53	-7.3	-14.2	150.4	144.1	1.04	110.25	3.43	3.31	-11.2	-10.2	153.0	148.5	1.03
FVN3	06/23/23	109-25.25	4.19	3.74	111.8	-21.5	109.25	3.67	3.59	-7.5	-14.0	149.4	143.2	1.04	110.50	3.37	3.26	-11.1	-10.3	154.3	149.9	1.03
FVN3	06/23/23	109-25.25	4.19	3.74	111.8	-21.5	109.00	3.73	3.65	-7.9	-13.5	149.3	142.4	1.05	110.75	3.31	3.20	-10.6	-10.9	156.6	151.2	1.04
FVN3	06/23/23	109-25.25	4.19	3.74	111.8	-21.5	108.75	3.79	3.70	-8.3	-13.2	148.9	141.7	1.05	111.00	3.25	3.14	-10.6	-10.9	157.7	152.7	1.03
FVN3	06/23/23	109-25.25	4.19	3.74	111.8	-21.5	108.50	3.85	3.76	-8.5	-12.9	148.3	141.1	1.05	111.25	3.19	3.08	-10.7	-10.8	158.9	154.4	1.03
TYM3	05/26/23	114-15	6.54	5.71	114.5	-29.8	114.00	3.68	3.43	-24.6	-5.2	121.8	127.5	0.96	114.50	3.60	3.31	-29.7	0.0	123.6	129.6	0.95
TYM3	05/26/23	114-15	6.54	5.71	114.5	-29.8	113.50	3.76	3.51	-24.9	-4.9	122.8	126.6	0.97	115.00	3.53	3.23	-30.2	0.4	124.5	131.3	0.95
TYM3	05/26/23	114-15	6.54	5.71	114.5	-29.8	113.00	3.83	3.58	-25.0	-4.7	123.2	126.0	0.98	115.50	3.45	3.14	-31.2	1.4	124.6	133.4	0.93
TYM3	05/26/23	114-15	6.54	5.71	114.5	-29.8	112.50	3.91	3.66	-25.0	-4.8	123.4	125.7	0.98	116.00	3.38	3.06	-31.8	2.0	126.4	135.6	0.93
TYM3	05/26/23	114-15	6.54	5.71	114.5	-29.8	112.00	3.99	3.73	-26.5	-3.2	126.9	125.7	1.01	116.50	3.30	2.98	-32.3	2.5	128.5	138.0	0.93
TYM3	05/26/23	114-15	6.54	5.71	114.5	-29.8	111.50	4.07	3.79	-27.9	-1.9	129.7	125.9	1.03	117.00	3.23	2.90	-32.1	2.4	132.0	140.3	0.94
TYN3	06/23/23	115-06+	6.63	5.66	117.0	-29.2	115.00	3.54	3.35	-18.8	-10.4	130.8	129.3	1.01	115.50	3.47	3.23	-23.5	-5.7	130.3	130.9	1.00
TYN3	06/23/23	115-06+	6.63	5.66	117.0	-29.2	114.50	3.62	3.43	-18.9	-10.3	130.2	128.6	1.01	116.00	3.39	3.16	-23.4	-5.7	131.5	132.1	1.00
TYN3	06/23/23	115-06+	6.63	5.66	117.0	-29.2	114.00	3.69	3.50	-19.0	-10.2	129.9	128.1	1.01	116.50	3.32	3.08	-24.0	-5.2	131.8	133.6	0.99
TYN3	06/23/23	115-06+	6.63	5.66	117.0	-29.2	113.50	3.77	3.58	-19.2	-10.0	130.0	127.9	1.02	117.00	3.24	3.01	-23.5	-5.7	134.3	135.0	0.99
TYN3	06/23/23	115-06+	6.63	5.66	117.0	-29.2	113.00	3.85	3.65	-19.7	-9.5	130.8	127.8	1.02	117.50	3.17	2.94	-23.4	-5.8	136.2	136.7	1.00
TYN3	06/23/23	115-06+	6.63	5.66	117.0	-29.2	112.50	3.93	3.72	-20.7	-8.5	132.7	128.0	1.04	118.00	3.10	2.86	-23.6	-5.6	137.7	138.5	0.99
USM3	05/26/23	129-29	15.15	12.29	123.3	-37.5	129.00	3.73	3.38	-35.3	-2.1	97.8	98.7	0.99	130.00	3.66	3.29	-37.4	-0.1	97.6	100.0	0.98
USM3	05/26/23	129-29	15.15	12.29	123.3	-37.5	128.00	3.80	3.44	-35.4	-2.1	97.7	98.0	1.00	131.00	3.60	3.22	-37.8	0.4	97.9	101.4	0.97
USM3	05/26/23	129-29	15.15	12.29	123.3	-37.5	127.00	3.86	3.50	-36.1	-1.4	99.4	97.6	1.02	132.00	3.53	3.15	-38.2	0.7	98.9	102.9	0.96
USM3	05/26/23	129-29	15.15	12.29	123.3	-37.5	126.00	3.93	3.57	-36.4	-1.1	99.7	97.5	1.02	133.00	3.47	3.08	-39.0	1.5	99.2	104.7	0.95
USM3	05/26/23	129-29	15.15	12.29	123.3	-37.5	125.00	4.00	3.63	-37.4	-0.1	101.8	97.5	1.04	134.00	3.40	3.01	-39.6	2.1	100.6	106.7	0.94
USM3	05/26/23	129-29	15.15	12.29	123.3	-37.5	124.00	4.07	3.68	-38.8	1.4	104.4	97.8	1.07	135.00	3.34	2.94	-40.4	3.0	101.8	108.8	0.94
USN3	06/23/23	130-4	15.00	11.90	126.0	-37.4	130.00	3.64	3.31	-32.6	-4.8	103.7	102.3	1.01	131.00	3.57	3.23	-34.4	-3.0	103.6	103.2	1.00
USN3	06/23/23	130-4	15.00	11.90	126.0	-37.4	129.00	3.70	3.38	-32.7	-4.7	103.8	101.7	1.02	132.00	3.51	3.16	-34.8	-2.6	103.6	104.3	0.99
USN3	06/23/23	130-4	15.00	11.90	126.0	-37.4	128.00	3.77	3.44	-33.2	-4.2	104.6	101.4	1.03	133.00	3.44	3.09	-34.9	-2.5	104.5	105.4	0.99
USN3	06/23/23	130-4	15.00	11.90	126.0	-37.4	127.00	3.84	3.51	-33.4	-4.0	104.9	101.3	1.04	134.00	3.38	3.02	-35.3	-2.1	105.1	106.8	0.98
USN3	06/23/23	130-4	15.00	11.90	126.0	-37.4	126.00	3.91	3.57	-33.8	-3.6	105.6	101.4	1.04	135.00	3.31	2.95	-35.9	-1.5	105.8	108.3	0.98
USN3	06/23/23	130-4	15.00	11.90	126.0	-37.4	125.00	3.98	3.63	-34.6	-2.8	107.0	101.6	1.05	136.00	3.25	2.89	-35.9	-1.5	107.6	109.8	0.98

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Note: Close 6/9/23. Swaption notional is for 1000 Board contracts. The costless swaption strike makes the DV01-neutral trade costless. The BP pick-up is the spread between the costless strike spread and invoice spread. Futures trading involves substantial risk of loss. Source: Citi Research

USD midcurve swaptions

We report levels and z-scores of at-the-money vols, implied correlations, and implied/realized ratios of correlations. We also compute the relative richness/cheapness of midcurve vols to principal components of the plain vanilla surface.

- Midcurve vols on 5y5y continue to stand out as being cheap to plain vanilla swaption vols when evaluated by both the implied correlations (Figure 51) and the multi-variable regressions (Figure 53).

Figure 50. Implied vols, normals (3m z-score)

ex\tl	1y1y	2y1y	3y1y	1y2y	2y2y	3y2y	5y5y	10y10y
1m	175.2 (-1.6)	146.7 (-1.3)	149.8 (-0.5)	154.1 (-1.5)	144.2 (-1.0)	146.1 (-0.4)	86.0 (-1.4)	72.5 (-1.1)
3m	164.4 (-1.2)	138.9 (-1.1)	145.6 (-0.5)	146.6 (-1.2)	138.4 (-0.9)	142.3 (-0.4)	92.5 (-1.4)	76.8 (-1.0)
6m	158.4 (-0.8)	135.5 (-0.6)	140.5 (-0.4)	144.0 (-0.8)	134.8 (-0.5)	139.3 (-0.3)	96.5 (-1.4)	78.8 (-1.1)
1y	160.9 (-0.5)	145.5 (-0.4)	138.2 (-0.3)	151.0 (-0.4)	139.6 (-0.4)	138.1 (-0.3)	97.3 (-1.3)	78.3 (-1.4)
2y	122.1 (-1.4)	109.8 (-1.2)	97.3 (-1.2)	114.9 (-1.3)	102.2 (-1.2)	94.9 (-1.1)	93.9 (-1.3)	76.6 (-1.3)

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Close 6/9/23. The table shows ATM implied vols and 3m z-scores in parentheses. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

Figure 51. Implied correlations, % (3y z-score)

ex\tl	1y1y	2y1y	3y1y	1y2y	2y2y	3y2y	5y5y	10y10y
1m	105.7 (1.0)	97.9 (0.7)	98.0 (0.3)	102.2 (1.0)	92.3 (0.6)	95.2 (0.2)	95.3 (1.2)	97.7 (1.1)
3m	102.0 (0.9)	99.3 (0.7)	98.4 (0.2)	103.5 (1.1)	95.4 (0.5)	95.8 (-0.2)	95.7 (1.3)	97.7 (0.9)
6m	99.7 (0.8)	99.3 (0.4)	98.3 (-0.3)	99.1 (0.7)	94.8 (0.1)	95.0 (-0.6)	95.9 (1.3)	97.7 (0.9)
1y	90.4 (-1.0)	97.2 (-0.5)	98.0 (-0.8)	78.5 (-0.8)	90.6 (-0.7)	93.5 (-0.8)	95.9 (1.5)	97.5 (0.5)
2y	98.6 (0.3)	100.0 (1.6)	100.1 (1.3)	99.1 (1.0)	100.8 (1.6)	100.7 (1.4)	96.3 (1.6)	97.5 (-0.4)

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Close 6/9/23. The table shows implied correlations (3y z-scores in parentheses). Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

Prepared for Christopher Lee

Figure 52. Implied/realized correlation ratios (1y z-score)

ex\tl	1y1y	2y1y	3y1y	1y2y	2y2y	3y2y	5y5y	10y10y
1m	1.09 (1.4)	0.98 (-0.3)	0.98 (-1.4)	1.08 (0.9)	0.93 (-0.8)	0.96 (-0.9)	0.98 (-0.6)	1.01 (-0.1)
3m	1.04 (1.7)	1.00 (0.1)	0.99 (-1.2)	1.08 (1.5)	0.96 (-0.5)	0.97 (-0.5)	0.98 (-0.4)	1.01 (-0.2)
6m	1.01 (0.9)	1.00 (0.2)	0.99 (-1.1)	1.02 (0.9)	0.96 (-0.4)	0.96 (-0.6)	0.98 (-0.1)	1.01 (0.0)
1y	0.91 (-0.7)	0.98 (-0.7)	0.98 (-0.8)	0.80 (-0.6)	0.92 (-0.7)	0.94 (-0.5)	0.99 (0.7)	1.01 (0.1)
2y	1.00 (-0.9)	1.00 (1.3)	1.00 (1.6)	1.02 (0.7)	1.02 (1.8)	1.02 (1.8)	0.99 (1.7)	1.01 (0.3)

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Close 6/9/23. The table shows the ratios of implied to 3m realized correlations (in daily changes). The 1y z-scores of ratios are shown in parentheses. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

Figure 53. Rich/cheap to plain vanilla swaptions (z-score)

ex\tl	1y1y	2y1y	3y1y	1y2y	2y2y	3y2y	5y5y	10y10y
1m	-23.0 (-2.9)	6.9 (0.8)	12.4 (1.4)	-12.7 (-1.7)	8.2 (1.0)	2.8 (0.3)	-1.3 (-0.6)	-0.9 (-0.4)
3m	-15.2 (-2.7)	-5.4 (-0.9)	1.0 (0.2)	-12.9 (-2.7)	-3.5 (-0.7)	-3.3 (-0.5)	-2.3 (-1.2)	-0.7 (-0.3)
6m	-12.9 (-2.0)	-7.5 (-1.3)	2.2 (0.4)	-10.9 (-2.0)	-3.2 (-0.7)	-0.7 (-0.1)	-2.5 (-1.4)	-0.3 (-0.1)
1y	4.5 (0.6)	3.5 (0.6)	5.4 (0.9)	3.8 (0.6)	3.4 (0.6)	4.2 (0.6)	-2.6 (-1.7)	0.7 (0.5)
2y	-0.1 (-0.1)	-8.4 (-2.2)	-11.7 (-2.6)	-3.3 (-1.5)	-10.2 (-2.6)	-13.4 (-2.8)	-3.2 (-2.5)	0.6 (0.5)

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Close 6/9/23. We regress midcurve vols on two corresponding plain vanilla vols using last 3y of data. For example, 1y2y1y is regressed against 1y3y and 1y2y vols. We calculate rich/cheap and z-scores in brackets. Source: Citi Research

USD curve (spread) CMS options

We report levels and z-scores of at-the-money vols, rich/cheap on PCA, implied correlations, and implied/realized vol ratios for curve options.

- Implied curve correlations remain at their multi-year lows for 5s30s. This reflects the persistent demand for intermediate curve caps as investors position for a long term 5s30s re-steepening (Figure 56).
- Implied CMS curve vol continues to trade well above realized, especially for curves with the 2y leg (Figure 57). While the implied/realized vol ratio is not necessarily a good leading indicator, it does reflect the current richness of CMS curve options.

Figure 54. Implied vols, normals (3m z-score)

ex\crv	2s5s	2s10s	2s30s	5s10s	5s30s	10s30s
1m	71.4 (-0.8)	108.1 (-0.8)	136.9 (-0.8)	52.0 (-0.9)	92.6 (-1.1)	46.0 (-1.4)
3m	67.9 (-0.8)	102.9 (-0.9)	132.0 (-0.9)	50.6 (-0.8)	92.9 (-0.9)	48.5 (-1.4)
6m	66.3 (-0.8)	102.1 (-0.8)	129.3 (-0.9)	47.3 (-0.9)	89.0 (-0.9)	48.5 (-1.5)
1y	60.2 (-1.0)	96.5 (-1.0)	120.0 (-1.0)	42.5 (-1.1)	83.2 (-1.1)	44.9 (-1.3)
2y	49.8 (-1.4)	77.1 (-1.5)	99.2 (-1.2)	36.6 (-0.9)	71.4 (-0.1)	41.0 (-1.2)
3y	42.9 (-1.3)	66.9 (-1.3)	84.5 (-1.3)	31.4 (-0.7)	60.1 (-0.8)	36.0 (-1.1)
5y	27.5 (-0.6)	55.2 (-0.7)	64.3 (-0.6)	22.5 (-0.9)	45.3 (-1.0)	28.8 (-1.3)

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Close 6/9/23. The table shows ATM implied vols and 3m z-scores in parentheses. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

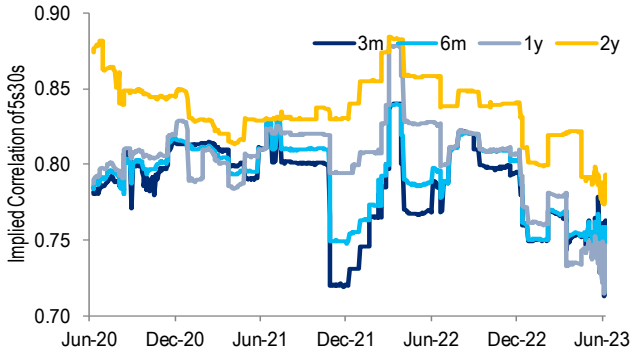
Figure 55. Rich/cheap to PCA, normals (z-score)

ex\crv	2s5s	2s10s	2s30s	5s10s	5s30s	10s30s
1m	-1.7 (-0.2)	2.4 (0.3)	3.3 (0.4)	-1.3 (-0.5)	-4.0 (-1.1)	-7.6 (-2.6)
3m	-1.6 (-0.2)	2.2 (0.4)	4.7 (0.9)	0.3 (0.2)	1.6 (0.5)	-2.5 (-0.9)
6m	-0.6 (-0.1)	3.9 (0.9)	6.5 (1.7)	0.2 (0.1)	2.6 (1.1)	-0.2 (-0.1)
1y	-0.6 (-0.2)	2.1 (0.8)	3.9 (1.4)	-1.6 (-0.9)	2.6 (1.0)	1.1 (0.6)
2y	-0.6 (-0.2)	-3.1 (-1.8)	0.3 (0.1)	-1.3 (-1.0)	3.5 (1.4)	0.8 (0.5)
3y	-0.4 (-0.2)	-3.7 (-1.4)	-0.9 (-0.4)	-0.8 (-0.7)	0.5 (0.3)	0.3 (0.3)
5y	-0.2 (-0.3)	1.5 (1.1)	0.3 (0.3)	-0.4 (-0.8)	0.3 (0.4)	0.3 (0.2)

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Close 6/9/23. We calculate rich/cheap and z-scores in parentheses from PCA on curve vols using two principal components and 3y of data. By design, this framework only yields relative value. Source: Citi Research

Figure 56. Implied correlations of 5s30s CMS options



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Note: The chart shows implied correlation of options on the 5s30s curve with different expirations. Source: Citi Research

Figure 57. Implied/realized ratios

implied/1m RV (1y z-score)

ex\crv	2s5s	2s10s	2s30s	5s10s	5s30s	10s30s
1m	2.55 (2.6)	1.82 (2.1)	1.45 (1.5)	1.45 (0.7)	1.23 (0.5)	1.09 (0.0)
3m	2.18 (2.0)	1.59 (1.7)	1.31 (1.2)	1.37 (0.6)	1.21 (0.7)	1.14 (0.3)
6m	1.82 (1.5)	1.42 (1.3)	1.19 (0.9)	1.27 (0.5)	1.16 (0.6)	1.15 (0.4)
1y	1.53 (1.3)	1.32 (0.9)	1.10 (0.6)	1.21 (0.1)	1.14 (0.5)	1.13 (0.5)

implied/3m RV (1y z-score)

ex\crv	2s5s	2s10s	2s30s	5s10s	5s30s	10s30s
1m	1.48 (0.5)	1.17 (0.1)	0.97 (-0.4)	1.04 (-1.1)	0.91 (-0.8)	0.85 (-0.9)
3m	1.25 (-0.1)	1.02 (-0.4)	0.88 (-0.7)	0.98 (-1.2)	0.90 (-0.7)	0.89 (-0.4)
6m	1.14 (-0.4)	0.97 (-0.6)	0.84 (-0.7)	0.93 (-1.3)	0.87 (-0.7)	0.92 (-0.1)
1y	1.10 (-0.7)	1.00 (-0.9)	0.86 (-0.9)	0.92 (-1.6)	0.91 (-0.5)	0.94 (0.3)

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Close 6/9/23. Close-to-close realized vol is used. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

Prepared for Christopher Lee

SOFR futures convexity adjustments

We calculate SOFR/FRA convexity adjustment for 1y SOFR packs and compare them to model values calibrated to cap/floor volatilities.

- Convexity adjustments on both Blues (Figure 59) and Golds (Figure 60) have finally started to decline and converge towards our model fair values. With that said, both convexity adjustments still look rich and has room to cheapen further, especially given that implied vols have declined to the low end of their 1-year range. We continue to favor shorting Blues convexity adjustment as a short vol proxy (Rates Vol Lab - Short Blues convexity via futures/swaps).
- *Note: Futures trading involves substantial risk of loss*

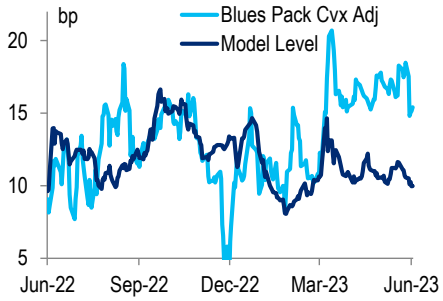
Figure 58. Convexity adjustments for 1y SOFR futures packs (vs CME swaps)

	convexity adjustment					vs the model				implied vs realized vol			
ED Pack	Cvx Adj (bp)	1 Week Chg (bp)	3m Z-Score	1Y Z-Score	Model	Vs Model	3m Z-Score	1Y Z-Score	3m Roll (short cvx,	Implied Vol	Realized Vol	Implied/ Realized	Cap vol Impl/Rlzd
M4-H5	4.03	1.23	0.78	0.45	2.94	1.09	0.92	0.54	0.88	199.5	229.7	0.9	0.7
U4-M5	4.41	0.98	**0.79**	0.27	3.79	0.62	**0.98**	0.43	0.38	178.1	201.4	0.9	0.8
Z4-U5	5.16	0.71	0.63	0.19	4.66	0.49	0.93	0.42	0.74	167.7	180.3	0.9	0.8
H5-Z5	6.10	0.29	0.40	0.24	5.53	0.58	0.83	0.55	0.95	161.6	166.3	1.0	0.9
M5-H6	8.24	-0.40	-0.52	0.21	6.36	1.88	0.05	0.51	**2.14**	168.5	156.4	1.1	0.9
U5-M6	9.77	-1.22	-0.73	0.36	7.16	2.60	0.06	0.82	1.52	166.3	148.8	1.1	0.9
Z5-U6	11.70	-1.89	-0.82	0.55	8.02	3.67	0.13	**1.10**	1.93	166.5	142.2	1.2	0.9
H6-Z6	13.70	-2.37	-0.80	**0.70**	8.96	4.74	0.22	**1.25**	2.00	166.0	136.1	1.2	0.9
M6-H7	15.40	-3.08	-0.96	**0.73**	9.98	5.42	-0.07	**1.18**	1.70	163.1	130.9	1.2	1.0
U6-M7	16.84	-2.97	-1.30	**0.63**	11.10	5.74	-0.42	1.05	1.44	159.0	126.2	1.3	1.0
Z6-U7	18.27	-2.72	-1.55	0.47	12.23	**6.04**	-0.77	0.90	1.43	155.1	121.9	**1.3**	1.0
H7-Z7	20.08	-2.33	-1.54	0.40	13.37	**6.71**	-0.88	0.82	1.81	152.8	118.0	**1.3**	1.0
M7-H8	22.29	-1.78	-1.35	0.45	14.50	**7.79**	-0.69	0.90	**2.21**	151.9	113.8	**1.3**	1.0

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Note: Close 6/9/23. Convexity adjustments for 1y SOFR packs are computed as the spread between the pack's rate (the average of 4 SOFR rates in the pack) and matched-maturity forward 1y CME swap rate. The model for convexity adjustment is the Ho-Lee model calibrated to cap/floor vols. Implied vol is calculated by matching the model to the observed convexity adjustment. Realized vol is 3m realized vol of the corresponding pack. For each valuation metric, we mark three best short convexity trades in **bold**. Note that these are the most attractive in the chart but not necessarily trades that we would recommend at this point in time. Source: Citi Research

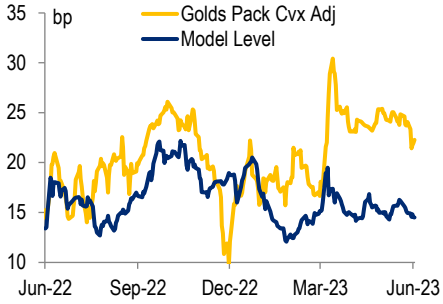
Figure 59. Blues pack (rolling) vs cap/floor-based model



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The chart shows the convexity adjustment of the Blues pack vs the (Ho-Lee) model. Source: Citi Research

Figure 60. Golds pack (rolling) vs cap/floor-based model



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The chart shows the convexity adjustment of the Golds pack vs the (Ho-Lee) model. Source: Citi Research

USD conditional curves

We analyze conditional curves based on the entry point, implied/realized betas, and carry.

Prepared for Christopher Lee

Figure 61. Conditional curves: bear trades

		spt and fwd crvs			costless strk curve				imp vs rlzd beta			bear steepener				bear flattener			
structure	ntlns, $mn	spot crv, bp	fwd crv, bp	3m ZS	strk crv, bp	1m chg	3m ZS	hist %	imp beta	rlzd beta	rlzd-imp	crv pick-up, bp	vol pick-up, bp	tot pick-up, bp	3m roll, bp	crv pick-up, bp	vol pick-up, bp	tot pick-up, bp	3m roll, bp
3m 1y2y	100/50.8	-66.8	-67.4	-0.4	-62.5	5.4	0.1	1	0.09	-0.13	-0.22	0.6	-4.9	-4.3	-4.0	-0.6	4.9	4.3	4.0
3m 1y3y	100/34.5	-109.1	-103.0	-0.8	-101.2	-1.7	-0.5	1	0.03	-0.28	-0.32	-6.1	-1.8	-7.9	-8.1	6.1	1.8	7.9	8.1
3m 1y5y	100/21.4	-147.7	-134.1	-1.2	-137.0	-12.3	-1.0	1	-0.06	-0.46	-0.40	-13.6	2.9	-10.7	-10.9	13.6	-2.9	10.7	10.9
3m 2y5y	100/42.1	-80.9	-66.7	-1.6	-74.8	-17.5	-1.8	1	-0.15	-0.32	-0.17	-14.2	8.1	-6.2	-6.7	14.2	-8.1	6.2	6.7
3m 2y10y	100/22.7	-102.3	-82.0	-1.5	-102.3	-37.2	-1.7	1	-0.39	-0.61	-0.21	-20.2	20.2	0.0	-0.5	20.2	-20.2	0.0	0.5
3m 2y30y	100/10.1	-129.7	-107.0	-1.3	-138.6	-56.8	-1.3	0	-0.69	-1.02	-0.34	-22.7	31.6	8.9	3.4	22.7	-31.6	-8.9	-3.4
3m 3y5y	100/62.0	-38.6	-31.1	-1.5	-36.0	-10.6	-0.8	1	-0.09	-0.17	-0.08	-7.5	4.9	-2.6	-2.6	7.5	-4.9	2.6	2.6
3m 3y10y	100/33.4	-59.9	-46.4	-1.2	-63.4	-30.4	-1.1	0	-0.33	-0.46	-0.12	-13.5	17.0	3.5	3.4	13.5	-17.0	-3.5	-3.4
3m 3y30y	100/14.8	-87.4	-71.4	-1.1	-99.6	-50.2	-1.0	0	-0.62	-0.88	-0.26	-16.0	28.2	12.2	3.4	16.0	-28.2	-12.2	-3.4
3m 5y10y	100/54.0	-21.3	-15.3	-1.1	-27.5	-20.3	-1.5	0	-0.24	-0.29	-0.05	-6.0	12.2	6.2	5.9	6.0	-12.2	-6.2	-5.9
3m 5y30y	100/23.9	-48.8	-40.3	-1.0	-64.1	-40.5	-1.2	1	-0.51	-0.70	-0.19	-8.5	23.8	15.3	3.4	8.5	-23.8	-15.3	-3.4
3m 10y30y	100/44.3	-27.5	-25.0	-0.8	-35.0	-20.0	-0.9	1	-0.25	-0.40	-0.15	-2.5	10.1	7.6	3.4	2.5	-10.1	-7.6	-3.4
6m 1y2y	100/50.8	-66.8	-59.1	-1.1	-58.0	-6.7	-0.9	1	0.01	-0.22	-0.23	-7.7	-1.1	-8.8	-4.8	7.7	1.1	8.8	4.8
6m 1y3y	100/34.5	-109.1	-87.3	-1.3	-90.7	-15.8	-1.5	1	-0.05	-0.37	-0.33	-21.7	3.4	-18.3	-8.3	21.7	-3.4	18.3	8.3
6m 1y5y	100/21.3	-147.7	-110.7	-1.5	-122.2	-27.0	-1.7	1	-0.16	-0.55	-0.39	-37.0	11.5	-25.5	-10.4	37.0	-11.5	25.5	10.4
6m 2y5y	100/42.0	-80.9	-51.6	-1.6	-64.9	-20.2	-1.9	2	-0.17	-0.33	-0.15	-29.4	13.3	-16.1	-5.9	29.4	-13.3	16.1	5.9
6m 2y10y	100/22.7	-102.3	-60.9	-1.4	-88.7	-39.0	-1.5	1	-0.38	-0.61	-0.23	-41.4	27.8	-13.6	-6.6	41.4	-27.8	13.6	6.6
6m 2y30y	100/10.0	-129.7	-83.4	-1.2	-127.0	-58.5	-1.3	1	-0.66	-1.02	-0.36	-46.3	43.6	-2.8	-4.6	46.3	-43.6	2.8	4.6
6m 3y5y	100/61.9	-38.6	-23.3	-1.3	-31.8	-11.3	-0.7	1	-0.11	-0.18	-0.06	-15.3	8.4	-6.8	-2.6	15.3	-8.4	6.8	2.6
6m 3y10y	100/33.4	-59.9	-32.6	-1.2	-55.3	-30.3	-0.9	1	-0.32	-0.46	-0.15	-27.3	22.7	-4.6	-3.7	27.3	-22.7	4.6	3.7
6m 3y30y	100/14.8	-87.4	-55.2	-1.0	-93.4	-49.9	-0.9	1	-0.59	-0.88	-0.29	-32.2	38.2	6.0	-2.2	32.2	-38.2	-6.0	2.2
6m 5y10y	100/54.0	-21.3	-9.3	-1.0	-23.5	-19.4	-1.3	1	-0.20	-0.28	-0.08	-12.0	14.2	2.1	-1.6	12.0	-14.2	-2.1	1.6
6m 5y30y	100/23.9	-48.8	-31.8	-0.9	-61.6	-39.6	-1.1	1	-0.46	-0.69	-0.23	-16.9	29.7	12.8	-0.5	16.9	-29.7	-12.8	0.5
6m 10y30y	100/44.3	-27.5	-22.5	-0.8	-36.4	-19.8	-0.9	1	-0.24	-0.39	-0.15	-4.9	13.9	8.9	0.9	4.9	-13.9	-8.9	-0.9
1y 1y2y	100/50.7	-66.8	-31.3	-1.7	-41.0	-14.7	-2.9	1	-0.08	-0.21	-0.12	-35.4	9.6	-25.8	-5.7	35.4	-9.6	25.8	5.7
1y 1y3y	100/34.4	-109.1	-46.0	-1.8	-61.7	-22.4	-2.9	2	-0.14	-0.32	-0.18	-63.0	15.6	-47.4	-8.6	63.0	-15.6	47.4	8.6
1y 1y5y	100/21.3	-147.7	-55.6	-1.6	-82.7	-31.6	-2.2	2	-0.25	-0.48	-0.23	-92.1	27.1	-65.0	-10.6	92.1	-27.1	65.0	10.6
1y 2y5y	100/42.0	-80.9	-24.2	-1.2	-41.8	-16.9	-1.2	2	-0.16	-0.27	-0.11	-56.7	17.5	-39.2	-5.6	56.7	-17.5	39.2	5.6
1y 2y10y	100/22.7	-102.3	-23.3	-1.1	-57.6	-31.9	-1.1	2	-0.33	-0.53	-0.20	-78.9	34.3	-44.7	-6.6	78.9	-34.3	44.7	6.6
1y 2y30y	100/10.0	-129.7	-42.2	-1.0	-98.7	-50.3	-1.0	2	-0.60	-0.90	-0.31	-87.5	56.5	-31.0	-4.8	87.5	-56.5	31.0	4.8
1y 3y5y	100/61.9	-38.6	-9.5	-0.9	-20.7	-9.4	-0.6	3	-0.11	-0.16	-0.06	-29.1	11.2	-17.9	-2.7	29.1	-11.2	17.9	2.7
1y 3y10y	100/33.4	-59.9	-8.6	-0.9	-36.0	-24.7	-0.8	2	-0.27	-0.42	-0.15	-51.3	27.4	-23.9	-4.1	51.3	-27.4	23.9	4.1
1y 3y30y	100/14.8	-87.4	-27.5	-0.8	-76.4	-43.2	-0.8	1	-0.53	-0.79	-0.26	-59.9	48.9	-11.0	-2.7	59.9	-48.9	11.0	2.7
1y 5y10y	100/54.0	-21.3	0.9	-0.9	-14.4	-15.8	-0.9	2	-0.16	-0.25	-0.09	-22.2	15.4	-6.9	-1.8	22.2	-15.4	6.9	1.8
1y 5y30y	100/23.9	-48.8	-18.0	-0.8	-53.7	-34.9	-0.9	1	-0.41	-0.62	-0.21	-30.8	35.8	5.0	-1.1	30.8	-35.8	-5.0	1.1
1y 10y30y	100/44.2	-27.5	-18.9	-0.7	-37.7	-19.0	-0.9	1	-0.23	-0.35	-0.12	-8.6	18.8	10.2	0.5	8.6	-18.8	-10.2	-0.5
2y 1y2y	100/51.0	-66.8	-6.6	-1.0	-16.0	-6.9	-1.0	6	-0.06	-0.06	0.00	-60.1	9.4	-50.7	-1.2	60.1	-9.4	50.7	1.2
2y 1y3y	100/34.5	-109.1	-7.8	-0.7	-23.8	-11.6	-0.7	6	-0.11	-0.13	-0.02	-101.3	16.1	-85.3	-2.1	101.3	-16.1	85.3	2.1
2y 1y5y	100/21.4	-147.7	-5.9	-0.6	-33.5	-19.2	-0.4	3	-0.18	-0.26	-0.08	-141.8	27.6	-114.2	-2.9	141.8	-27.6	114.2	2.9
2y 2y5y	100/41.9	-80.9	0.7	-0.4	-16.5	-12.3	-0.1	3	-0.12	-0.19	-0.08	-81.6	17.2	-64.4	-1.9	81.6	-17.2	64.4	1.9
2y 2y10y	100/22.6	-102.3	10.2	-0.5	-27.1	-21.9	-0.3	3	-0.26	-0.40	-0.14	-112.5	37.4	-75.2	-1.9	112.5	-37.4	75.2	1.9
2y 2y30y	100/10.0	-129.7	-9.1	-0.5	-77.0	-35.9	-0.4	2	-0.51	-0.70	-0.19	-120.6	67.8	-52.8	-0.7	120.6	-67.8	52.8	0.7
2y 3y5y	100/61.9	-38.6	1.8	-0.5	-8.6	-7.7	0.0	3	-0.07	-0.13	-0.06	-40.5	10.5	-30.0	-1.0	40.5	-10.5	30.0	1.0
2y 3y10y	100/33.4	-59.9	11.4	-0.6	-18.4	-17.4	-0.3	3	-0.21	-0.33	-0.12	-71.3	29.8	-41.5	-1.1	71.3	-29.8	41.5	1.1
2y 3y30y	100/14.8	-87.4	-8.0	-0.5	-67.0	-31.4	-0.5	2	-0.45	-0.62	-0.17	-79.4	59.0	-20.4	0.0	79.4	-59.0	20.4	0.0
2y 5y10y	100/54.0	-21.3	9.5	-0.7	-8.5	-10.0	-0.7	2	-0.14	-0.20	-0.06	-30.9	18.1	-12.8	-0.3	30.9	-18.1	12.8	0.3
2y 5y30y	100/23.9	-48.8	-9.8	-0.6	-55.1	-24.1	-0.7	1	-0.36	-0.48	-0.11	-38.9	45.3	6.4	0.7	38.9	-45.3	-6.4	-0.7
2y 10y30y	100/44.2	-27.5	-19.4	-0.5	-43.8	-13.9	-0.5	0	-0.21	-0.26	-0.05	-8.1	24.5	16.4	0.8	8.1	-24.5	-16.4	-0.8
3y 1y2y	100/50.8	-66.8	1.7	0.0	-6.4	-4.8	-0.1	11	-0.04	-0.07	-0.02	-68.4	8.0	-60.4	-0.9	68.4	-8.0	60.4	0.9
3y 1y3y	100/34.4	-109.1	3.9	-0.1	-10.6	-9.4	0.0	10	-0.08	-0.15	-0.07	-113.0	14.6	-98.5	-1.3	113.0	-14.6	98.5	1.3
3y 1y5y	100/21.3	-147.7	8.3	-0.2	-15.5	-15.7	0.1	8	-0.13	-0.26	-0.12	-156.0	23.8	-132.2	-1.8	156.0	-23.8	132.2	1.8
3y 2y5y	100/41.9	-80.9	6.6	-0.3	-8.6	-10.9	0.2	4	-0.08	-0.18	-0.10	-87.6	15.2	-72.4	-0.9	87.6	-15.2	72.4	0.9
3y 2y10y	100/22.6	-102.3	17.8	-0.6	-19.2	-17.8	-0.2	3	-0.22	-0.36	-0.14	-120.0	36.9	-83.1	-0.8	120.0	-36.9	83.1	0.8
3y 2y30y	100/10.0	-129.7	-6.7	-0.5	-78.3	-29.7	-0.3	2	-0.45	-0.61	-0.16	-123.1	71.6	-51.5	0.2	123.1	-71.6	51.5	-0.2
3y 3y5y	100/61.8	-38.6	4.3	-0.3	-4.2	-6.3	0.3	4	-0.05	-0.11	-0.06	-42.9	8.5	-34.4	-0.5	42.9	-8.5	34.4	0.5
3y 3y10y	100/33.4	-59.9	15.5	-0.6	-14.0	-13.3	-0.3	3	-0.18	-0.28	-0.10	-75.4	29.4	-46.0	-0.4	75.4	-29.4	46.0	0.4
3y 3y30y	100/14.7	-87.4	-9.0	-0.5	-71.6	-25.1	-0.5	2	-0.40	-0.52	-0.11	-78.4	62.6	-15.8	0.4	78.4	-62.6	15.8	-0.4
3y 5y10y	100/54.1	-21.3	11.1	-0.9	-8.7	-7.1	-0.9	2	-0.13	-0.17	-0.04	-32.4	19.8	-12.7	0.1	32.4	-19.8	12.7	-0.1
3y 5y30y	100/23.8	-48.8	-13.3	-0.6	-64.4	-18.9	-0.8	0	-0.34	-0.39	-0.05	-35.5	51.1	15.6	0.9	35.5	-51.1	-15.6	-0.9
3y 10y30y	100/44.1	-27.5	-24.4	-0.4	-52.3	-11.5	-0.5	0	-0.20	-0.21	-0.01	-3.0	27.9	24.8	0.8	3.0	-27.9	-24.8	-0.8

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Note: Close 6/9/23. The leg with the smaller vol is struck ATM, while the other leg is struck OTM for zero cost. We compute historical % of the strike curve using data starting in 1990. Implied beta is the beta between the curve and the average of end points implied by vols. Realized beta is empirical beta between the curve and the average of end points computed over the last 3m. The curve pick-up is the spread between spot and forward curves for steepeners, and the spread between forward and spot curve for flatteners. The vol pick-up is the spread between forward and strike curves for steepeners, and the spread between strike and forward curves for flatteners. Total pick-up is the sum of curve and vol pick-ups. 3m carry is computed assuming unchanged curve and vol cube and is expressed in bp of running yield. For each valuation metric, we mark three best **bear steepeners in red**, and three **best bear flatteners in green**. Note that the "trades" shown are backward looking, and suggest which actions would have been preferable at the pricing when the screen was run. These are illustrative only and not intended as open trade ideas. Source: Citi Research

Prepared for Christopher Lee

Figure 62. Conditional curves: bull trades

		spt and fwd crvs			costless strk curve				imp vs rlzd beta			bull steepener				bull flattener			
structure	ntlns, $mn	spot crv, bp	fwd crv, bp	3m ZS	strk crv, bp	1m chg	3m ZS	hist %	imp beta	rlzd beta	rlzd-imp	crv pick-up, bp	vol pick-up, bp	tot pick-up, bp	3m roll, bp	crv pick-up, bp	vol pick-up, bp	tot pick-up, bp	3m roll, bp
3m 1y2y	100/50.8	-66.8	-67.4	-0.4	-73.5	-3.9	-0.8	0	0.09	-0.13	-0.22	0.6	6.2	6.8	0.0	-0.6	-6.2	-6.8	0.0
3m 1y3y	100/34.5	-109.1	-103.0	-0.8	-105.2	-13.0	-1.0	1	0.03	-0.28	-0.32	-6.1	2.2	-3.9	0.0	6.1	-2.2	3.9	0.0
3m 1y5y	100/21.4	-147.7	-134.1	-1.2	-130.2	-31.8	-1.4	1	-0.06	-0.46	-0.40	-13.6	-3.9	-17.5	0.0	13.6	3.9	17.5	0.0
3m 2y5y	100/42.1	-80.9	-66.7	-1.6	-56.6	-27.2	-2.0	2	-0.15	-0.32	-0.17	-14.2	-10.1	-24.4	0.0	14.2	10.1	24.4	0.0
3m 2y10y	100/22.7	-102.3	-82.0	-1.5	-55.7	-49.0	-1.9	2	-0.39	-0.61	-0.21	-20.2	-26.3	-46.5	0.0	20.2	26.3	46.5	0.0
3m 2y30y	100/10.1	-129.7	-107.0	-1.3	-64.6	-69.2	-2.0	3	-0.69	-1.02	-0.34	-22.7	-42.4	-65.1	0.0	22.7	42.4	65.1	0.0
3m 3y5y	100/62.0	-38.6	-31.1	-1.5	-25.2	-17.3	-2.0	2	-0.09	-0.17	-0.08	-7.5	-5.9	-13.4	0.0	7.5	5.9	13.4	0.0
3m 3y10y	100/33.4	-59.9	-46.4	-1.2	-25.5	-38.8	-1.8	3	-0.33	-0.46	-0.12	-13.5	-20.9	-34.4	0.0	13.5	20.9	34.4	0.0
3m 3y30y	100/14.8	-87.4	-71.4	-1.1	-35.7	-59.0	-1.8	3	-0.62	-0.88	-0.26	-16.0	-35.8	-51.7	0.0	16.0	35.8	51.7	0.0
3m 5y10y	100/54.0	-21.3	-15.3	-1.1	-2.4	-21.0	-1.2	3	-0.24	-0.29	-0.05	-6.0	-12.9	-18.9	0.0	6.0	12.9	18.9	0.0
3m 5y30y	100/23.9	-48.8	-40.3	-1.0	-14.9	-40.7	-1.4	3	-0.51	-0.70	-0.19	-8.5	-25.4	-33.9	0.0	8.5	25.4	33.9	0.0
3m 10y30y	100/44.3	-27.5	-25.0	-0.8	-14.9	-19.8	-1.4	3	-0.25	-0.40	-0.15	-2.5	-10.0	-12.5	0.0	2.5	10.0	12.5	0.0
6m 1y2y	100/50.8	-66.8	-59.1	-1.1	-60.4	-13.6	-1.2	1	0.01	-0.22	-0.23	-7.7	1.3	-6.3	-3.9	7.7	-1.3	6.3	3.9
6m 1y3y	100/34.5	-109.1	-87.3	-1.3	-82.9	-25.7	-1.5	1	-0.05	-0.37	-0.33	-21.7	-4.5	-26.2	-6.1	21.7	4.5	26.2	6.1
6m 1y5y	100/21.3	-147.7	-110.7	-1.5	-95.2	-46.1	-1.7	2	-0.16	-0.55	-0.39	-37.0	-15.5	-52.5	-8.6	37.0	15.5	52.5	8.6
6m 2y5y	100/42.0	-80.9	-51.6	-1.6	-35.4	-30.9	-2.0	3	-0.17	-0.33	-0.15	-29.4	-16.1	-45.5	-5.3	29.4	16.1	45.5	5.3
6m 2y10y	100/22.7	-102.3	-60.9	-1.4	-26.3	-53.2	-1.7	3	-0.38	-0.61	-0.23	-41.4	-34.5	-75.9	-6.2	41.4	34.5	75.9	6.2
6m 2y30y	100/10.0	-129.7	-83.4	-1.2	-27.9	-72.4	-1.7	3	-0.66	-1.02	-0.36	-46.3	-55.5	-101.8	-6.2	46.3	55.5	101.8	6.2
6m 3y5y	100/61.9	-38.6	-23.3	-1.3	-13.5	-19.0	-1.9	3	-0.11	-0.18	-0.06	-15.3	-9.8	-25.1	-3.3	15.3	9.8	25.1	3.3
6m 3y10y	100/33.4	-59.9	-32.6	-1.2	-5.7	-41.1	-1.6	3	-0.32	-0.46	-0.15	-27.3	-26.9	-54.2	-4.7	27.3	26.9	54.2	4.7
6m 3y30y	100/14.8	-87.4	-55.2	-1.0	-8.9	-60.2	-1.6	4	-0.59	-0.88	-0.29	-32.2	-46.3	-78.5	-5.1	32.2	46.3	78.5	5.1
6m 5y10y	100/54.0	-21.3	-9.3	-1.0	5.3	-21.5	-1.2	5	-0.20	-0.28	-0.08	-12.0	-14.6	-26.6	-2.2	12.0	14.6	26.6	2.2
6m 5y30y	100/23.9	-48.8	-31.8	-0.9	-1.1	-40.4	-1.3	4	-0.46	-0.69	-0.23	-16.9	-30.7	-47.7	-3.4	16.9	30.7	47.7	3.4
6m 10y30y	100/44.3	-27.5	-22.5	-0.8	-9.1	-18.7	-1.2	4	-0.24	-0.39	-0.15	-4.9	-13.4	-18.4	-1.9	4.9	13.4	18.4	1.9
1y 1y2y	100/50.7	-66.8	-31.3	-1.7	-20.1	-16.0	-1.5	5	-0.08	-0.21	-0.12	-35.4	-11.3	-46.7	-5.9	35.4	11.3	46.7	5.9
1y 1y3y	100/34.4	-109.1	-46.0	-1.8	-28.1	-28.8	-1.9	4	-0.14	-0.32	-0.18	-63.0	-17.9	-81.0	-7.7	63.0	17.9	81.0	7.7
1y 1y5y	100/21.3	-147.7	-55.6	-1.6	-23.8	-48.4	-1.9	5	-0.25	-0.48	-0.23	-92.1	-31.7	-123.9	-9.0	92.1	31.7	123.9	9.0
1y 2y5y	100/42.0	-80.9	-24.2	-1.2	-6.3	-31.3	-1.7	6	-0.16	-0.27	-0.11	-56.7	-17.9	-74.6	-3.9	56.7	17.9	74.6	3.9
1y 2y10y	100/22.7	-102.3	-23.3	-1.1	12.2	-51.8	-1.4	11	-0.33	-0.53	-0.20	-78.9	-35.5	-114.5	-4.7	78.9	35.5	114.5	4.7
1y 2y30y	100/10.0	-129.7	-42.2	-1.0	17.1	-67.2	-1.3	8	-0.60	-0.90	-0.31	-87.5	-59.3	-146.8	-4.3	87.5	59.3	146.8	4.3
1y 3y5y	100/61.9	-38.6	-9.5	-0.9	1.9	-18.5	-1.4	11	-0.11	-0.16	-0.06	-29.1	-11.5	-40.5	-2.1	29.1	11.5	40.5	2.1
1y 3y10y	100/33.4	-59.9	-8.6	-0.9	19.5	-38.7	-1.3	18	-0.27	-0.42	-0.15	-51.3	-28.1	-79.5	-3.2	51.3	28.1	79.5	3.2
1y 3y30y	100/14.8	-87.4	-27.5	-0.8	23.0	-54.1	-1.1	11	-0.53	-0.79	-0.26	-59.9	-50.6	-110.4	-3.2	59.9	50.6	110.4	3.2
1y 5y10y	100/54.0	-21.3	0.9	-0.9	16.0	-19.6	-1.1	21	-0.16	-0.25	-0.09	-22.2	-15.0	-37.3	-1.6	22.2	15.0	37.3	1.6
1y 5y30y	100/23.9	-48.8	-18.0	-0.8	17.1	-34.7	-1.0	10	-0.41	-0.62	-0.21	-30.8	-35.1	-65.9	-2.3	30.8	35.1	65.9	2.3
1y 10y30y	100/44.2	-27.5	-18.9	-0.7	-1.8	-15.2	-0.8	5	-0.23	-0.35	-0.12	-8.6	-17.1	-25.6	-1.1	8.6	17.1	25.6	1.1
2y 1y2y	100/51.0	-66.8	-6.6	-1.0	1.3	-8.0	-1.2	18	-0.06	-0.06	0.00	-60.1	-7.9	-68.0	-0.6	60.1	7.9	68.0	0.6
2y 1y3y	100/34.5	-109.1	-7.8	-0.7	5.7	-14.7	-0.9	17	-0.11	-0.13	-0.02	-101.3	-13.5	-114.8	-1.3	101.3	13.5	114.8	1.3
2y 1y5y	100/21.4	-147.7	-5.9	-0.6	17.2	-26.5	-1.1	18	-0.18	-0.26	-0.08	-141.8	-23.1	-164.9	-1.2	141.8	23.1	164.9	1.2
2y 2y5y	100/41.9	-80.9	0.7	-0.4	15.3	-18.5	-1.1	22	-0.12	-0.19	-0.08	-81.6	-14.5	-96.2	-0.8	81.6	14.5	96.2	0.8
2y 2y10y	100/22.6	-102.3	10.2	-0.5	41.5	-31.9	-1.0	26	-0.26	-0.40	-0.14	-112.5	-31.2	-143.7	-1.0	112.5	31.2	143.7	1.0
2y 2y30y	100/10.0	-129.7	-9.1	-0.5	46.5	-42.8	-1.0	21	-0.51	-0.70	-0.19	-120.6	-55.7	-176.3	-0.3	120.6	55.7	176.3	0.3
2y 3y5y	100/61.9	-38.6	1.8	-0.5	10.8	-11.7	-1.3	25	-0.07	-0.13	-0.06	-40.5	-9.0	-49.4	-0.1	40.5	9.0	49.4	0.1
2y 3y10y	100/33.4	-59.9	11.4	-0.6	36.6	-25.0	-1.1	28	-0.21	-0.33	-0.12	-71.3	-25.2	-96.6	-0.5	71.3	25.2	96.6	0.5
2y 3y30y	100/14.8	-87.4	-8.0	-0.5	41.1	-35.9	-1.0	21	-0.45	-0.62	-0.17	-79.4	-49.1	-128.5	0.1	79.4	49.1	128.5	-0.1
2y 5y10y	100/54.0	-21.3	9.5	-0.7	25.2	-13.2	-0.9	31	-0.14	-0.20	-0.06	-30.9	-15.6	-46.5	-0.5	30.9	15.6	46.5	0.5
2y 5y30y	100/23.9	-48.8	-9.8	-0.6	28.8	-23.9	-0.8	18	-0.36	-0.48	-0.11	-38.9	-38.6	-77.5	-0.1	38.9	38.6	77.5	0.1
2y 10y30y	100/44.2	-27.5	-19.4	-0.5	1.4	-10.8	-0.6	7	-0.21	-0.26	-0.05	-8.1	-20.8	-28.9	0.2	8.1	20.8	28.9	-0.2
3y 1y2y	100/50.8	-66.8	1.7	0.0	8.0	-5.2	-0.1	29	-0.04	-0.07	-0.02	-68.4	-6.4	-74.8	-0.7	68.4	6.4	74.8	0.7
3y 1y3y	100/34.4	-109.1	3.9	-0.1	15.5	-11.0	-0.5	25	-0.08	-0.15	-0.07	-113.0	-11.6	-124.6	-0.8	113.0	11.6	124.6	0.8
3y 1y5y	100/21.3	-147.7	8.3	-0.2	27.0	-17.8	-0.9	23	-0.13	-0.26	-0.12	-156.0	-18.8	-174.7	-0.6	156.0	18.8	174.7	0.6
3y 2y5y	100/41.9	-80.9	6.6	-0.3	18.7	-12.5	-1.4	25	-0.08	-0.18	-0.10	-87.6	-12.1	-99.6	0.1	87.6	12.1	99.6	-0.1
3y 2y10y	100/22.6	-102.3	17.8	-0.6	46.7	-22.0	-1.3	29	-0.22	-0.36	-0.14	-120.0	-28.9	-148.9	0.0	120.0	28.9	148.9	0.0
3y 2y30y	100/10.0	-129.7	-6.7	-0.5	47.9	-29.3	-1.1	21	-0.45	-0.61	-0.16	-123.1	-54.6	-177.7	0.7	123.1	54.6	177.7	-0.7
3y 3y5y	100/61.8	-38.6	4.3	-0.3	11.2	-6.8	-1.6	25	-0.05	-0.11	-0.06	-42.9	-6.9	-49.8	0.2	42.9	6.9	49.8	-0.2
3y 3y10y	100/33.4	-59.9	15.5	-0.6	39.0	-16.3	-1.3	30	-0.18	-0.28	-0.10	-75.4	-23.5	-98.9	0.1	75.4	23.5	98.9	-0.1
3y 3y30y	100/14.7	-87.4	-9.0	-0.5	39.9	-23.7	-1.1	21	-0.40	-0.52	-0.11	-78.4	-48.8	-127.2	0.7	78.4	48.8	127.2	-0.7
3y 5y10y	100/54.1	-21.3	11.1	-0.9	27.4	-9.5	-1.0	33	-0.13	-0.17	-0.04	-32.4	-16.2	-48.7	-0.1	32.4	16.2	48.7	0.1
3y 5y30y	100/23.8	-48.8	-13.3	-0.6	27.7	-17.0	-0.8	17	-0.34	-0.39	-0.05	-35.5	-41.0	-76.5	0.5	35.5	41.0	76.5	-0.5
3y 10y30y	100/44.1	-27.5	-24.4	-0.4	-1.5	-7.5	-0.4	5	-0.20	-0.21	-0.01	-3.0	-22.9	-26.0	0.5	3.0	22.9	26.0	-0.5

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Note: Close 6/9/23. The leg with the smaller vol is struck ATM, while the other leg is struck OTM for zero cost. We compute historical % of the strike curve using data starting in 1990. Implied beta is the beta between the curve and the average of end points implied by vols. Realized beta is empirical beta between the curve and the average of end points computed over the last 3m. The curve pick-up is the spread between spot and forward curves for steepeners, and the spread between forward and spot curve for flatteners. The vol pick-up is the spread between forward and strike curves for steepeners, and the spread between strike and forward curves for flatteners. Total pick-up is the sum of curve and vol pick-ups. 3m roll is computed assuming unchanged curve and vol cube and is expressed in bp of running yield. For each valuation metric, we mark three best **bull steepeners in red**, and three **best bull flatteners in green**. Note that the "trades" shown are backward looking, and suggest which actions would have been preferable at the pricing when the screen was run. These are illustrative only and not intended as open trade ideas. Source: Citi Research

Prepared for Christopher Lee

USD conditional flies

We analyze conditional fly trades based on the entry point, directionality with rates and carry.

Figure 63. Conditional payer flies (50/50 DV01-weighted)

		spt and fwd crvs			fwd fly vs the belly					costless strike				long payer in the belly				short payer in the belly			
curve	notionals, $mn	spt fly, bp	fwd fly, bp	3m ZS	3m corr (chgs)	1y corr (lvls)	1y RC, bp	1y RC, ZC	belly OTM, bp	strk fly, bp	1m chg	3m ZS	hist %	crv pick-up, bp	vol pick-up, bp	tot pick-up, bp	3m roll, bp	crv pick-up, bp	vol pick-up, bp	tot pick-up, bp	3m roll, bp
3m 1y3y5y	144.8/100/31.0	-35.2	-35.9	-0.2	19	-20	-8.8	-0.8	3.4	-32.6	4.9	-0.1	0	0.7	-3.4	-2.7	-2.7	-0.7	3.4	2.7	2.7
3m 1y3y7y	144.8/100/22.8	-28.1	-30.5	0.2	51	-5	-9.3	-0.8	6.4	-24.1	9.6	0.2	16	2.4	-6.4	-4.0	-4.1	-2.4	6.4	4.0	4.1
3m 2y3y5y	73.6/100/31.0	-1.9	-2.2	0.3	44	20	-0.9	-0.4	0.8	-1.4	1.8	-0.4	79	0.4	-0.8	-0.5	-0.7	-0.4	0.8	0.5	0.7
3m 2y5y7y	118.8/100/36.8	-33.4	-27.9	-2.0	-46	-42	-3.1	-0.6	-1.1	-29.0	-3.1	-1.6	1	-5.4	1.1	-4.3	-4.6	5.4	-1.1	4.3	4.6
3m 2y5y10y	118.8/100/27.0	-29.8	-25.7	-1.2	20	-14	-3.7	-0.7	1.5	-24.2	1.2	-0.5	3	-4.1	-1.5	-5.6	-5.8	4.1	1.5	5.6	5.8
3m 2y10y30y	220.1/100/22.2	-37.4	-28.5	-1.7	-39	-22	-5.3	-0.6	-5.2	-33.8	-7.3	-2.0	1	-8.9	5.2	-3.6	-3.9	8.9	-5.2	3.6	3.9
3m 3y5y7y	80.7/100/36.8	-12.2	-10.1	-2.3	-8	-31	-1.3	-0.6	0.6	-9.6	-0.1	0.2	2	-2.0	-0.6	-2.6	-2.6	2.0	0.6	2.6	2.6
3m 3y5y10y	80.7/100/27.0	-8.6	-7.9	0.2	68	46	-1.9	-0.9	3.2	-4.7	4.2	1.6	57	-0.7	-3.2	-3.9	-3.9	0.7	3.2	3.9	3.9
3m 3y7y10y	109.6/100/36.6	-22.9	-18.7	-1.9	-57	-48	-1.2	-0.3	-3.0	-21.8	-5.2	-1.0	0	-4.1	3.0	-1.1	-1.1	4.1	-3.0	1.1	1.1
3m 3y7y15y	109.6/100/26.4	-21.9	-18.6	-1.4	9	-2	-1.7	-0.5	-1.3	-19.9	-1.2	-0.3	2	-3.3	1.3	-2.0	-2.1	3.3	-1.3	2.0	2.1
3m 3y10y30y	149.5/100/22.2	-16.2	-10.7	-1.5	-27	-7	-3.5	-0.6	-3.7	-14.4	-4.2	-1.5	0	-5.5	3.7	-1.8	-1.9	5.5	-3.7	1.8	1.9
3m 5y10y15y	92.6/100/36.0	-9.7	-7.5	-1.1	-57	-49	0.5	0.3	-3.8	-11.3	-4.9	-1.5	0	-2.2	3.8	1.6	1.6	2.2	-3.8	-1.6	-1.6
3m 5y10y30y	92.6/100/22.2	3.1	4.8	0.0	40	48	-2.5	-0.9	-1.2	3.7	0.6	-0.5	43	-1.8	1.2	-0.6	-0.6	1.8	-1.2	0.6	0.6
3m 10y20y30y	86.0/100/38.1	6.9	7.6	-1.4	-62	-61	-0.3	-0.2	-2.2	5.5	-4.2	-1.4	13	-0.8	2.2	1.4	1.4	0.8	-2.2	-1.4	-1.4
6m 1y3y5y	145.1/100/31.0	-35.2	-32.0	-0.9	-20	-25	-5.3	-0.6	2.3	-29.7	-1.5	-0.8	0	-3.2	-2.3	-5.6	-3.0	3.2	2.3	5.6	3.0
6m 1y3y7y	145.1/100/22.8	-28.1	-28.3	-0.4	26	-5	-6.1	-0.7	5.9	-22.4	3.1	-0.4	17	0.2	-5.9	-5.7	-2.6	-0.2	5.9	5.7	2.6
6m 2y3y5y	73.7/100/31.0	-1.9	-2.5	0.0	48	31	-0.8	-0.5	1.7	-0.7	1.1	-0.5	85	0.6	-1.7	-1.1	-0.6	-0.6	1.7	1.1	0.6
6m 2y5y7y	118.9/100/36.8	-33.4	-22.1	-2.4	-56	-43	-1.6	-0.3	-3.1	-25.2	-4.8	-2.1	2	-11.3	3.1	-8.2	-2.9	11.3	-3.1	8.2	2.9
6m 2y5y10y	118.9/100/27.0	-29.8	-21.1	-1.8	3	-10	-2.3	-0.5	-0.2	-21.3	-0.7	-1.2	4	-8.7	0.2	-8.5	-2.6	8.7	-0.2	8.5	2.6
6m 2y10y30y	220.4/100/22.1	-37.4	-19.2	-1.9	-46	-20	-3.1	-0.4	-7.1	-26.3	-8.5	-2.2	2	-18.2	7.1	-11.1	-4.9	18.2	-7.1	11.1	4.9
6m 3y5y7y	80.7/100/36.8	-12.2	-7.9	-2.3	-20	-28	-0.8	-0.5	-0.7	-8.6	-0.6	0.1	2	-4.3	0.7	-3.6	-1.0	4.3	-0.7	3.6	1.0
6m 3y5y10y	80.7/100/27.0	-8.6	-7.0	0.3	66	55	-1.6	-0.9	2.3	-4.7	3.4	1.3	58	-1.6	-2.3	-4.0	-0.7	1.6	2.3	4.0	0.7
6m 3y7y10y	109.6/100/36.6	-22.9	-14.5	-1.7	-62	-48	0.0	0.0	-4.9	-19.4	-5.8	-0.9	1	-8.4	4.9	-3.5	-1.6	8.4	-4.9	3.5	1.6
6m 3y7y15y	109.6/100/26.4	-21.9	-15.1	-1.2	-3	2	-0.6	-0.2	-2.3	-17.4	-1.8	-0.3	2	-6.8	2.3	-4.5	-1.8	6.8	-2.3	4.5	1.8
6m 3y10y30y	149.6/100/22.1	-16.2	-5.0	-1.5	-32	-3	-2.3	-0.5	-4.7	-9.8	-4.3	-1.1	2	-11.2	4.7	-6.5	-2.9	11.2	-4.7	6.5	2.9
6m 5y10y15y	92.7/100/36.0	-9.7	-5.3	-1.1	-58	-48	1.0	0.5	-4.0	-9.3	-4.6	-1.2	1	-4.4	4.0	-0.4	-1.1	4.4	-4.0	0.4	1.1
6m 5y10y30y	92.7/100/22.1	3.1	6.6	-0.1	36	53	-2.2	-0.9	-0.4	6.2	0.9	-0.3	56	-3.5	0.4	-3.1	-1.4	3.5	-0.4	3.1	1.4
6m 10y20y30y	86.0/100/38.1	6.9	8.3	-1.3	-62	-58	-0.1	-0.1	-2.9	5.5	-4.2	-1.3	13	-1.5	2.9	1.4	0.1	1.5	-2.9	-1.4	-0.1
1y 1y3y5y	145.4/100/31.0	-35.2	-18.3	-2.0	-44	-32	-0.8	-0.2	-2.5	-20.8	-6.1	-2.2	2	-17.0	2.5	-14.5	-3.3	17.0	-2.5	14.5	3.3
1y 1y3y7y	145.4/100/22.8	-28.1	-17.4	-1.2	19	0	-1.9	-0.4	1.4	-16.0	-2.4	-1.4	23	-10.7	-1.4	-12.1	-2.9	10.7	1.4	12.1	2.9
1y 2y3y5y	73.7/100/31.0	-1.9	-2.6	-0.2	65	59	-0.8	-0.9	2.5	-0.1	0.7	-0.6	89	0.7	-2.5	-1.7	-0.3	-0.7	2.5	1.7	0.3
1y 2y5y7y	119.1/100/36.8	-33.4	-11.3	-2.0	-52	-43	0.2	0.0	-4.4	-15.7	-4.3	-1.3	3	-22.1	4.4	-17.7	-2.7	22.1	-4.4	17.7	2.7
1y 2y5y10y	119.1/100/27.0	-29.8	-12.6	-2.2	0	-1	-0.6	-0.2	-1.3	-13.8	-1.0	-1.0	8	-17.2	1.3	-16.0	-2.3	17.2	-1.3	16.0	2.3
1y 2y10y30y	220.7/100/22.1	-37.4	-2.2	-1.6	-42	-13	-0.7	-0.1	-7.5	-9.7	-6.0	-1.3	3	-35.2	7.5	-27.7	-4.6	35.2	-7.5	27.7	4.6
1y 3y5y7y	80.8/100/36.8	-12.2	-3.9	-1.4	-35	-29	-0.1	-0.1	-1.4	-5.4	-0.6	0.1	3	-8.2	1.4	-6.8	-1.0	8.2	-1.4	6.8	1.0
1y 3y5y10y	80.8/100/27.0	-8.6	-5.2	0.6	65	68	-0.9	-0.7	1.7	-3.5	2.7	1.1	70	-3.4	-1.7	-5.2	-0.6	3.4	1.7	5.2	0.6
1y 3y7y10y	109.7/100/36.6	-22.9	-6.9	-1.1	-64	-46	1.5	0.5	-6.4	-13.2	-4.9	-0.7	2	-16.0	6.4	-9.7	-1.7	16.0	-6.4	9.7	1.7
1y 3y7y15y	109.7/100/26.4	-21.9	-8.7	-0.6	-13	11	0.8	0.3	-3.0	-11.8	-1.0	0.0	4	-13.2	3.0	-10.1	-1.7	13.2	-3.0	10.1	1.7
1y 3y10y30y	149.7/100/22.1	-16.2	5.1	-1.2	-33	5	-0.9	-0.2	-4.5	0.6	-2.3	-0.5	12	-21.4	4.5	-16.9	-2.9	21.4	-4.5	16.9	2.9
1y 5y10y15y	92.7/100/36.0	-9.7	-1.4	-1.0	-54	-43	1.3	0.7	-3.9	-5.3	-3.1	-0.6	3	-8.3	3.9	-4.3	-1.0	8.3	-3.9	4.3	1.0
1y 5y10y30y	92.7/100/22.1	3.1	9.9	0.0	35	61	-2.0	-1.0	1.1	11.0	2.1	0.3	75	-6.8	-1.1	-7.9	-1.3	6.8	1.1	7.9	1.3
1y 10y20y30y	86.0/100/38.0	6.9	9.4	-1.1	-57	-52	0.2	0.1	-3.9	5.5	-4.0	-1.2	13	-2.6	3.9	1.3	0.1	2.6	-3.9	-1.3	-0.1

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Note: Close 6/9/23. Conditional flies are structured by buying (selling) payers in the belly vs selling (buying) payers on the wings with 50/50 DV01 weights. The wings are struck ATM, while the strike on the belly is solved for zero cost. Spot, forward and costless strike rates flies are reported using -0.5/1/-0.5 weights. We compute correlations of the forward fly with the belly (using a 3m window in changes and 1y window in levels) and rich/cheap of the forward fly to the belly based on 1y regression in levels. We also compute historical % of the strike fly using data starting in 1990. 3m roll is computed assuming unchanged curve and vol cube and is expressed in bp of running yield. For each valuation metric, we mark three best long belly payer trades in red, and three best short belly payer trades in green. Note that the "trades" shown are backward looking, and suggest which actions would have been preferable at the pricing when the screen was run. These are illustrative only and not intended as open trade ideas. Source: Citi Research

Prepared for Christopher Lee

Figure 64. Conditional receiver flies (50/50 DV01-weighted)

		spt and fwd crvs			fwd fly vs the belly					costless strike				long receiver in the belly				short receiver in the belly			
curve	ntl, $mn	spt fly, bp	fwd fly, bp	3m ZS	3m corr (chgs)	1y corr (lvls)	1y RC, bp	1y RC, zs	belly OTM, bp	strk fly, bp	1m chg	3m ZS	hist %	crv pick-up, bp	vol pick-up, bp	tot pick-up, bp	3m roll, bp	crv pick-up, bp	vol pick-up, bp	tot pick-up, bp	3m roll, bp
3m 1y3y5y	144.8/100/31.0	-35.2	-35.9	-0.2	19	-20	-8.8	-0.8	4.0	-39.9	2.2	-0.3	0	-0.7	-4.0	-4.7	0.0	0.7	4.0	4.7	0.0
3m 1y3y7y	144.8/100/22.8	-28.1	-30.5	0.2	51	-5	-9.3	-0.8	7.7	-38.2	6.8	-0.1	5	-2.4	-7.7	-10.1	0.0	2.4	7.7	10.1	0.0
3m 2y3y5y	73.6/100/31.0	-1.9	-2.2	0.3	44	20	-0.9	-0.4	1.0	-3.2	3.6	1.2	62	-0.4	-1.0	-1.4	0.0	0.4	1.0	1.4	0.0
3m 2y5y7y	118.8/100/36.8	-33.4	-27.9	-2.0	-46	-42	-3.1	-0.6	-1.1	-26.8	-7.1	-1.8	0	5.4	1.1	6.5	0.0	-5.4	-1.1	-6.5	0.0
3m 2y5y10y	118.8/100/27.0	-29.8	-25.7	-1.2	20	-14	-3.7	-0.7	1.6	-27.3	-2.5	-1.3	1	4.1	-1.6	2.5	0.0	-4.1	1.6	-2.5	0.0
3m 2y10y30y	220.1/100/22.2	-37.4	-28.5	-1.7	-39	-22	-5.3	-0.6	-5.2	-23.3	-11.3	-1.4	2	8.9	5.2	14.1	0.0	-8.9	-5.2	-14.1	0.0
3m 3y5y7y	80.7/100/36.8	-12.2	-10.1	-2.3	-8	-31	-1.3	-0.6	0.6	-10.7	-3.0	-1.7	1	2.0	-0.6	1.5	0.0	-2.0	0.6	-1.5	0.0
3m 3y5y10y	80.7/100/27.0	-8.6	-7.9	0.2	68	46	-1.9	-0.9	3.3	-11.2	1.6	-0.7	20	0.7	-3.3	-2.6	0.0	-0.7	3.3	2.6	0.0
3m 3y7y10y	109.6/100/36.6	-22.9	-18.7	-1.9	-57	-48	-1.2	-0.3	-3.1	-15.6	-7.6	-2.0	2	4.1	3.1	7.2	0.0	-4.1	-3.1	-7.2	0.0
3m 3y7y15y	109.6/100/26.4	-21.9	-18.6	-1.4	9	-2	-1.7	-0.5	-1.3	-17.3	-3.8	-1.3	2	3.3	1.3	4.6	0.0	-3.3	-1.3	-4.6	0.0
3m 3y10y30y	149.5/100/22.2	-16.2	-10.7	-1.5	-27	-7	-3.5	-0.6	-3.6	-7.1	-7.3	-1.2	2	5.5	3.6	9.1	0.0	-5.5	-3.6	-9.1	0.0
3m 5y10y15y	92.6/100/36.0	-9.7	-7.5	-1.1	-57	-49	0.5	0.3	-3.8	-3.7	-5.4	-0.8	3	2.2	3.8	6.0	0.0	-2.2	-3.8	-6.0	0.0
3m 5y10y30y	92.6/100/22.2	3.1	4.8	0.0	40	48	-2.5	-0.9	-1.2	6.0	0.2	0.2	55	1.8	1.2	2.9	0.0	-1.8	-1.2	-2.9	0.0
3m 10y20y30y	86.0/100/38.1	6.9	7.6	-1.4	-62	-61	-0.3	-0.2	-2.2	9.8	-3.5	-1.4	37	0.8	2.2	2.9	0.0	-0.8	-2.2	-2.9	0.0
6m 1y3y5y	145.1/100/31.0	-35.2	-32.0	-0.9	-20	-25	-5.3	-0.6	2.7	-34.7	-3.3	-0.9	0	3.2	-2.7	0.5	1.6	-3.2	2.7	-0.5	-1.6
6m 1y3y7y	145.1/100/22.8	-28.1	-28.3	-0.4	26	-5	-6.1	-0.7	6.8	-35.1	1.4	-0.5	8	-0.2	-6.8	-7.0	1.1	0.2	6.8	7.0	-1.1
6m 2y3y5y	73.7/100/31.0	-1.9	-2.5	0.0	48	31	-0.8	-0.5	2.0	-4.5	3.0	1.0	50	-0.6	-2.0	-2.6	-0.4	0.6	2.0	2.6	0.4
6m 2y5y7y	118.9/100/36.8	-33.4	-22.1	-2.4	-56	-43	-1.6	-0.3	-3.2	-18.9	-8.8	-2.1	2	11.3	3.2	14.5	2.6	-11.3	-3.2	-14.5	-2.6
6m 2y5y10y	118.9/100/27.0	-29.8	-21.1	-1.8	3	-10	-2.3	-0.5	-0.2	-20.9	-4.1	-1.7	4	8.7	0.2	8.9	2.0	-8.7	-0.2	-8.9	-2.0
6m 2y10y30y	220.4/100/22.1	-37.4	-19.2	-1.9	-46	-20	-3.1	-0.4	-6.9	-12.3	-13.7	-1.5	3	18.2	6.9	25.1	3.8	-18.2	-6.9	-25.1	-3.8
6m 3y5y7y	80.7/100/36.8	-12.2	-7.9	-2.3	-20	-28	-0.8	-0.5	-0.7	-7.3	-3.5	-1.9	2	4.3	0.7	4.9	1.2	-4.3	-0.7	-4.9	-1.2
6m 3y5y10y	80.7/100/27.0	-8.6	-7.0	0.3	66	55	-1.6	-0.9	2.4	-9.4	1.3	-0.7	26	1.6	-2.4	-0.8	0.6	-1.6	2.4	0.8	-0.6
6m 3y7y10y	109.6/100/36.6	-22.9	-14.5	-1.7	-62	-48	0.0	0.0	-4.9	-9.5	-8.2	-1.7	3	8.4	4.9	13.3	2.1	-8.4	-4.9	-13.3	-2.1
6m 3y7y15y	109.6/100/26.4	-21.9	-15.1	-1.2	-3	2	-0.6	-0.2	-2.3	-12.8	-4.5	-1.2	3	6.8	2.3	9.1	1.5	-6.8	-2.3	-9.1	-1.5
6m 3y10y30y	149.6/100/22.1	-16.2	-5.0	-1.5	-32	-3	-2.3	-0.5	-4.6	-0.5	-8.5	-1.3	10	11.2	4.6	15.8	2.3	-11.2	-4.6	-15.8	-2.3
6m 5y10y15y	92.7/100/36.0	-9.7	-5.3	-1.1	-58	-48	1.0	0.5	-3.9	-1.4	-5.8	-0.9	4	4.4	3.9	8.3	0.8	-4.4	-3.9	-8.3	-0.8
6m 5y10y30y	92.7/100/22.1	3.1	6.6	-0.1	36	53	-2.2	-0.9	-0.4	7.0	-0.4	0.0	60	3.5	0.4	3.9	0.3	-3.5	-0.4	-3.9	-0.3
6m 10y20y30y	86.0/100/38.1	6.9	8.3	-1.3	-62	-58	-0.1	-0.1	-2.8	11.2	-3.5	-1.3	45	1.5	2.8	4.3	0.6	-1.5	-2.8	-4.3	-0.6
1y 1y3y5y	145.4/100/31.0	-35.2	-18.3	-2.0	-44	-32	-0.8	-0.2	-2.4	-15.9	-5.0	-1.4	5	17.0	2.4	19.4	3.3	-17.0	-2.4	-19.4	-3.3
1y 1y3y7y	145.4/100/22.8	-28.1	-17.4	-1.2	19	0	-1.9	-0.4	1.7	-19.1	-0.6	-0.9	19	10.7	-1.7	9.0	3.0	-10.7	1.7	-9.0	-3.0
1y 2y3y5y	73.7/100/31.0	-1.9	-2.6	-0.2	65	59	-0.8	-0.9	2.7	-5.3	2.4	0.4	44	-0.7	-2.7	-3.4	0.1	0.7	2.7	3.4	-0.1
1y 2y5y7y	119.1/100/36.8	-33.4	-11.3	-2.0	-52	-43	0.2	0.0	-4.1	-7.2	-9.4	-2.1	3	22.1	4.1	26.2	2.0	-22.1	-4.1	-26.2	-2.0
1y 2y5y10y	119.1/100/27.0	-29.8	-12.6	-2.2	0	-1	-0.6	-0.2	-1.1	-11.5	-5.1	-2.2	10	17.2	1.1	18.3	1.6	-17.2	-1.1	-18.3	-1.6
1y 2y10y30y	220.7/100/22.1	-37.4	-2.2	-1.6	-42	-13	-0.7	-0.1	-6.7	4.5	-14.6	-1.6	16	35.2	6.7	41.9	2.9	-35.2	-6.7	-41.9	-2.9
1y 3y5y7y	80.8/100/36.8	-12.2	-3.9	-1.4	-35	-29	-0.1	-0.1	-1.4	-2.5	-3.7	-1.7	8	8.2	1.4	9.7	0.7	-8.2	-1.4	-9.7	-0.7
1y 3y5y10y	80.8/100/27.0	-8.6	-5.2	0.6	65	68	-0.9	-0.7	1.7	-6.9	0.7	-0.5	39	3.4	-1.7	1.7	0.4	-3.4	1.7	-1.7	-0.4
1y 3y7y10y	109.7/100/36.6	-22.9	-6.9	-1.1	-64	-46	1.5	0.5	-6.0	-0.8	-8.1	-1.0	8	16.0	6.0	22.1	1.4	-16.0	-6.0	-22.1	-1.4
1y 3y7y15y	109.7/100/26.4	-21.9	-8.7	-0.6	-13	11	0.8	0.3	-2.9	-5.9	-5.0	-0.8	12	13.2	2.9	16.0	1.1	-13.2	-2.9	-16.0	-1.1
1y 3y10y30y	149.7/100/22.1	-16.2	5.1	-1.2	-33	5	-0.9	-0.2	-4.1	9.2	-9.0	-1.2	34	21.4	4.1	25.5	1.6	-21.4	-4.1	-25.5	-1.6
1y 5y10y15y	92.7/100/36.0	-9.7	-1.4	-1.0	-54	-43	1.3	0.7	-3.6	2.2	-5.7	-0.9	19	8.3	3.6	11.8	0.6	-8.3	-3.6	-11.8	-0.6
1y 5y10y30y	92.7/100/22.1	3.1	9.9	0.0	35	61	-2.0	-1.0	1.1	8.8	-1.1	-0.3	68	6.8	-1.1	5.8	0.3	-6.8	1.1	-5.8	-0.3
1y 10y20y30y	86.0/100/38.0	6.9	9.4	-1.1	-57	-52	0.2	0.1	-3.6	13.0	-2.9	-0.8	53	2.6	3.6	6.2	0.4	-2.6	-3.6	-6.2	-0.4

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Note: Close 6/9/23. Conditional flies are structured by buying (selling) receivers in the belly vs selling (buying) receivers on the wings with 50/50 DV01 weights. The wings are struck ATM, while the strike on the belly is solved for zero cost. Spot, forward and costless strike rates flies are reported using -0.5/1/-0.5 weights. We compute correlations of the forward fly with the belly (using a 3m window in changes and 1y window in levels) and rich/cheap of the forward fly to the belly based on 1y regression in levels. We also compute historical % of the strike fly using data starting in 1990. 3m roll is computed assuming unchanged curve and vol cube and is expressed in bp of running yield. For each valuation metric, we mark three best **long belly receiver trades in red**, and three **best short belly receiver trades in green**. Note that the "trades" shown are backward looking, and suggest which actions would have been preferable at the pricing when the screen was run. These are illustrative only and not intended as open trade ideas.
Source: Citi Research

Prepared for Christopher Lee

EUR at-the-money swaption vols

EUR relative value on the swaption surface

We report levels and z-scores of at-the-money vols, as well as the relative richness/cheapness of vols to rates (based on principal components of the curve).

- Top-left drove the cheapening in implied vols over the last month, with 1y tails down by double-digit in short expiries and supporting the 1y2y vol ratio at 3m expiry around the highest in four years (Figure 66). Short-expiry vols look among the cheapest vols based on our macro-vol (Figure 67) but not according to our PCA (Figure 68).

Figure 65. Implied vols, normals (3m z-score)

exp\tails	1y	2y	3y	5y	7y	10y	30y
1m	60.5 (-1.4)	98.6 (-1.1)	101.0 (-1.1)	97.2 (-1.2)	93.2 (-1.2)	86.1 (-1.3)	78.6 (-1.0)
3m	75.2 (-1.4)	105.6 (-1.1)	105.5 (-1.1)	100.1 (-1.2)	96.6 (-1.3)	90.8 (-1.3)	82.9 (-0.9)
6m	93.7 (-1.4)	110.0 (-1.2)	109.0 (-1.2)	102.7 (-1.4)	99.4 (-1.4)	93.3 (-1.5)	84.3 (-1.1)
1y	109.9 (-1.2)	113.8 (-1.3)	111.1 (-1.2)	105.7 (-1.4)	102.2 (-1.5)	97.1 (-1.5)	86.9 (-1.2)
2y	110.2 (-1.5)	110.8 (-1.5)	108.4 (-1.5)	104.7 (-1.6)	102.1 (-1.5)	98.2 (-1.5)	87.2 (-1.2)
3y	105.8 (-1.7)	106.2 (-1.6)	104.1 (-1.7)	101.0 (-1.8)	99.1 (-1.7)	95.7 (-1.5)	84.1 (-1.2)
5y	99.6 (-1.6)	99.5 (-1.6)	97.7 (-1.6)	94.4 (-1.7)	92.3 (-1.5)	89.2 (-1.3)	77.4 (-0.9)
10y	86.5 (-1.4)	86.4 (-1.4)	84.0 (-1.6)	80.7 (-1.7)	78.1 (-1.6)	74.8 (-1.3)	63.6 (-0.4)

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Close 6/9/23. The table shows ATM implied vols and 3m z-scores in parentheses. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

Figure 66. Change in implied vols since May 12th, normal

exp\tails	1y	2y	3y	5y	7y	10y	30y
1m	-22.0	-7.8	-4.3	-7.1	-6.6	-6.5	1.2
3m	-22.7	-9.6	-6.9	-8.4	-7.6	-6.8	-0.4
6m	-15.1	-8.4	-6.1	-8.4	-7.6	-6.9	-2.4
1y	-6.4	-6.3	-4.6	-6.3	-6.1	-5.4	-3.2
2y	-6.6	-6.6	-5.4	-5.3	-4.4	-3.4	-2.9
3y	-6.6	-6.6	-5.8	-4.5	-3.7	-2.6	-2.0
5y	-3.9	-3.8	-3.5	-3.0	-2.3	-1.6	-1.3
10y	-1.5	-1.5	-1.5	-1.6	-1.3	-0.9	-0.2

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Note: The table shows changes in ATM implied vols from 5/12/23 to 6/9/23. Source: Citi Research

Figure 67. Macro-vol (rich/cheap to rates), normals (z-score)

exp\tails	1y	2y	3y	5y	7y	10y	30y
1m	-184.0 (-2.9)	-207.2 (-2.7)	-181.0 (-2.7)	-150.4 (-2.8)	-146.1 (-2.9)	-133.1 (-3.0)	-140.7 (-3.0)
3m	-185.1 (-2.8)	-142.3 (-2.6)	-140.6 (-2.7)	-114.6 (-2.8)	-105.9 (-2.9)	-92.4 (-3.0)	-93.4 (-3.0)
6m	-121.6 (-2.7)	-93.2 (-2.5)	-94.2 (-2.6)	-77.4 (-2.7)	-74.8 (-2.8)	-73.0 (-3.0)	-79.3 (-3.0)
1y	-30.3 (-1.7)	-38.9 (-2.0)	-43.4 (-2.3)	-39.4 (-2.4)	-43.1 (-2.6)	-44.8 (-2.7)	-55.3 (-2.9)
2y	12.8 (1.0)	-4.0 (-0.4)	-7.0 (-0.7)	-12.4 (-1.2)	-15.7 (-1.6)	-21.5 (-2.1)	-30.5 (-2.5)
3y	22.4 (1.6)	12.9 (1.1)	9.6 (0.9)	1.8 (0.2)	-4.5 (-0.6)	-13.4 (-1.7)	-18.1 (-2.1)
5y	16.5 (1.4)	14.9 (1.4)	10.5 (1.1)	2.9 (0.4)	-1.5 (-0.3)	-8.7 (-1.6)	-3.8 (-0.7)
10y	-7.5 (-1.3)	-7.5 (-1.3)	-6.9 (-1.3)	-4.9 (-1.1)	-4.4 (-1.1)	-2.7 (-0.7)	-8.4 (-2.2)

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Close 6/9/23. We regress each ATM point on 3 principal components of the swap and inflation swap curves. We report rich/cheap for each vol as a residual from this regression together with z-scores in parentheses. We calibrate the model to the 6/1/2021-6/30/2022 sample history and test the model out-of-sample using data since 7/1/2022. Source: Citi Research

Figure 68. Rich/cheap to PCA of the vol surface, normals (z-score)

exp\tls	3m	1y	2y	3y	5y	7y	10y	30y
1m	4.4 (0.4)	-4.9 (-0.8)	8.2 (1.6)	5.5 (1.3)	1.1 (0.3)	-0.0 (-0.0)	-2.2 (-0.6)	-2.9 (-0.5)
3m	6.1 (0.7)	-1.8 (-0.3)	6.8 (1.7)	4.8 (1.7)	-0.7 (-0.3)	-1.1 (-0.5)	-2.6 (-1.0)	-3.7 (-0.8)
6m	10.1 (1.2)	1.6 (0.4)	4.3 (1.6)	3.2 (1.7)	-1.4 (-0.7)	-1.5 (-0.8)	-2.6 (-1.1)	-4.3 (-1.2)
1y	7.7 (2.0)	3.3 (1.3)	2.1 (0.9)	0.9 (0.5)	-0.7 (-0.4)	-1.0 (-0.6)	-0.6 (-0.3)	-2.7 (-1.0)
2y	-2.3 (-0.6)	-2.1 (-0.6)	-1.4 (-0.5)	-1.4 (-0.7)	-0.5 (-0.5)	0.2 (0.2)	1.6 (0.9)	-0.3 (-0.2)
3y	-6.4 (-1.5)	-4.7 (-1.4)	-3.7 (-1.5)	-2.9 (-1.6)	-1.2 (-1.3)	0.3 (0.3)	2.3 (1.3)	0.4 (0.4)
5y	-5.0 (-1.9)	-3.3 (-1.8)	-2.6 (-1.7)	-1.7 (-1.6)	0.1 (0.1)	1.1 (1.0)	2.8 (1.6)	1.2 (1.3)
10y	0.5 (0.4)	0.3 (0.2)	0.4 (0.3)	0.3 (0.2)	0.6 (0.6)	0.6 (0.6)	0.7 (0.6)	0.5 (0.6)

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Close 6/9/23. We calculate rich/cheap and z-scores in parentheses from PCA on the vol surface based on 3 principal components using 2y of data. By design, this framework only yields relative value. Source: Citi Research

EUR implied vs realized volatilities

We calculate the relative richness of short-dated implied vols to 1m and 3m realized vols. We calculate realized vols using both intraday (hourly) and close-to-close sampling.

- Intraday realized vols are broadly in line with close-to-close (Figure 69), but they look somewhat cheap to implied across one-year expiries (Figure 71).

Figure 69. 1m intraday/close-to-close ratio of realized vols (1y z-score)

exp\tails	1y	2y	3y	5y	7y	10y	30y
1m	1.11 (1.1)	1.03 (0.3)	1.01 (0.0)	1.05 (-0.2)	1.04 (-0.2)	1.03 (-0.3)	1.10 (-0.5)
3m	1.09 (0.8)	1.03 (0.3)	1.02 (0.1)	1.06 (-0.1)	1.04 (-0.2)	1.04 (-0.2)	1.10 (-0.5)
6m	1.07 (0.7)	1.03 (0.2)	1.03 (0.1)	1.06 (-0.1)	1.04 (-0.2)	1.04 (-0.3)	1.10 (-0.5)
1y	1.00 (-0.0)	1.00 (-0.2)	1.02 (-0.2)	1.05 (-0.3)	1.03 (-0.3)	1.04 (-0.3)	1.10 (-0.5)

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Note: Close 6/9/23. The 30m sampling between 7am and 4pm Lon time is used. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

Figure 70. 3m intraday/close-to-close ratio of realized vols (1y z-score)

exp\tails	1y	2y	3y	5y	7y	10y	30y
1m	1.00 (0.6)	1.08 (1.1)	1.12 (1.2)	1.16 (1.1)	1.15 (1.3)	1.14 (1.3)	1.14 (0.0)
3m	0.99 (0.5)	1.08 (1.1)	1.11 (1.1)	1.15 (1.1)	1.14 (1.3)	1.14 (1.2)	1.14 (-0.0)
6m	1.02 (0.8)	1.10 (1.1)	1.13 (1.2)	1.16 (1.2)	1.15 (1.3)	1.15 (1.3)	1.15 (0.1)
1y	1.12 (1.2)	1.15 (1.2)	1.16 (1.3)	1.19 (1.4)	1.18 (1.5)	1.17 (1.4)	1.16 (0.1)

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Note: Close 6/9/23. The 30m sampling between 7am and 4pm Lon time is used. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

Prepared for Christopher Lee

Figure 71. Intraday realized: implied/1m realized ratio (1y z-score)

exp\tails	1y	2y	3y	5y	7y	10y	30y
1m	1.05 (-0.2)	1.13 (0.2)	1.07 (0.2)	1.02 (0.4)	1.02 (0.2)	0.97 (-0.0)	0.94 (0.2)
3m	1.03 (0.3)	1.10 (0.8)	1.06 (0.7)	1.02 (0.9)	1.03 (0.8)	1.00 (0.7)	0.98 (0.7)
6m	1.00 (1.0)	1.03 (1.0)	1.03 (1.1)	1.02 (1.3)	1.04 (1.2)	1.02 (1.0)	0.99 (0.9)
1y	0.94 (1.2)	0.99 (1.3)	1.01 (1.5)	1.03 (1.7)	1.06 (1.6)	1.05 (1.5)	1.02 (1.4)

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Note: Close 6/9/23. The 30m sampling between 7am and 4pm Lon time is used. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

Figure 72. Intraday realized: implied/3m realized ratio (1y z-score)

exp\tails	1y	2y	3y	5y	7y	10y	30y
1m	0.38 (-1.8)	0.58 (-1.7)	0.62 (-1.6)	0.64 (-1.6)	0.68 (-1.6)	0.70 (-1.6)	0.71 (-1.3)
3m	0.40 (-1.8)	0.58 (-1.7)	0.62 (-1.6)	0.64 (-1.6)	0.69 (-1.6)	0.73 (-1.5)	0.74 (-1.2)
6m	0.47 (-1.8)	0.59 (-1.7)	0.64 (-1.6)	0.66 (-1.6)	0.72 (-1.4)	0.75 (-1.3)	0.75 (-1.1)
1y	0.57 (-1.6)	0.64 (-1.4)	0.68 (-1.2)	0.71 (-1.0)	0.77 (-0.8)	0.80 (-0.6)	0.78 (-0.7)

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Note: Close 6/9/23. The 30m sampling between 7am and 4pm Lon time is used. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

Figure 73. Close-to-close realized: implied/1m realized ratio (1y z-score)

exp\tails	1y	2y	3y	5y	7y	10y	30y
1m	1.16 (0.4)	1.16 (0.6)	1.08 (0.3)	1.07 (0.3)	1.05 (0.1)	1.00 (-0.2)	1.03 (-0.2)
3m	1.12 (0.8)	1.12 (1.0)	1.08 (0.8)	1.08 (0.7)	1.07 (0.5)	1.04 (0.3)	1.08 (0.2)
6m	1.08 (1.3)	1.06 (1.2)	1.06 (1.1)	1.07 (0.9)	1.08 (0.7)	1.05 (0.5)	1.09 (0.3)
1y	0.93 (1.1)	0.98 (1.1)	1.03 (1.1)	1.08 (1.1)	1.09 (0.9)	1.09 (0.8)	1.12 (0.6)

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Note: Close 6/9/23. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

Figure 74. Close-to-close realized: implied/3m realized ratio (1y z-score)

exp\tails	1y	2y	3y	5y	7y	10y	30y
1m	0.38 (-2.0)	0.63 (-1.8)	0.69 (-1.6)	0.74 (-1.5)	0.78 (-1.5)	0.80 (-1.5)	0.81 (-1.3)
3m	0.40 (-1.9)	0.62 (-1.8)	0.69 (-1.6)	0.74 (-1.5)	0.79 (-1.3)	0.83 (-1.2)	0.85 (-1.2)
6m	0.48 (-1.9)	0.65 (-1.6)	0.72 (-1.3)	0.77 (-1.2)	0.83 (-0.9)	0.86 (-0.6)	0.86 (-1.0)
1y	0.64 (-1.1)	0.74 (-0.6)	0.79 (-0.3)	0.85 (0.0)	0.90 (0.3)	0.93 (0.5)	0.91 (-0.6)

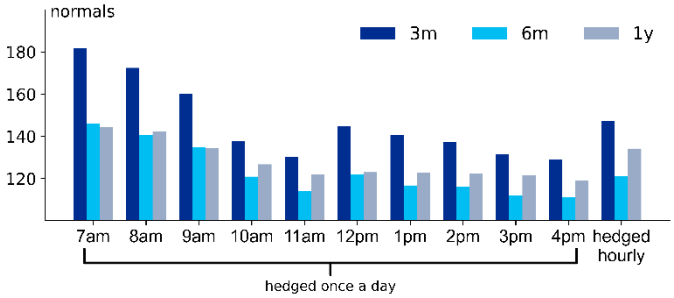
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Note: Close 6/9/23. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

EUR day-to-day realized volatilities for a given time snapshot

We calculate day-to-day realized volatilities measured at a given hour within a day for the last 3m, 6m and 1y (London time)

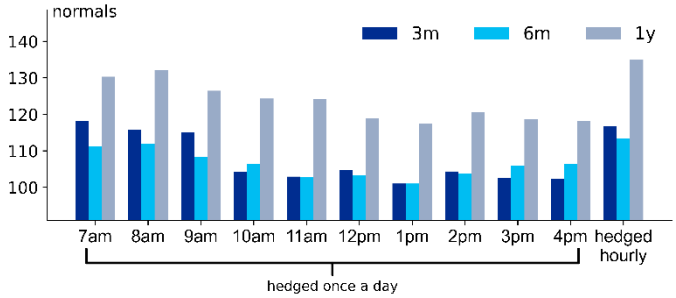
- Hedging at 1pm London has continued delivered the lowest realized vol of the 10y rate (Figure 76) over a three-month horizon.

Figure 75. Realized vol of the 2y rate



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Note: Close 6/9/23. Time is London. Source: Citi Research

Figure 76. Realized vol of the 10y rate



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Note: Close 6/9/23. Time is London. Source: Citi Research

EUR expiry and tail switches

We analyze ATM vol switches across expiries and tails.

- 3m1y looks some 1.5 sigma cheap to 6m1y (Figure 77) as well as 2 sigma cheap to 3m2y (Figure 78).

Figure 77. Short/long expiry ratio (3m z-score)

exp\tails	1y	2y	3y	5y	7y	10y	30y
1m/3m	0.80 (-1.6)	0.93 (-1.2)	0.96 (-1.1)	0.97 (-1.0)	0.97 (-1.2)	0.95 (-1.4)	0.95 (-1.0)
3m/6m	0.80 (-1.5)	0.96 (-1.0)	0.97 (-1.0)	0.97 (-0.9)	0.97 (-0.9)	0.97 (-0.9)	0.98 (-0.5)
6m/1y	0.85 (-1.5)	0.97 (-1.1)	0.98 (-1.2)	0.97 (-1.4)	0.97 (-1.4)	0.96 (-1.6)	0.97 (-1.0)
1y/3y	1.04 (-0.8)	1.07 (-0.8)	1.07 (-0.8)	1.05 (-1.1)	1.03 (-1.3)	1.01 (-1.4)	1.03 (-1.2)
3y/5y	1.06 (-1.6)	1.07 (-1.6)	1.07 (-1.9)	1.07 (-1.7)	1.07 (-1.9)	1.07 (-1.9)	1.09 (-1.4)
5y/10y	1.15 (-1.2)	1.15 (-1.1)	1.16 (-0.9)	1.17 (-1.1)	1.18 (-0.9)	1.19 (-0.7)	1.22 (-1.1)

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Note: Close 6/9/23. The table reports shorter expiry/longer expiry ratios with 3m z-scores in parentheses. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

Figure 78. Short/long tail ratio (3m z-score)

exp\tails	1y/2y	1y/3y	2y/5y	2y/10y	5y/10y	5y/30y	10y/30y
1m	0.61 (-2.0)	0.60 (-1.9)	1.01 (-0.9)	1.14 (-0.6)	1.13 (-0.1)	1.24 (-0.8)	1.10 (-1.2)
3m	0.71 (-2.0)	0.71 (-1.9)	1.05 (-0.6)	1.16 (-0.5)	1.10 (-0.4)	1.21 (-0.9)	1.10 (-1.0)
1y	0.97 (0.0)	0.99 (-0.3)	1.08 (-0.1)	1.17 (-0.2)	1.09 (-0.3)	1.22 (-0.7)	1.12 (-0.9)
2y	0.99 (0.3)	1.02 (0.2)	1.06 (-0.2)	1.13 (-0.1)	1.07 (-0.0)	1.20 (-0.4)	1.13 (-0.6)
5y	1.00 (-1.6)	1.02 (-0.9)	1.05 (-0.3)	1.12 (-0.5)	1.06 (-0.6)	1.22 (-0.9)	1.15 (-0.8)
10y	1.00 (1.4)	1.03 (0.8)	1.07 (0.2)	1.16 (-0.1)	1.08 (-0.3)	1.27 (-2.0)	1.18 (-1.8)

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Note: Close 6/9/23. The table reports shorter tail/longer tail ratios with 3m z-scores in parentheses. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

Prepared for Christopher Lee

EUR forward volatilities

We calculate forward 1y2y, 1y5y, 1y10y and 1y30y volatilities implied from the spot swaption surface. Forward volatilities reflect market expectations about future short-dated vols and may highlight dislocations on the spot vol surface.

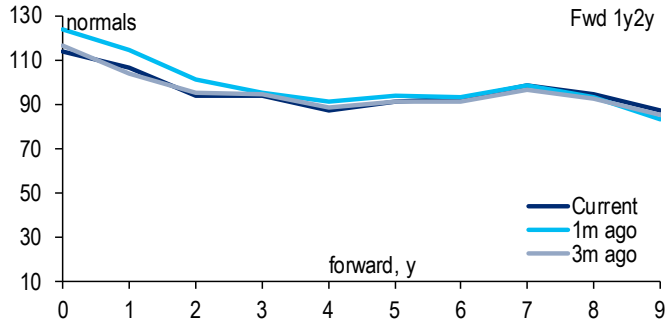
- Forward vols have generally declined over the last month, but less than spot (Figure 79). 1y2y vol is expected to decline by some 25bpv over the next four years while remaining some 7bpv richer than 1y10y.

Figure 79. Forward volatilities

	1y2y							1y5y							1y10y							1y30y						
Yrs		1w	1m	vs 1y		vs 10y			1w	1m	vs 1y		vs 10y			1w	1m	vs 1y		vs 10y			1w	1m	vs 1y		vs 10y	
Fwd	Vol	chg	chg	avg	z-scr	spt	z-scr	Vol	chg	chg	avg	z-scr	spt avg	z-scr	Vol	chg	chg	avg	z-scr	spt avg	z-scr	Vol	chg	chg	avg	z-scr	spt	z-scr
spt	113.8	(3.0)	(10.1)	(13.7)	(0.8)	71.2	2.1	105.7	(3.7)	(9.6)	(17.1)	(1.1)	52.2	1.9	97.1	(3.5)	(7.4)	(18.2)	(1.2)	35.7	1.6	86.9	(4.0)	(5.1)	(16.3)	(1.1)	23.1	1.3
1	106.4	(3.2)	(8.1)	(3.0)	(0.2)	63.9	1.9	103.9	(1.7)	(2.8)	(1.7)	(0.2)	50.4	1.8	99.5	(2.1)	(1.1)	(1.8)	(0.2)	38.1	1.7	87.2	(2.4)	(2.3)	(2.9)	(0.3)	23.3	1.3
2	93.9	(3.1)	(7.2)	(8.6)	(0.7)	51.4	1.5	94.2	(2.0)	(2.6)	(1.8)	(0.2)	40.7	1.5	92.4	(1.4)	(0.7)	2.5	0.4	30.9	1.4	77.3	(2.4)	0.8	2.0	0.4	13.4	0.7
3	93.8	1.0	(1.3)	(6.7)	(0.6)	51.2	1.5	88.6	(1.3)	(2.2)	(0.9)	(0.1)	35.1	1.3	85.1	(0.7)	(0.8)	2.9	0.7	23.7	1.1	67.2	(1.0)	(1.3)	3.2	0.9	3.4	0.2
4	87.0	(0.1)	(4.1)	(2.7)	(0.2)	44.5	1.3	83.4	0.8	(1.5)	2.8	0.5	30.0	1.1	80.3	1.9	2.1	4.6	1.5	18.9	0.9	66.2	2.8	3.2	8.2	2.3	2.3	0.1
5	91.0	(1.3)	(2.8)	(6.5)	(0.6)	48.5	1.5	83.6	(0.7)	(4.2)	(3.8)	(0.4)	30.1	1.1	76.1	(0.5)	(2.7)	(3.3)	(0.7)	14.7	0.7	58.2	0.5	(0.4)	0.4	0.1	(5.6)	(0.3)
6	92.7	(1.9)	(0.3)	(4.9)	(0.5)	50.2	1.5	82.2	(0.6)	(3.2)	(2.4)	(0.3)	28.7	1.0	71.7	(0.3)	(2.5)	(3.3)	(0.7)	10.2	0.5	52.0	1.4	0.3	1.9	0.6	(11.8)	(0.6)
7	98.7	(1.1)	(0.1)	(6.0)	(0.5)	56.2	1.7	85.6	(0.9)	(2.3)	(4.1)	(0.5)	32.1	1.1	75.0	0.2	(0.5)	(2.8)	(0.5)	13.6	0.6	54.4	1.8	0.9	0.4	0.1	(9.4)	(0.5)
8	94.4	(0.5)	1.0	(6.6)	(0.6)	51.9	1.6	81.0	(0.3)	(1.8)	(4.6)	(0.5)	27.5	1.0	70.6	0.6	(0.0)	(2.8)	(0.5)	9.2	0.4	48.8	2.7	1.6	0.6	0.3	(15.1)	(0.8)
9	86.9	1.2	3.8	(6.8)	(0.5)	44.4	1.3	77.4	0.7	(0.5)	(4.7)	(0.6)	24.0	0.9	67.8	1.2	0.8	(2.0)	(0.3)	6.4	0.3	44.3	3.6	2.4	1.9	0.6	(19.5)	(1.1)

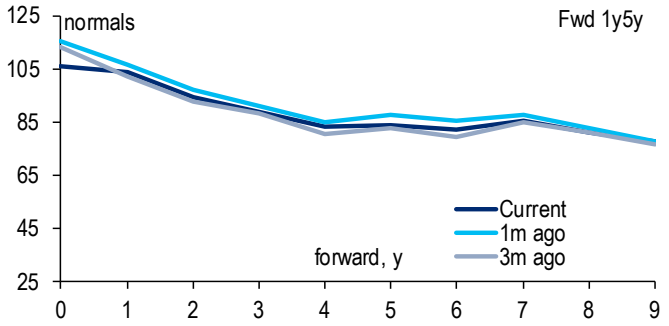
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Note: Close 6/9/23. Forward vols are computed using the standard triangular approximation from spot vols and using 6m realized correlations as a proxy for implied correlations. In addition to levels of forward vols, we calculate 1w and 1m changes, the difference from 1y average (with corresponding z-scores), and the difference from the 10y average of the corresponding 1y expiry spot vol (together with z-scores). Source: Citi Research

Figure 80. Forward vol of 1y2y



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Note: Close 6/9/23. Source: Citi Research

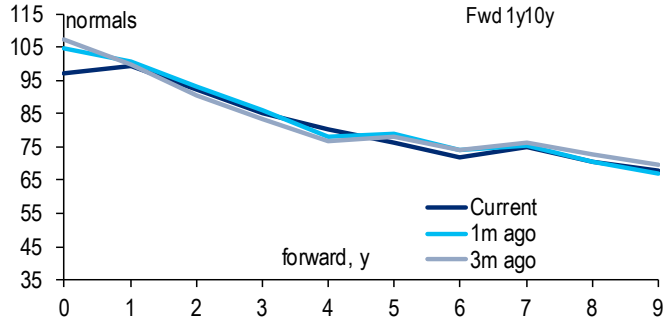
Figure 81. Forward vol of 1y5y



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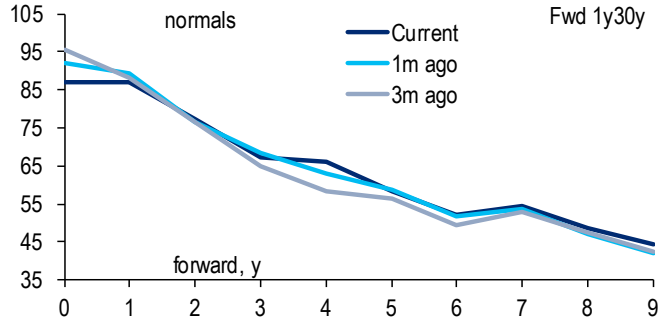
Prepared for Christopher Lee

Figure 82. Forward vol of 1y10y



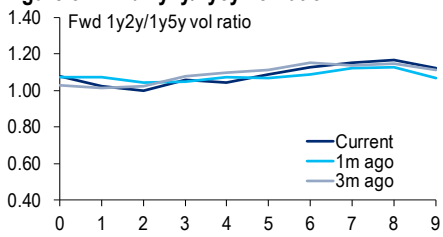
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Note: Close 6/9/23. Source: Citi Research

Figure 83. Forward vol of 1y30y



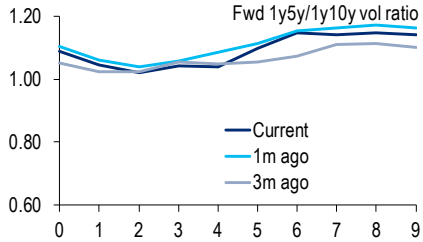
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Figure 84. Fwd 1y2y/1y5y vol ratio



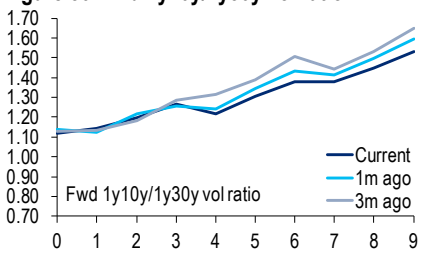
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Figure 85. Fwd 1y5y/1y10y vol ratio



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Note: Close 6/9/23. Source: Citi Research

Figure 86. Fwd 1y10y/1y30y vol ratio



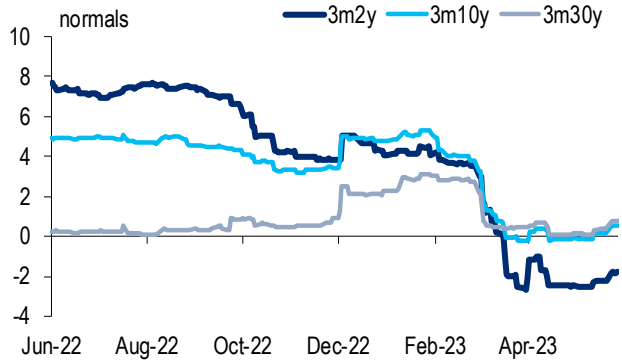
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Note: Close 6/9/23. Source: Citi Research

EUR swaption skew

We calculate z-scores of payer and receiver swaption skews across the surface.

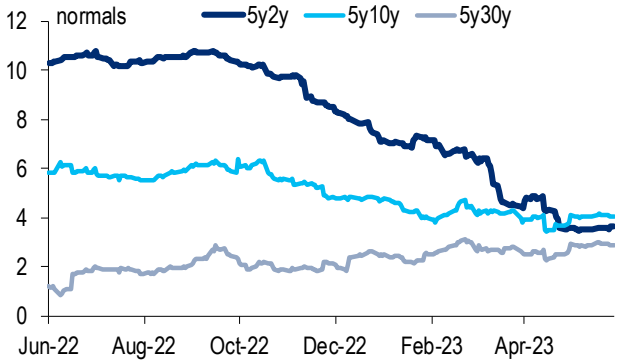
- Short-dated risk reversals have extended their slight rebound following their sharp retracement in March (Figure 87 and Figure 89). However, risk reversal on the 2y10y swap rate is still 2 sigma cheap on a one-year history (Figure 91).

Figure 87. 3m expiry payer skew



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Note: 25bp-out payer skews for 3m expiry payers. Source: Citi Research

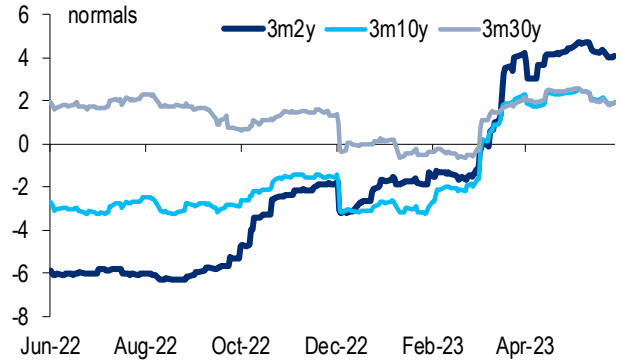
Figure 88. 5y expiry payer skew



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Note: 100bp-out payer skews for 5y expiry payers. Source: Citi Research

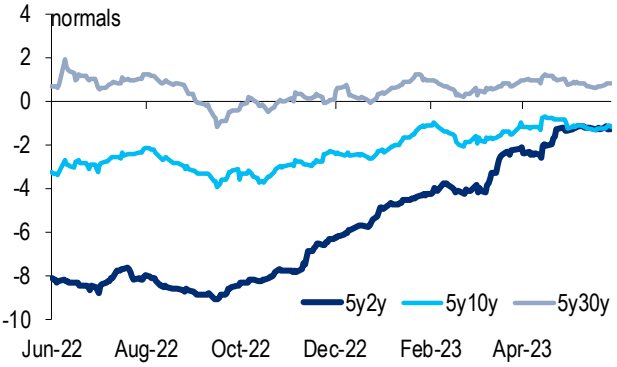
Prepared for Christopher Lee

Figure 89. 3m expiry receiver skew



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Note: 25bp-out receiver skews for 3m expiry receivers. Source: Citi Research

Figure 90. 5y expiry receiver skew



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Note: 100bp-out receiver skews for 5y expiry receivers. Source: Citi Research

Figure 91. Skews (1y z-scores) of risk reversals, OTM payer and OTM receivers

	Risk Reversals						Payer skew						Receiver skew					
expr\tail	1y	2y	5y	7y	10y	30y	1y	2y	5y	7y	10y	30y	1y	2y	5y	7y	10y	30y
3m	-8.1 (-1.7)	-5.9 (-1.7)	-3.6 (-1.7)	-2.7 (-1.7)	-1.4 (-1.7)	-1.2 (-0.4)	-2.3 (-1.6)	-1.8 (-1.7)	-0.6 (-1.7)	-0.1 (-1.6)	0.6 (-1.6)	0.8 (-0.2)	5.9 (1.8)	4.1 (1.8)	3.0 (1.8)	2.5 (1.8)	1.9 (1.8)	1.9 (0.7)
6m	-10.2 (-1.8)	-7.5 (-1.8)	-3.6 (-1.7)	-2.1 (-1.7)	-0.1 (-1.7)	-0.5 (-0.3)	-1.0 (-1.7)	-0.6 (-1.7)	1.4 (-1.6)	2.2 (-1.6)	3.4 (-1.5)	3.6 (0.2)	9.2 (1.9)	6.9 (1.8)	5.0 (1.8)	4.3 (1.9)	3.5 (1.9)	4.1 (0.9)
1y	-4.6 (-1.8)	-3.1 (-1.8)	-0.6 (-1.7)	0.4 (-1.8)	1.7 (-1.8)	0.8 (0.0)	-0.2 (-1.7)	0.3 (-1.7)	1.5 (-1.7)	2.0 (-1.7)	2.7 (-1.7)	2.5 (0.4)	4.4 (1.8)	3.4 (1.8)	2.1 (1.8)	1.6 (1.9)	1.0 (2.0)	1.7 (0.4)
2y	-0.1 (-1.8)	0.5 (-1.8)	1.6 (-1.8)	1.9 (-1.9)	2.4 (-2.0)	0.8 (0.0)	1.1 (-1.8)	1.2 (-1.8)	1.8 (-1.8)	2.0 (-1.8)	2.3 (-1.9)	1.5 (0.2)	1.2 (1.8)	0.8 (1.8)	0.2 (1.8)	0.1 (1.9)	-0.1 (2.0)	0.8 (0.2)
3y	3.3 (-1.8)	3.3 (-1.8)	4.3 (-1.9)	4.9 (-1.9)	5.5 (-1.7)	2.1 (0.3)	4.1 (-1.8)	3.9 (-1.8)	4.4 (-1.8)	4.8 (-1.8)	5.3 (-1.6)	3.8 (0.6)	0.9 (1.8)	0.6 (1.8)	0.1 (1.9)	-0.1 (1.9)	-0.3 (1.8)	1.7 (0.0)
5y	5.0 (-1.8)	4.9 (-1.8)	4.9 (-2.0)	5.1 (-1.8)	5.2 (-1.3)	2.1 (0.5)	3.8 (-1.8)	3.7 (-1.8)	3.6 (-2.0)	3.8 (-1.8)	4.1 (-1.2)	2.9 (1.3)	-1.3 (1.8)	-1.3 (1.8)	-1.3 (2.0)	-1.3 (1.8)	-1.2 (1.4)	0.8 (0.6)
10y	6.2 (-1.8)	6.2 (-1.8)	4.4 (-1.6)	3.9 (-1.5)	2.9 (-0.9)	0.9 (-0.6)	3.6 (-1.9)	3.6 (-1.9)	2.8 (-1.6)	2.7 (-1.4)	2.3 (-0.6)	1.7 (0.5)	-2.6 (1.8)	-2.6 (1.8)	-1.6 (1.5)	-1.2 (1.5)	-0.5 (1.1)	0.8 (1.2)

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Note: Close 6/9/23. The table reports skews and 1y z-scores in parentheses for risk reversals, payer and receiver skews, defined in normal vol terms. The following OTM conventions are used: 25bp out for 3m expiries, 50bp out for 6m-2y expiries, 100bp out for 3y+ expiries. Deep red (deep blue) indicates largest (smallest) z-scores.
Source: Citi Research

EUR midcurve swaptions

We report levels and z-scores of at-the-money vols, implied correlations, and implied/realized ratios of correlations. We also compute the relative richness/cheapness of midcurve vols to principal components of the plain vanilla surface.

- Midcurve vols on 3y1y and 3y2y forward rates continue looking cheap to swaptions when evaluated according to implied correlations (Figure 93) and linear regressions (Figure 95).

Figure 92. Implied vols, normals (3m z-score)

ex\tl	1y1y	2y1y	3y1y	1y2y	2y2y	3y2y	5y5y	10y10y
1m	146.5 (-0.7)	114.5 (-0.8)	102.3 (-1.3)	129.4 (-0.7)	106.2 (-1.0)	100.0 (-1.2)	86.3 (-1.2)	83.7 (-0.8)
3m	144.6 (-0.6)	113.7 (-0.9)	104.9 (-1.3)	128.0 (-0.7)	107.3 (-1.0)	102.2 (-1.3)	91.3 (-1.1)	88.4 (-0.6)
6m	131.7 (-0.9)	114.6 (-0.9)	106.3 (-1.4)	122.0 (-0.9)	109.1 (-1.1)	102.3 (-1.4)	92.3 (-1.3)	88.6 (-0.8)
1y	121.9 (-1.2)	112.0 (-0.9)	107.4 (-1.6)	115.6 (-1.1)	109.3 (-1.2)	105.3 (-1.5)	95.8 (-1.1)	89.6 (-0.9)
2y	115.5 (-1.2)	109.7 (-1.1)	108.2 (-1.4)	111.6 (-1.2)	108.5 (-1.3)	106.4 (-1.2)	98.0 (-1.0)	88.0 (-0.7)

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Close 6/9/23. The table shows ATM implied vols and 3m z-scores in parentheses. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

Figure 93. Implied correlations, % (3y z-score)

ex\tl	1y1y	2y1y	3y1y	1y2y	2y2y	3y2y	5y5y	10y10y
1m	96.5 (0.6)	99.1 (0.8)	99.4 (1.0)	92.8 (0.5)	97.8 (0.8)	98.2 (1.0)	96.6 (1.1)	97.3 (0.2)
3m	97.3 (0.8)	99.2 (0.9)	99.4 (1.1)	94.3 (0.7)	98.0 (0.9)	98.4 (1.1)	96.7 (1.1)	97.4 (0.3)
6m	98.0 (0.9)	99.3 (0.9)	99.6 (1.3)	95.6 (0.9)	98.1 (0.8)	98.7 (1.3)	97.1 (1.0)	97.4 (0.3)
1y	98.5 (0.9)	99.3 (0.9)	99.6 (1.4)	96.7 (1.0)	98.0 (0.9)	98.8 (1.4)	97.5 (1.2)	97.5 (0.4)
2y	98.7 (0.5)	99.3 (0.7)	99.6 (1.6)	96.6 (0.2)	97.9 (0.7)	98.7 (1.6)	97.8 (1.6)	97.6 (0.4)

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Close 6/9/23. The table shows implied correlations (3y z-scores in parentheses). Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

Prepared for Christopher Lee

Figure 94. Implied/realized correlation ratios (1y z-score)

ex\tl	1y1y	2y1y	3y1y	1y2y	2y2y	3y2y	5y5y	10y10y
1m	0.99 (-0.3)	1.00 (0.2)	1.00 (0.8)	1.00 (0.2)	1.01 (0.1)	1.00 (0.5)	0.99 (0.5)	0.99 (-1.8)
3m	1.00 (0.3)	1.00 (0.0)	1.00 (0.9)	1.00 (0.5)	1.01 (0.1)	1.00 (0.6)	0.99 (0.4)	0.99 (-1.7)
6m	1.00 (0.9)	1.00 (-0.1)	1.00 (0.9)	1.01 (0.9)	1.01 (0.1)	1.00 (0.8)	0.99 (0.0)	0.99 (-1.6)
1y	1.00 (0.9)	1.00 (0.7)	1.00 (1.1)	1.00 (0.9)	1.01 (0.4)	1.00 (1.1)	0.99 (0.4)	1.00 (-1.4)
2y	1.01 (1.3)	1.01 (1.7)	1.00 (1.1)	1.02 (1.2)	1.00 (1.4)	0.99 (0.9)	0.99 (0.9)	1.00 (-1.6)

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Close 6/9/23. The table shows the ratios of implied to 3m realized correlations (in daily changes). The 1y z-scores of ratios are shown in parentheses. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

Figure 95. Rich/cheap to plain vanilla swaptions (z-score)

ex\tl	1y1y	2y1y	3y1y	1y2y	2y2y	3y2y	5y5y	10y10y
1m	21.8 (3.3)	-2.0 (-0.3)	-13.0 (-2.0)	9.6 (2.1)	-6.8 (-1.2)	-11.9 (-1.4)	-8.6 (-1.2)	-0.2 (0.0)
3m	17.5 (3.9)	-0.2 (-0.1)	-11.0 (-1.9)	8.2 (3.2)	-4.9 (-1.2)	-12.5 (-1.8)	-7.4 (-1.4)	1.2 (0.3)
6m	7.3 (2.4)	0.1 (0.0)	-11.0 (-2.3)	3.2 (1.4)	-4.4 (-1.3)	-11.4 (-2.1)	-6.0 (-1.4)	0.2 (0.1)
1y	0.4 (0.2)	-2.0 (-0.9)	-6.3 (-1.7)	-1.5 (-0.8)	-3.0 (-1.2)	-6.4 (-1.5)	-2.9 (-0.7)	0.3 (0.1)
2y	-0.9 (-0.5)	-1.3 (-0.7)	-1.9 (-0.7)	-1.2 (-0.9)	-0.4 (-0.2)	-1.2 (-0.3)	1.5 (0.4)	1.7 (0.9)

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Close 6/9/23. We regress midcurve vols on two corresponding plain vanilla vols using last 3y of data. For example, 1y2y1y is regressed against 1y3y and 1y2y vols. We calculate rich/cheap and z-scores in brackets. Source: Citi Research

EUR curve (spread) CMS options

We report levels and z-scores of at-the-money vols, rich/cheap on PCA, implied correlations, and implied/realized vol ratios for curve options.

- Our PCA's three-year z-scores flag the richness of 6m implied vols on 2s5s, both outright and against 5s30s (Figure 97).

Figure 96. Implied vols, normals (3m z-score)

ex\crv	2s5s	2s10s	2s30s	5s10s	5s30s	10s30s
1m	35.9 (-1.3)	53.4 (-1.2)	75.6 (-1.1)	26.5 (-1.0)	52.1 (-1.2)	34.0 (-0.9)
3m	41.2 (-1.3)	60.7 (-1.2)	83.8 (-1.1)	30.7 (-1.1)	59.2 (-1.2)	38.1 (-1.1)
6m	44.2 (-1.4)	65.2 (-1.4)	89.0 (-1.3)	33.5 (-1.3)	63.5 (-1.4)	40.3 (-1.6)
1y	46.4 (-1.5)	68.6 (-1.6)	93.3 (-1.4)	37.1 (-1.4)	68.3 (-1.7)	43.8 (-1.7)
2y	44.8 (-1.6)	65.6 (-1.7)	91.3 (-1.6)	36.5 (-1.7)	69.8 (-1.8)	45.4 (-1.5)
3y	41.9 (-1.8)	61.3 (-1.7)	86.6 (-1.8)	35.3 (-1.9)	67.7 (-1.9)	44.3 (-1.6)
5y	38.8 (-1.8)	55.4 (-1.3)	79.1 (-1.6)	33.1 (-1.7)	63.2 (-1.5)	41.8 (-0.9)

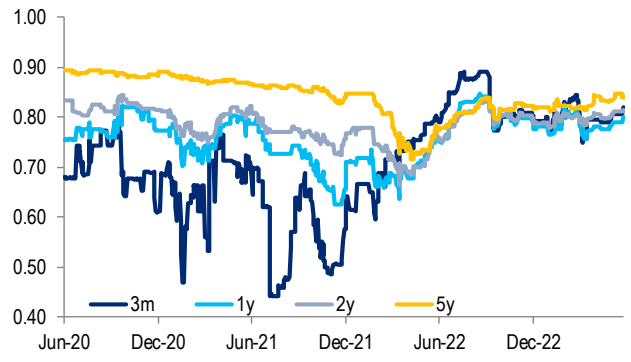
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Close 6/9/23. The table shows ATM implied vols and 3m z-scores in parentheses. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

Figure 97. Rich/cheap to PCA, normals (z-score)

ex\crv	2s5s	2s10s	2s30s	5s10s	5s30s	10s30s
1m	2.6 (0.9)	-5.5 (-1.4)	-12.7 (-2.1)	-8.6 (-2.0)	-20.0 (-2.5)	-13.8 (-1.8)
3m	4.9 (2.2)	-0.9 (-0.4)	-6.6 (-1.6)	-6.4 (-1.9)	-15.0 (-2.7)	-11.0 (-2.1)
6m	6.4 (2.9)	2.4 (1.1)	-1.9 (-0.6)	-3.8 (-1.6)	-10.8 (-2.9)	-9.0 (-2.5)
1y	8.5 (3.3)	5.3 (1.7)	3.8 (1.3)	0.1 (0.0)	-4.6 (-2.6)	-4.9 (-2.5)
2y	7.2 (3.1)	4.8 (1.5)	6.9 (2.5)	1.0 (0.6)	0.9 (0.5)	-0.8 (-0.5)
3y	5.4 (2.6)	3.9 (1.3)	7.1 (2.3)	1.2 (0.6)	2.9 (1.3)	0.7 (0.5)
5y	4.7 (2.0)	3.8 (1.5)	8.1 (2.2)	2.0 (1.0)	5.1 (1.9)	3.1 (1.6)

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Close 6/9/23. We calculate rich/cheap and z-scores in parentheses from PCA on curve vols using two principal components and 3y of data. By design, this framework only yields relative value. Source: Citi Research

Figure 98. Implied correlations of 2s10s CMS options



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Note: The chart shows implied correlation of options on the 2s10s curve with different expirations. Source: Citi Research

Figure 99. Implied/realized ratios

implied/1m RV (1y z-score)

ex\crv	2s5s	2s10s	2s30s	5s10s	5s30s	10s30s
1m	1.75 (1.4)	1.38 (0.2)	1.30 (-0.1)	1.24 (-0.6)	1.16 (-0.7)	1.27 (-0.4)
3m	2.02 (1.8)	1.55 (0.9)	1.39 (0.4)	1.45 (0.0)	1.32 (-0.2)	1.41 (-0.1)
6m	1.71 (1.1)	1.44 (0.6)	1.32 (0.4)	1.57 (0.3)	1.40 (0.0)	1.49 (0.0)
1y	1.44 (0.3)	1.36 (0.3)	1.29 (0.2)	1.84 (0.8)	1.56 (0.3)	1.64 (0.3)

implied/3m RV (1y z-score)

ex\crv	2s5s	2s10s	2s30s	5s10s	5s30s	10s30s
1m	0.81 (-1.5)	0.81 (-1.6)	0.96 (-1.2)	1.00 (-1.0)	1.06 (-0.8)	1.19 (-0.3)
3m	0.87 (-1.4)	0.87 (-1.5)	0.99 (-1.0)	1.14 (-0.8)	1.18 (-0.3)	1.31 (0.1)
6m	0.90 (-1.4)	0.91 (-1.2)	1.02 (-0.5)	1.26 (-0.3)	1.27 (0.1)	1.38 (0.3)
1y	1.02 (-1.0)	1.04 (-0.6)	1.13 (0.2)	1.56 (0.6)	1.45 (0.8)	1.50 (0.8)

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Close 6/9/23. Close-to-close realized vol is used. Deep red (deep blue) indicates largest (smallest) z-scores. Source: Citi Research

EUR conditional curves

We analyze conditional trades based on the entry point, implied/realized betas, and carry.

Prepared for Christopher Lee

Figure 100. Conditional curves: bear trades

		spt and fwd crvs			costless strk curve				imp vs rlzd beta			bear steepener				bear flattener			
structure	ntlns, $mn	spot crv, bp	fwd crv, bp	3m ZS	strk crv, bp	1m chg	3m ZS	hist %	imp beta	rlzd beta	rlzd-imp	crv pick-up, bp	vol pick-up, bp	tot pick-up, bp	3m roll, bp	crv pick-up, bp	vol pick-up, bp	tot pick-up, bp	3m roll, bp
3m 1y2y	100/50.6	-19.5	-25.7	-0.4	-12.6	11.8	0.1	10	0.35	-0.04	-0.38	6.2	-13.1	-6.9	-2.2	-6.2	13.1	6.9	2.2
3m 1y3y	100/34.2	-43.2	-48.8	-0.6	-35.6	11.3	-0.1	7	0.35	-0.09	-0.44	5.6	-13.1	-7.6	-2.2	-5.6	13.1	7.6	2.2
3m 1y5y	100/21.1	-69.6	-72.1	-1.0	-61.3	1.0	-0.6	4	0.31	-0.28	-0.59	2.5	-10.8	-8.3	-2.2	-2.5	10.8	8.3	2.2
3m 2y5y	100/41.7	-50.1	-46.4	-1.4	-48.5	-10.8	-1.4	2	-0.06	-0.23	-0.17	-3.7	2.2	-1.5	-1.6	3.7	-2.2	1.5	1.6
3m 2y10y	100/22.2	-61.0	-54.1	-1.1	-60.2	-18.7	-0.9	4	-0.16	-0.35	-0.19	-6.8	6.0	-0.8	-0.9	6.8	-6.0	0.8	0.9
3m 2y30y	100/9.4	-97.0	-90.2	-0.3	-99.7	-17.2	0.2	2	-0.27	-0.43	-0.16	-6.8	9.5	2.7	1.9	6.8	-9.5	-2.7	-1.9
3m 3y5y	100/61.6	-26.3	-23.3	-1.4	-25.4	-10.2	-1.3	2	-0.05	-0.17	-0.12	-3.0	2.1	-0.9	-0.9	3.0	-2.1	0.9	0.9
3m 3y10y	100/32.8	-37.3	-31.1	-1.0	-37.1	-18.2	-0.6	4	-0.16	-0.29	-0.14	-6.2	6.0	-0.2	-0.1	6.2	-6.0	0.2	0.1
3m 3y30y	100/13.9	-73.3	-67.1	0.0	-76.6	-16.9	0.3	2	-0.26	-0.37	-0.11	-6.2	9.5	3.3	1.9	6.2	-9.5	-3.3	-1.9
3m 5y10y	100/53.2	-10.9	-7.8	-0.5	-11.6	-8.1	-0.1	6	-0.10	-0.12	-0.02	-3.2	3.8	0.6	0.6	3.2	-3.8	-0.6	-0.6
3m 5y30y	100/22.6	-47.0	-43.8	0.4	-51.0	-6.7	0.7	2	-0.20	-0.18	0.02	-3.2	7.2	4.1	1.9	3.2	-7.2	-4.1	-1.9
3m 10y30y	100/42.4	-36.0	-36.0	0.9	-39.3	1.1	1.1	2	-0.09	-0.06	0.03	0.0	3.3	3.3	1.9	0.0	-3.3	-3.3	-1.9
6m 1y2y	100/50.6	-19.5	-23.6	-0.3	-14.2	7.4	0.1	10	0.16	-0.07	-0.23	4.1	-9.3	-5.3	-1.4	-4.1	9.3	5.3	1.4
6m 1y3y	100/34.2	-43.2	-43.7	-0.6	-34.9	4.6	-0.1	7	0.16	-0.14	-0.30	0.5	-8.8	-8.3	-2.8	-0.5	8.8	8.3	2.8
6m 1y5y	100/21.0	-69.6	-62.5	-1.0	-57.3	-7.0	-0.8	5	0.10	-0.31	-0.41	-7.1	-5.2	-12.3	-4.4	7.1	5.2	12.3	4.4
6m 2y5y	100/41.6	-50.1	-38.9	-1.2	-42.9	-14.2	-1.3	3	-0.07	-0.24	-0.17	-11.2	4.1	-7.1	-3.2	11.2	-4.1	7.1	3.2
6m 2y10y	100/22.2	-61.0	-42.7	-0.9	-52.3	-22.3	-0.7	5	-0.18	-0.36	-0.19	-18.3	9.6	-8.7	-4.1	18.3	-9.6	8.7	4.1
6m 2y30y	100/9.4	-97.0	-78.2	-0.1	-93.5	-20.4	0.3	2	-0.29	-0.44	-0.15	-18.8	15.3	-3.5	-2.9	18.8	-15.3	3.5	2.9
6m 3y5y	100/61.5	-26.3	-18.7	-1.2	-22.2	-11.4	-1.3	3	-0.06	-0.16	-0.10	-7.6	3.5	-4.1	-1.8	7.6	-3.5	4.1	1.8
6m 3y10y	100/32.8	-37.3	-22.5	-0.7	-31.6	-19.6	-0.5	5	-0.16	-0.28	-0.12	-14.8	9.1	-5.7	-2.8	14.8	-9.1	5.7	2.8
6m 3y30y	100/13.9	-73.3	-58.0	0.1	-72.8	-17.9	0.4	2	-0.28	-0.36	-0.08	-15.3	14.8	-0.5	-1.7	15.3	-14.8	0.5	1.7
6m 5y10y	100/53.2	-10.9	-3.8	-0.3	-9.2	-8.0	0.1	6	-0.10	-0.12	-0.02	-7.2	5.4	-1.7	-1.2	7.2	-5.4	1.7	1.2
6m 5y30y	100/22.6	-47.0	-39.3	0.5	-50.3	-6.3	0.8	2	-0.21	-0.18	0.03	-7.7	11.0	3.3	-0.2	7.7	-11.0	-3.3	0.2
6m 10y30y	100/42.4	-36.0	-35.5	1.0	-40.8	1.6	1.2	2	-0.10	-0.06	0.04	-0.5	5.3	4.8	0.8	0.5	-5.3	-4.8	-0.8
1y 1y2y	100/50.5	-19.5	-13.0	-0.5	-10.2	-3.9	-0.7	11	0.03	-0.06	-0.09	-6.5	-2.8	-9.3	-1.8	6.5	2.8	9.3	1.8
1y 1y3y	100/34.2	-43.2	-26.2	-0.8	-25.4	-9.6	-0.9	10	0.01	-0.16	-0.17	-17.0	-0.9	-17.9	-3.7	17.0	0.9	17.9	3.7
1y 1y5y	100/21.0	-69.6	-35.9	-0.9	-39.4	-21.1	-1.0	8	-0.04	-0.29	-0.25	-33.6	3.4	-30.2	-5.3	33.6	-3.4	30.2	5.3
1y 2y5y	100/41.6	-50.1	-22.9	-0.9	-29.3	-17.3	-0.9	6	-0.08	-0.22	-0.14	-27.2	6.4	-20.8	-3.6	27.2	-6.4	20.8	3.6
1y 2y10y	100/22.2	-61.0	-19.3	-0.6	-33.0	-24.6	-0.4	7	-0.17	-0.33	-0.17	-41.6	13.7	-27.9	-4.7	41.6	-13.7	27.9	4.7
1y 2y30y	100/9.4	-97.0	-54.2	0.1	-77.2	-22.9	0.3	5	-0.30	-0.40	-0.10	-42.8	23.0	-19.8	-3.5	42.8	-23.0	19.8	3.5
1y 3y5y	100/61.5	-26.3	-9.7	-0.9	-14.1	-11.6	-0.8	6	-0.05	-0.12	-0.07	-16.6	4.4	-12.2	-1.7	16.6	-4.4	12.2	1.7
1y 3y10y	100/32.8	-37.3	-6.2	-0.4	-17.8	-18.9	-0.3	7	-0.14	-0.23	-0.09	-31.1	11.6	-19.5	-2.9	31.1	-11.6	19.5	2.9
1y 3y30y	100/13.9	-73.3	-41.1	0.3	-61.8	-17.4	0.5	4	-0.27	-0.29	-0.03	-32.2	20.8	-11.5	-1.9	32.2	-20.8	11.5	1.9
1y 5y10y	100/53.3	-10.9	3.5	0.0	-3.5	-7.3	0.2	9	-0.09	-0.10	-0.02	-14.5	7.1	-7.4	-1.3	14.5	-7.1	7.4	1.3
1y 5y30y	100/22.5	-47.0	-31.3	0.7	-47.4	-5.5	0.8	3	-0.21	-0.16	0.05	-15.6	16.1	0.4	-0.5	15.6	-16.1	-0.4	0.5
1y 10y30y	100/42.3	-36.0	-34.9	1.0	-43.5	1.6	1.1	2	-0.11	-0.05	0.06	-1.2	8.6	7.4	0.8	1.2	-8.6	-7.4	-0.8
2y 1y2y	100/50.8	-19.5	-2.1	-0.8	-1.0	-9.0	-0.9	15	0.01	-0.09	-0.10	-17.4	-1.1	-18.6	-1.1	17.4	1.1	18.6	1.1
2y 1y3y	100/34.3	-43.2	-5.5	-0.9	-6.9	-14.4	-1.0	14	-0.02	-0.11	-0.09	-37.8	1.4	-36.4	-2.2	37.8	-1.4	36.4	2.2
2y 1y5y	100/21.1	-69.6	-4.7	-0.7	-10.5	-19.9	-0.7	13	-0.06	-0.22	-0.16	-64.9	5.9	-59.0	-3.0	64.9	-5.9	59.0	3.0
2y 2y5y	100/41.5	-50.1	-2.6	-0.6	-9.6	-10.8	-0.4	10	-0.06	-0.12	-0.06	-47.5	7.0	-40.4	-2.0	47.5	-7.0	40.4	2.0
2y 2y10y	100/22.2	-61.0	8.9	-0.2	-6.0	-12.4	0.0	10	-0.13	-0.20	-0.07	-69.9	14.9	-55.0	-2.6	69.9	-14.9	55.0	2.6
2y 2y30y	100/9.4	-97.0	-28.7	0.5	-58.2	-9.2	0.7	6	-0.27	-0.24	0.02	-68.3	29.4	-38.8	-1.4	68.3	-29.4	38.8	1.4
2y 3y5y	100/61.5	-26.3	0.8	-0.1	-3.7	-5.5	0.0	10	-0.04	-0.10	-0.06	-27.1	4.4	-22.7	-0.9	27.1	-4.4	22.7	0.9
2y 3y10y	100/32.8	-37.3	12.3	0.2	0.1	-7.2	0.3	10	-0.10	-0.18	-0.08	-49.6	12.2	-37.4	-1.5	49.6	-12.2	37.4	1.5
2y 3y30y	100/13.9	-73.3	-25.4	0.8	-51.9	-4.1	0.9	5	-0.24	-0.22	0.02	-47.9	26.5	-21.4	-0.4	47.9	-26.5	21.4	0.4
2y 5y10y	100/53.4	-10.9	11.5	0.4	3.9	-1.7	0.6	10	-0.06	-0.08	-0.01	-22.4	7.6	-14.8	-0.7	22.4	-7.6	14.8	0.7
2y 5y30y	100/22.5	-47.0	-26.2	1.0	-47.8	1.5	1.1	3	-0.19	-0.11	0.08	-20.8	21.7	0.9	0.3	20.8	-21.7	-0.9	-0.3
2y 10y30y	100/42.2	-36.0	-37.7	1.2	-51.1	2.9	1.3	1	-0.12	-0.03	0.09	1.6	13.5	15.1	1.0	-1.6	-13.5	-15.1	-1.0
3y 1y2y	100/50.6	-19.5	5.8	-0.5	6.6	-5.7	-0.6	25	0.00	0.03	0.03	-25.3	-0.8	-26.1	-0.6	25.3	0.8	26.1	0.6
3y 1y3y	100/34.2	-43.2	7.7	-0.2	5.3	-6.0	-0.3	19	-0.02	0.03	0.05	-51.0	2.5	-48.5	-1.0	51.0	-2.5	48.5	1.0
3y 1y5y	100/21.0	-69.6	12.7	0.0	5.5	-6.4	0.2	17	-0.05	-0.07	-0.02	-82.3	7.2	-75.1	-1.3	82.3	-7.2	75.1	1.3
3y 2y5y	100/41.5	-50.1	6.9	0.4	-1.1	-0.7	0.7	14	-0.05	-0.10	-0.04	-56.9	8.0	-49.0	-0.8	56.9	-8.0	49.0	0.8
3y 2y10y	100/22.2	-61.0	20.8	0.7	5.0	1.2	1.1	12	-0.11	-0.16	-0.05	-81.8	15.8	-66.0	-0.9	81.8	-15.8	66.0	0.9
3y 2y30y	100/9.4	-97.0	-22.6	1.1	-57.2	7.5	1.4	6	-0.26	-0.19	0.07	-74.4	34.6	-39.8	0.3	74.4	-34.6	39.8	-0.3
3y 3y5y	100/61.5	-26.3	5.0	0.5	0.3	-0.4	0.9	13	-0.03	-0.09	-0.06	-31.3	4.7	-26.7	-0.4	31.3	-4.7	26.7	0.4
3y 3y10y	100/32.9	-37.3	18.9	0.8	6.5	1.3	1.2	11	-0.09	-0.16	-0.07	-56.1	12.4	-43.8	-0.4	56.1	-12.4	43.8	0.4
3y 3y30y	100/13.9	-73.3	-24.5	1.2	-55.5	7.5	1.4	5	-0.23	-0.18	0.05	-48.8	31.0	-17.8	0.7	48.8	-31.0	17.8	-0.7
3y 5y10y	100/53.5	-10.9	13.9	1.1	6.3	1.7	1.4	11	-0.05	-0.06	-0.01	-24.8	7.6	-17.2	-0.1	24.8	-7.6	17.2	0.1
3y 5y30y	100/22.5	-47.0	-29.5	1.3	-55.3	7.6	1.5	2	-0.19	-0.08	0.11	-17.5	25.8	8.4	1.0	17.5	-25.8	-8.4	-1.0
3y 10y30y	100/42.1	-36.0	-43.4	1.3	-61.1	5.6	1.4	0	-0.13	-0.02	0.12	7.4	17.7	25.0	1.1	-7.4	-17.7	-25.0	-1.1

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Note: Close 6/9/23. The leg with the smaller vol is struck ATM, while the other leg is struck OTM for zero cost. We compute historical % of the strike curve using data starting in 1990. Implied beta is the beta between the curve and the average of end points implied by vols. Realized beta is empirical beta between the curve and the average of end points computed over the last 3m. The curve pick-up is the spread between spot and forward curves for steepeners, and the spread between forward and spot curve for flatteners. The vol pick-up is the spread between forward and strike curves for steepeners, and the spread between strike and forward curves for flatteners. Total pick-up is the sum of curve and vol pick-ups. 3m carry is computed assuming unchanged curve and vol cube and is expressed in bp of running yield. For each valuation metric, we mark three best **bear steepeners in red**, and three **best bear flatteners in green**. Note that the "trades" shown are backward looking, and suggest which actions would have been preferable at the pricing when the screen was run. These are illustrative only and not intended as open trade ideas. Source: Citi Research

Prepared for Christopher Lee

Figure 101. Conditional curves: bull trades

		spt and fwd crvs			costless strk curve				imp vs rlzd beta			bull steepener				bull flattener			
structure	ntlns, $mn	spot crv, bp	fwd crv, bp	3m ZS	strk crv, bp	1m chg	3m ZS	hist %	imp beta	rlzd beta	rlzd-imp	crv pick-up, bp	vol pick-up, bp	tot pick-up, bp	3m roll, bp	crv pick-up, bp	vol pick-up, bp	tot pick-up, bp	3m roll, bp
3m 1y2y	100/50.6	-19.5	-25.7	-0.4	-40.6	-1.1	-0.9	5	0.35	-0.04	-0.38	6.2	14.8	21.0	0.0	-6.2	-14.8	-21.0	0.0
3m 1y3y	100/34.2	-43.2	-48.8	-0.6	-63.4	-4.8	-1.0	4	0.35	-0.09	-0.44	5.6	14.6	20.1	0.0	-5.6	-14.6	-20.1	0.0
3m 1y5y	100/21.1	-69.6	-72.1	-1.0	-83.7	-14.2	-1.3	3	0.31	-0.28	-0.59	2.5	11.6	14.1	0.0	-2.5	-11.6	-14.1	0.0
3m 2y5y	100/41.7	-50.1	-46.4	-1.4	-44.0	-13.6	-1.2	3	-0.06	-0.23	-0.17	-3.7	-2.4	-6.1	0.0	3.7	2.4	6.1	0.0
3m 2y10y	100/22.2	-61.0	-54.1	-1.1	-47.4	-24.7	-1.0	5	-0.16	-0.35	-0.19	-6.8	-6.7	-13.6	0.0	6.8	6.7	13.6	0.0
3m 2y30y	100/9.4	-97.0	-90.2	-0.3	-79.5	-30.0	-0.6	4	-0.27	-0.43	-0.16	-6.8	-10.7	-17.5	0.0	6.8	10.7	17.5	0.0
3m 3y5y	100/61.6	-26.3	-23.3	-1.4	-21.0	-9.9	-1.3	3	-0.05	-0.17	-0.12	-3.0	-2.3	-5.3	0.0	3.0	2.3	5.3	0.0
3m 3y10y	100/32.8	-37.3	-31.1	-1.0	-24.5	-20.9	-1.0	6	-0.16	-0.29	-0.14	-6.2	-6.6	-12.8	0.0	6.2	6.6	12.8	0.0
3m 3y30y	100/13.9	-73.3	-67.1	0.0	-56.7	-25.9	-0.4	5	-0.26	-0.37	-0.11	-6.2	-10.5	-16.6	0.0	6.2	10.5	16.6	0.0
3m 5y10y	100/53.2	-10.9	-7.8	-0.5	-3.7	-10.9	-0.6	9	-0.10	-0.12	-0.02	-3.2	-4.1	-7.2	0.0	3.2	4.1	7.2	0.0
3m 5y30y	100/22.6	-47.0	-43.8	0.4	-36.1	-15.8	0.1	5	-0.20	-0.18	0.02	-3.2	-7.7	-10.9	0.0	3.2	7.7	10.9	0.0
3m 10y30y	100/42.4	-36.0	-36.0	0.9	-32.7	-4.4	0.7	2	-0.09	-0.06	0.03	0.0	-3.3	-3.3	0.0	0.0	3.3	3.3	0.0
6m 1y2y	100/50.6	-19.5	-23.6	-0.3	-34.0	-1.0	-0.7	6	0.16	-0.07	-0.23	4.1	10.4	14.5	-3.0	-4.1	-10.4	-14.5	3.0
6m 1y3y	100/34.2	-43.2	-43.7	-0.6	-53.3	-6.3	-0.9	5	0.16	-0.14	-0.30	0.5	9.6	10.1	-4.0	-0.5	-9.6	-10.1	4.0
6m 1y5y	100/21.0	-69.6	-62.5	-1.0	-67.9	-15.4	-1.1	4	0.10	-0.31	-0.41	-7.1	5.4	-1.7	-6.0	7.1	-5.4	1.7	6.0
6m 2y5y	100/41.6	-50.1	-38.9	-1.2	-34.4	-14.8	-1.0	5	-0.07	-0.24	-0.17	-11.2	-4.5	-15.6	-3.3	11.2	4.5	15.6	3.3
6m 2y10y	100/22.2	-61.0	-42.7	-0.9	-32.0	-24.7	-0.8	7	-0.18	-0.36	-0.19	-18.3	-10.7	-29.0	-5.1	18.3	10.7	29.0	5.1
6m 2y30y	100/9.4	-97.0	-78.2	-0.1	-61.1	-29.5	-0.4	6	-0.29	-0.44	-0.15	-18.8	-17.1	-36.0	-5.5	18.8	17.1	36.0	5.5
6m 3y5y	100/61.5	-26.3	-18.7	-1.2	-14.9	-9.4	-0.9	6	-0.06	-0.16	-0.10	-7.6	-3.8	-11.4	-2.2	7.6	3.8	11.4	2.2
6m 3y10y	100/32.8	-37.3	-22.5	-0.7	-12.6	-19.2	-0.7	8	-0.16	-0.28	-0.12	-14.8	-9.9	-24.6	-4.1	14.8	9.9	24.6	4.1
6m 3y30y	100/13.9	-73.3	-58.0	0.1	-41.8	-23.7	-0.2	6	-0.28	-0.36	-0.08	-15.3	-16.2	-31.5	-4.6	15.3	16.2	31.5	4.6
6m 5y10y	100/53.2	-10.9	-3.8	-0.3	1.9	-9.9	-0.5	10	-0.10	-0.12	-0.02	-7.2	-5.7	-12.8	-2.2	7.2	5.7	12.8	2.2
6m 5y30y	100/22.6	-47.0	-39.3	0.5	-27.7	-14.3	0.2	6	-0.21	-0.18	0.03	-7.7	-11.6	-19.3	-2.9	7.7	11.6	19.3	2.9
6m 10y30y	100/42.4	-36.0	-35.5	1.0	-30.2	-4.1	0.7	3	-0.10	-0.06	0.04	-0.5	-5.3	-5.9	-1.0	0.5	5.3	5.9	1.0
1y 1y2y	100/50.5	-19.5	-13.0	-0.5	-15.9	-3.5	-0.3	9	0.03	-0.06	-0.09	-6.5	2.9	-3.6	-3.6	6.5	-2.9	3.6	3.6
1y 1y3y	100/34.2	-43.2	-26.2	-0.8	-27.3	-12.6	-0.7	9	0.01	-0.16	-0.17	-17.0	1.1	-15.9	-4.8	17.0	-1.1	15.9	4.8
1y 1y5y	100/21.0	-69.6	-35.9	-0.9	-32.5	-20.9	-0.8	9	-0.04	-0.29	-0.25	-33.6	-3.5	-37.1	-6.7	33.6	3.5	37.1	6.7
1y 2y5y	100/41.6	-50.1	-22.9	-0.9	-16.5	-17.3	-0.9	8	-0.08	-0.22	-0.14	-27.2	-6.4	-33.6	-3.2	27.2	6.4	33.6	3.2
1y 2y10y	100/22.2	-61.0	-19.3	-0.6	-5.2	-25.5	-0.6	10	-0.17	-0.33	-0.17	-41.6	-14.1	-55.8	-4.5	41.6	14.1	55.8	4.5
1y 2y30y	100/9.4	-97.0	-54.2	0.1	-30.1	-28.0	-0.1	8	-0.30	-0.40	-0.10	-42.8	-24.1	-66.9	-4.6	42.8	24.1	66.9	4.6
1y 3y5y	100/61.5	-26.3	-9.7	-0.9	-5.2	-8.3	-0.8	9	-0.05	-0.12	-0.07	-16.6	-4.5	-21.1	-2.0	16.6	4.5	21.1	2.0
1y 3y10y	100/32.8	-37.3	-6.2	-0.4	5.9	-16.4	-0.5	11	-0.14	-0.23	-0.09	-31.1	-12.0	-43.1	-3.4	31.1	12.0	43.1	3.4
1y 3y30y	100/13.9	-73.3	-41.1	0.3	-19.4	-18.7	0.1	8	-0.27	-0.29	-0.03	-32.2	-21.7	-53.9	-3.7	32.2	21.7	53.9	3.7
1y 5y10y	100/53.3	-10.9	3.5	0.0	10.7	-8.2	-0.2	13	-0.09	-0.10	-0.02	-14.5	-7.1	-21.6	-1.6	14.5	7.1	21.6	1.6
1y 5y30y	100/22.5	-47.0	-31.3	0.7	-15.1	-10.6	0.4	8	-0.21	-0.16	0.05	-15.6	-16.2	-31.9	-2.1	15.6	16.2	31.9	2.1
1y 10y30y	100/42.3	-36.0	-34.9	1.0	-26.5	-2.2	0.9	3	-0.11	-0.05	0.06	-1.2	-8.4	-9.5	-0.7	1.2	8.4	9.5	0.7
2y 1y2y	100/50.8	-19.5	-2.1	-0.8	-3.4	-10.0	-0.8	14	0.01	-0.09	-0.10	-17.4	1.3	-16.1	-1.2	17.4	-1.3	16.1	1.2
2y 1y3y	100/34.3	-43.2	-5.5	-0.9	-4.3	-17.9	-0.8	14	-0.02	-0.11	-0.09	-37.8	-1.1	-38.9	-1.8	37.8	1.1	38.9	1.8
2y 1y5y	100/21.1	-69.6	-4.7	-0.7	0.9	-23.0	-0.7	15	-0.06	-0.22	-0.16	-64.9	-5.5	-70.4	-2.7	64.9	5.5	70.4	2.7
2y 2y5y	100/41.5	-50.1	-2.6	-0.6	4.2	-13.0	-0.6	17	-0.06	-0.12	-0.06	-47.5	-6.8	-54.3	-1.5	47.5	6.8	54.3	1.5
2y 2y10y	100/22.2	-61.0	8.9	-0.2	23.5	-19.2	-0.2	18	-0.13	-0.20	-0.07	-69.9	-14.6	-84.5	-1.9	69.9	14.6	84.5	1.9
2y 2y30y	100/9.4	-97.0	-28.7	0.5	0.2	-17.3	0.3	11	-0.27	-0.24	0.02	-68.3	-28.9	-97.2	-1.5	68.3	28.9	97.2	1.5
2y 3y5y	100/61.5	-26.3	0.8	-0.1	5.1	-5.1	-0.2	19	-0.04	-0.10	-0.06	-27.1	-4.3	-31.4	-0.9	27.1	4.3	31.4	0.9
2y 3y10y	100/32.8	-37.3	12.3	0.2	24.2	-11.1	0.0	17	-0.10	-0.18	-0.08	-49.6	-11.9	-61.5	-1.4	49.6	11.9	61.5	1.4
2y 3y30y	100/13.9	-73.3	-25.4	0.8	0.6	-9.2	0.6	11	-0.24	-0.22	0.02	-47.9	-25.9	-73.9	-1.0	47.9	25.9	73.9	1.0
2y 5y10y	100/53.4	-10.9	11.5	0.4	18.8	-6.0	0.3	17	-0.06	-0.08	-0.01	-22.4	-7.3	-29.8	-0.5	22.4	7.3	29.8	0.5
2y 5y30y	100/22.5	-47.0	-26.2	1.0	-5.3	-4.1	0.8	10	-0.19	-0.11	0.08	-20.8	-20.8	-41.6	-0.3	20.8	20.8	41.6	0.3
2y 10y30y	100/42.2	-36.0	-37.7	1.2	-24.9	2.0	1.1	3	-0.12	-0.03	0.09	1.6	-12.8	-11.1	0.1	-1.6	12.8	11.1	-0.1
3y 1y2y	100/50.6	-19.5	5.8	-0.5	5.0	-5.4	-0.4	22	0.00	0.03	0.03	-25.3	0.8	-24.5	-0.8	25.3	-0.8	24.5	0.8
3y 1y3y	100/34.2	-43.2	7.7	-0.2	10.0	-8.5	-0.2	24	-0.02	0.03	0.05	-51.0	-2.2	-53.2	-1.2	51.0	2.2	53.2	1.2
3y 1y5y	100/21.0	-69.6	12.7	0.0	19.4	-11.7	-0.1	24	-0.05	-0.07	-0.02	-82.3	-6.7	-88.9	-1.5	82.3	6.7	88.9	1.5
3y 2y5y	100/41.5	-50.1	6.9	0.4	14.3	-6.2	0.1	25	-0.05	-0.10	-0.04	-56.9	-7.5	-64.4	-0.7	56.9	7.5	64.4	0.7
3y 2y10y	100/22.2	-61.0	20.8	0.7	35.7	-9.0	0.3	21	-0.11	-0.16	-0.05	-81.8	-14.9	-96.7	-0.8	81.8	14.9	96.7	0.8
3y 2y30y	100/9.4	-97.0	-22.6	1.1	10.3	-5.5	0.8	11	-0.26	-0.19	0.07	-74.4	-32.9	-107.3	-0.2	74.4	32.9	107.3	0.2
3y 3y5y	100/61.5	-26.3	5.0	0.5	9.3	-3.1	0.1	23	-0.03	-0.09	-0.06	-31.3	-4.4	-35.7	-0.3	31.3	4.4	35.7	0.3
3y 3y10y	100/32.9	-37.3	18.9	0.8	30.6	-5.8	0.3	20	-0.09	-0.16	-0.07	-56.1	-11.7	-67.8	-0.3	56.1	11.7	67.8	0.3
3y 3y30y	100/13.9	-73.3	-24.5	1.2	4.8	-2.2	0.9	11	-0.23	-0.18	0.05	-48.8	-29.4	-78.1	0.1	48.8	29.4	78.1	-0.1
3y 5y10y	100/53.5	-10.9	13.9	1.1	21.0	-2.6	0.5	19	-0.05	-0.06	-0.01	-24.8	-7.1	-32.0	-0.1	24.8	7.1	32.0	0.1
3y 5y30y	100/22.5	-47.0	-29.5	1.3	-5.2	1.3	1.1	10	-0.19	-0.08	0.11	-17.5	-24.3	-41.8	0.4	17.5	24.3	41.8	-0.4
3y 10y30y	100/42.1	-36.0	-43.4	1.3	-27.1	3.9	1.3	3	-0.13	-0.02	0.12	7.4	-16.4	-9.0	0.4	-7.4	16.4	9.0	-0.4

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Note: Close 6/9/23. The leg with the smaller vol is struck ATM, while the other leg is struck OTM for zero cost. We compute historical % of the strike curve using data starting in 1990. Implied beta is the beta between the curve and the average of end points implied by vols. Realized beta is empirical beta between the curve and the average of end points computed over the last 3m. The curve pick-up is the spread between spot and forward curves for steepeners, and the spread between forward and spot curve for flatteners. The vol pick-up is the spread between forward and strike curves for steepeners, and the spread between strike and forward curves for flatteners. Total pick-up is the sum of curve and vol pick-ups. 3m roll is computed assuming unchanged curve and vol cube and is expressed in bp of running yield. For each valuation metric, we mark three best **bull steepeners in red**, and three **best bull flatteners in green**. Note that the "trades" shown are backward looking, and suggest which actions would have been preferable at the pricing when the screen was run. These are illustrative only and not intended as open trade ideas. Source: Citi Research

Prepared for Christopher Lee

EUR conditional flies

We analyze conditional fly trades based on the entry point, directionality with rates and carry.

Figure 102. Conditional payer flies (50/50 DV01-weighted)

		spt and fwd crvs			fwd fly vs the belly					costless strike				long payer in the belly				short payer in the belly			
curve	notionals, $mn	spt fly, bp	fwd fly, bp	3m ZS	3m corr (chgs)	1y corr (lvls)	1y RC, bp	1y RC, ZC	belly OTM, bp	strk fly, bp	1m chg	3m ZS	hist %	crv pick-up, bp	vol pick-up, bp	tot pick-up, bp	3m roll, bp	crv pick-up, bp	vol pick-up, bp	tot pick-up, bp	3m roll, bp
3m 1y3y5y	146.1/100/30.8	-8.4	-12.7	0.1	39	-55	-5.8	-0.7	7.3	-5.4	12.4	0.3	16	4.3	-7.3	-3.1	-3.0	-4.3	7.3	3.1	3.0
3m 1y3y7y	146.1/100/22.5	-3.9	-9.1	0.3	53	-27	-5.1	-0.7	8.1	-1.0	14.7	0.3	74	5.2	-8.1	-3.0	-2.7	-5.2	8.1	3.0	2.7
3m 2y3y5y	73.9/100/30.8	1.3	0.1	1.0	58	67	0.8	0.7	1.0	1.2	5.3	0.5	93	1.2	-1.0	0.2	0.1	-1.2	1.0	-0.2	-0.1
3m 2y5y7y	120.0/100/36.6	-20.5	-19.6	-1.5	-43	-71	-7.7	-1.2	-0.4	-19.9	-2.9	-1.4	2	-1.0	0.4	-0.6	-0.6	1.0	-0.4	0.6	0.6
3m 2y5y10y	120.0/100/26.6	-19.6	-19.3	-1.2	-25	-55	-7.0	-1.4	0.8	-18.5	-1.3	-1.2	5	-0.3	-0.8	-1.0	-1.1	0.3	0.8	1.0	1.1
3m 2y10y30y	225.5/100/21.2	-12.5	-9.0	-1.8	-32	-44	-22.4	-1.6	-1.4	-10.4	-9.4	-2.0	6	-3.4	1.4	-2.0	-2.1	3.4	-1.4	2.0	2.1
3m 3y5y7y	81.2/100/36.6	-8.7	-8.0	-2.0	-41	-68	-3.7	-1.6	-0.3	-8.4	-3.0	-1.7	1	-0.6	0.3	-0.3	-0.3	0.6	-0.3	0.3	0.3
3m 3y5y10y	81.2/100/26.6	-7.7	-7.8	-1.2	-4	40	-3.0	-1.8	0.8	-7.0	-1.3	-1.3	69	0.1	-0.8	-0.7	-0.7	-0.1	0.8	0.7	0.7
3m 3y7y10y	111.0/100/36.4	-16.7	-15.0	-2.4	-51	-72	-6.3	-1.3	-0.6	-15.7	-5.9	-1.5	4	-1.7	0.6	-1.1	-1.0	1.7	-0.6	1.1	1.0
3m 3y7y15y	111.0/100/25.9	-17.5	-15.9	-2.4	-35	-44	-7.8	-1.8	-0.1	-16.0	-5.1	-2.4	6	-1.6	0.1	-1.5	-1.5	1.6	-0.1	1.5	1.5
3m 3y10y30y	152.6/100/21.2	-0.6	2.5	-2.1	-33	-33	-17.6	-1.7	-1.3	1.1	-9.4	-2.3	18	-3.1	1.3	-1.7	-1.8	3.1	-1.3	1.7	1.8
3m 5y10y15y	93.9/100/35.6	-6.2	-4.7	-2.1	-35	-62	-4.3	-1.3	-1.3	-6.0	-3.2	-1.0	6	-1.5	1.3	-0.2	-0.2	1.5	-1.3	0.2	0.2
3m 5y10y30y	93.9/100/21.2	12.6	14.1	-1.6	-10	15	-11.7	-1.9	-0.3	13.8	-4.2	-1.9	75	-1.6	0.3	-1.3	-1.3	1.6	-0.3	1.3	1.3
3m 10y20y30y	87.8/100/37.2	8.9	8.7	-0.1	-25	-55	0.4	0.1	-0.6	8.1	-1.1	0.2	16	0.2	0.6	0.8	0.8	-0.2	-0.6	-0.8	-0.8
6m 1y3y5y	146.2/100/30.8	-8.4	-12.5	0.1	5	-56	-3.2	-0.6	6.1	-6.4	9.2	0.4	14	4.1	-6.1	-2.0	-0.5	-4.1	6.1	2.0	0.5
6m 1y3y7y	146.2/100/22.5	-3.9	-10.0	0.3	26	-22	-2.7	-0.5	7.1	-2.9	11.4	0.4	69	6.0	-7.1	-1.0	-0.1	-6.0	7.1	1.0	0.1
6m 2y3y5y	73.9/100/30.8	1.3	-0.7	0.9	34	69	0.7	0.7	1.4	0.7	4.4	0.8	93	2.0	-1.4	0.6	0.1	-2.0	1.4	-0.6	-0.1
6m 2y5y7y	120.1/100/36.6	-20.5	-16.9	-1.9	-51	-70	-6.5	-1.3	-1.1	-18.0	-4.8	-1.6	2	-3.6	1.1	-2.5	-1.2	3.6	-1.1	2.5	1.2
6m 2y5y10y	120.1/100/26.6	-19.6	-17.6	-1.6	-37	-52	-6.1	-1.5	0.6	-17.0	-3.3	-1.5	10	-2.0	-0.6	-2.6	-1.1	2.0	0.6	2.6	1.1
6m 2y10y30y	225.6/100/21.2	-12.5	-3.6	-2.0	-38	-39	-19.9	-1.6	-2.1	-5.7	-11.1	-2.1	8	-8.9	2.1	-6.8	-2.8	8.9	-2.1	6.8	2.8
6m 3y5y7y	81.2/100/36.6	-8.7	-6.8	-2.1	-38	-64	-3.2	-1.6	-0.8	-7.7	-3.6	-2.1	1	-1.8	0.8	-1.0	-0.5	1.8	-0.8	1.0	0.5
6m 3y5y10y	81.2/100/26.6	-7.7	-7.5	-1.5	4	52	-2.8	-2.0	0.9	-6.6	-2.1	-1.6	71	-0.2	-0.9	-1.1	-0.4	0.2	0.9	1.1	0.4
6m 3y7y10y	111.0/100/36.4	-16.7	-12.5	-2.2	-51	-71	-5.1	-1.3	-1.0	-13.5	-6.4	-1.5	4	-4.2	1.0	-3.2	-1.3	4.2	-1.0	3.2	1.3
6m 3y7y15y	111.0/100/25.9	-17.5	-13.6	-2.6	-35	-38	-7.0	-1.9	0.0	-13.6	-5.7	-2.5	11	-3.9	0.0	-3.9	-1.5	3.9	0.0	3.9	1.5
6m 3y10y30y	152.6/100/21.2	-0.6	6.5	-2.1	-32	-26	-16.1	-1.8	-1.9	4.6	-9.8	-2.5	24	-7.1	1.9	-5.3	-2.1	7.1	-1.9	5.3	2.1
6m 5y10y15y	93.9/100/35.6	-6.2	-2.9	-1.8	-36	-59	-3.8	-1.3	-1.7	-4.6	-3.0	-0.6	6	-3.3	1.7	-1.6	-0.8	3.3	-1.7	1.6	0.8
6m 5y10y30y	93.9/100/21.2	12.6	15.9	-1.6	-11	20	-11.3	-1.9	-0.1	15.8	-4.1	-2.0	79	-3.3	0.1	-3.2	-1.1	3.3	-0.1	3.2	1.1
6m 10y20y30y	87.8/100/37.2	8.9	8.6	0.1	-24	-51	0.9	0.4	-1.0	7.6	-1.0	0.4	15	0.2	1.0	1.2	0.3	-0.2	-1.0	-1.2	-0.3
1y 1y3y5y	146.4/100/30.8	-8.4	-8.2	-0.6	-44	-57	-1.7	-0.6	2.6	-5.6	1.0	-0.3	16	-0.2	-2.6	-2.8	-1.0	0.2	2.6	2.8	1.0
1y 1y3y7y	146.4/100/22.5	-3.9	-7.8	-0.2	-24	0	-1.5	-0.6	4.0	-3.8	3.1	-0.2	67	3.8	-4.0	-0.2	-0.7	-3.8	4.0	0.2	0.7
1y 2y3y5y	74.0/100/30.8	1.3	-1.7	0.4	-30	71	-0.1	-0.1	1.2	-0.5	2.6	0.3	87	3.0	-1.2	1.9	-0.1	-3.0	1.2	-1.9	0.1
1y 2y5y7y	120.2/100/36.6	-20.5	-11.0	-1.9	-53	-67	-4.9	-1.3	-1.7	-12.7	-6.6	-1.2	4	-9.5	1.7	-7.8	-1.6	9.5	-1.7	7.8	1.6
1y 2y5y10y	120.2/100/26.6	-19.6	-13.2	-2.2	-41	-44	-4.9	-1.8	0.3	-12.9	-5.3	-1.7	19	-6.3	-0.3	-6.6	-1.2	6.3	0.3	6.6	1.2
1y 2y10y30y	225.6/100/21.2	-12.5	7.8	-2.3	-39	-28	-16.9	-1.7	-2.6	5.2	-12.4	-2.1	24	-20.2	2.6	-17.7	-3.1	20.2	-2.6	17.7	3.1
1y 3y5y7y	81.3/100/36.6	-8.7	-4.4	-2.0	-15	-60	-2.2	-1.5	-0.8	-5.2	-3.5	-1.4	5	-4.3	0.8	-3.5	-0.6	4.3	-0.8	3.5	0.6
1y 3y5y10y	81.3/100/26.6	-7.7	-6.6	-2.0	30	65	-2.2	-2.0	1.3	-5.4	-2.3	-1.9	78	-1.1	-1.3	-2.3	-0.2	1.1	1.3	2.3	0.2
1y 3y7y10y	111.0/100/36.4	-16.7	-7.6	-1.7	-38	-67	-3.4	-1.1	-1.6	-9.1	-6.5	-1.0	7	-9.2	1.6	-7.6	-0.9	9.2	-1.6	7.6	0.9
1y 3y7y15y	111.0/100/25.9	-17.5	-8.8	-2.8	-20	-20	-5.8	-2.0	-0.2	-8.9	-5.9	-2.3	27	-8.7	0.2	-8.5	-1.0	8.7	-0.2	8.5	1.0
1y 3y10y30y	152.5/100/21.2	-0.6	14.3	-2.2	-23	-12	-13.9	-1.8	-1.6	12.8	-9.4	-2.4	40	-15.0	1.6	-13.4	-2.1	15.0	-1.6	13.4	2.1
1y 5y10y15y	93.9/100/35.6	-6.2	0.6	-1.5	-33	-52	-3.1	-1.3	-2.1	-1.5	-2.7	-0.5	9	-6.8	2.1	-4.7	-0.8	6.8	-2.1	4.7	0.8
1y 5y10y30y	93.9/100/21.2	12.6	19.2	-1.6	-7	27	-10.9	-1.9	0.6	19.8	-3.7	-1.7	86	-6.7	-0.6	-7.3	-1.1	6.7	0.6	7.3	1.1
1y 10y20y30y	87.8/100/37.2	8.9	8.3	0.6	-20	-41	1.6	0.7	-1.7	6.6	-1.0	0.7	12	0.5	1.7	2.3	0.3	-0.5	-1.7	-2.3	-0.3

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Note: Close 6/9/23. Conditional flies are structured by buying (selling) payers in the belly vs selling (buying) payers on the wings with 50/50 DV01 weights. The wings are struck ATM, while the strike on the belly is solved for zero cost. Spot, forward and costless strike rates flies are reported using -0.5/1/-0.5 weights. We compute correlations of the forward fly with the belly (using a 3m window in changes and 1y window in levels) and rich/cheap of the forward fly to the belly based on 1y regression in levels. We also compute historical % of the strike fly using data starting in 1990. 3m roll is computed assuming unchanged curve and vol cube and is expressed in bp of running yield. For each valuation metric, we mark three best **long belly payer trades in red**, and three **best short belly payer trades in green**. Note that the "trades" shown are backward looking, and suggest which actions would have been preferable at the pricing when the screen was run. These are illustrative only and not intended as open trade ideas. Source: Citi Research

Prepared for Christopher Lee

Figure 103. Conditional receiver flies (50/50 DV01-weighted)

		spt and fwd crvs			fwd fly vs the belly					costless strike				long receiver in the belly				short receiver in the belly			
curve	ntl, $mn	spt fly, bp	fwd fly, bp	3m ZS	3m corr (chgs)	1y corr (lvls)	1y RC, bp	1y RC, zs	belly OTM, bp	strk fly, bp	1m chg	3m ZS	hist %	crv pick-up, bp	vol pick-up, bp	tot pick-up, bp	3m roll, bp	crv pick-up, bp	vol pick-up, bp	tot pick-up, bp	3m roll, bp
3m 1y3y5y	146.1/100/30.8	-8.4	-12.7	0.1	39	-55	-5.8	-0.7	8.1	-20.8	4.7	-0.5	5	-4.3	-8.1	-12.4	0.0	4.3	8.1	12.4	0.0
3m 1y3y7y	146.1/100/22.5	-3.9	-9.1	0.3	53	-27	-5.1	-0.7	9.0	-18.1	7.3	-0.3	22	-5.2	-9.0	-14.1	0.0	5.2	9.0	14.1	0.0
3m 2y3y5y	73.9/100/30.8	1.3	0.1	1.0	58	67	0.8	0.7	1.1	-0.9	3.6	0.9	86	-1.2	-1.1	-2.3	0.0	1.2	1.1	2.3	0.0
3m 2y5y7y	120.0/100/36.6	-20.5	-19.6	-1.5	-43	-71	-7.7	-1.2	-0.4	-19.2	-3.2	-1.2	2	1.0	0.4	1.4	0.0	-1.0	-0.4	-1.4	0.0
3m 2y5y10y	120.0/100/26.6	-19.6	-19.3	-1.2	-25	-55	-7.0	-1.4	0.8	-20.1	-1.2	-1.1	3	0.3	-0.8	-0.6	0.0	-0.3	0.8	0.6	0.0
3m 2y10y30y	225.5/100/21.2	-12.5	-9.0	-1.8	-32	-44	-22.4	-1.6	-1.4	-7.6	-8.3	-1.3	7	3.4	1.4	4.8	0.0	-3.4	-1.4	-4.8	0.0
3m 3y5y7y	81.2/100/36.6	-8.7	-8.0	-2.0	-41	-68	-3.7	-1.6	-0.4	-7.7	-2.1	-1.3	1	0.6	0.4	1.0	0.0	-0.6	-0.4	-1.0	0.0
3m 3y5y10y	81.2/100/26.6	-7.7	-7.8	-1.2	-4	40	-3.0	-1.8	0.9	-8.6	0.0	-1.0	59	-0.1	-0.9	-0.9	0.0	0.1	0.9	0.9	0.0
3m 3y7y10y	111.0/100/36.4	-16.7	-15.0	-2.4	-51	-72	-6.3	-1.3	-0.6	-14.4	-5.4	-1.5	4	1.7	0.6	2.3	0.0	-1.7	-0.6	-2.3	0.0
3m 3y7y15y	111.0/100/25.9	-17.5	-15.9	-2.4	-35	-44	-7.8	-1.8	-0.1	-15.8	-3.8	-1.7	6	1.6	0.1	1.6	0.0	-1.6	-0.1	-1.6	0.0
3m 3y10y30y	152.6/100/21.2	-0.6	2.5	-2.1	-33	-33	-17.6	-1.7	-1.4	3.8	-7.1	-1.4	23	3.1	1.4	4.5	0.0	-3.1	-1.4	-4.5	0.0
3m 5y10y15y	93.9/100/35.6	-6.2	-4.7	-2.1	-35	-62	-4.3	-1.3	-1.3	-3.4	-3.2	-1.0	7	1.5	1.3	2.8	0.0	-1.5	-1.3	-2.8	0.0
3m 5y10y30y	93.9/100/21.2	12.6	14.1	-1.6	-10	15	-11.7	-1.9	-0.3	14.4	-2.3	-1.3	77	1.6	0.3	1.9	0.0	-1.6	-0.3	-1.9	0.0
3m 10y20y30y	87.8/100/37.2	8.9	8.7	-0.1	-25	-55	0.4	0.1	-0.6	9.3	-1.6	0.1	21	-0.2	0.6	0.5	0.0	0.2	-0.6	-0.5	0.0
6m 1y3y5y	146.2/100/30.8	-8.4	-12.5	0.1	5	-56	-3.2	-0.6	6.6	-19.1	2.8	-0.5	6	-4.1	-6.6	-10.7	0.9	4.1	6.6	10.7	-0.9
6m 1y3y7y	146.2/100/22.5	-3.9	-10.0	0.3	26	-22	-2.7	-0.5	7.7	-17.7	5.5	-0.2	23	-6.0	-7.7	-13.7	0.4	6.0	7.7	13.7	-0.4
6m 2y3y5y	73.9/100/30.8	1.3	-0.7	0.9	34	69	0.7	0.7	1.6	-2.3	2.1	0.4	78	-2.0	-1.6	-3.6	-0.5	2.0	1.6	3.6	0.5
6m 2y5y7y	120.1/100/36.6	-20.5	-16.9	-1.9	-51	-70	-6.5	-1.3	-1.2	-15.8	-4.6	-1.1	3	3.6	1.2	4.8	1.3	-3.6	-1.2	-4.8	-1.3
6m 2y5y10y	120.1/100/26.6	-19.6	-17.6	-1.6	-37	-52	-6.1	-1.5	0.6	-18.2	-2.6	-1.3	6	2.0	-0.6	1.4	0.7	-2.0	0.6	-1.4	-0.7
6m 2y10y30y	225.6/100/21.2	-12.5	-3.6	-2.0	-38	-39	-19.9	-1.6	-2.2	-1.4	-9.8	-1.2	14	8.9	2.2	11.1	2.5	-8.9	-2.2	-11.1	-2.5
6m 3y5y7y	81.2/100/36.6	-8.7	-6.8	-2.1	-38	-64	-3.2	-1.6	-0.9	-6.0	-2.0	-1.1	4	1.8	0.9	2.7	0.6	-1.8	-0.9	-2.7	-0.6
6m 3y5y10y	81.2/100/26.6	-7.7	-7.5	-1.5	4	52	-2.8	-2.0	0.9	-8.4	0.0	-1.0	61	0.2	-0.9	-0.7	0.1	-0.2	0.9	0.7	-0.1
6m 3y7y10y	111.0/100/36.4	-16.7	-12.5	-2.2	-51	-71	-5.1	-1.3	-1.0	-11.5	-5.4	-1.1	6	4.2	1.0	5.2	1.1	-4.2	-1.0	-5.2	-1.1
6m 3y7y15y	111.0/100/25.9	-17.5	-13.6	-2.6	-35	-38	-7.0	-1.9	0.0	-13.5	-3.9	-1.6	11	3.9	0.0	3.9	0.9	-3.9	0.0	-3.9	-0.9
6m 3y10y30y	152.6/100/21.2	-0.6	6.5	-2.1	-32	-26	-16.1	-1.8	-1.9	8.4	-7.2	-1.3	32	7.1	1.9	9.0	1.9	-7.1	-1.9	-9.0	-1.9
6m 5y10y15y	93.9/100/35.6	-6.2	-2.9	-1.8	-36	-59	-3.8	-1.3	-1.7	-1.2	-3.2	-0.9	9	3.3	1.7	5.0	0.9	-3.3	-1.7	-5.0	-0.9
6m 5y10y30y	93.9/100/21.2	12.6	15.9	-1.6	-11	20	-11.3	-1.9	-0.1	16.0	-2.4	-1.2	80	3.3	0.1	3.4	0.7	-3.3	-0.1	-3.4	-0.7
6m 10y20y30y	87.8/100/37.2	8.9	8.6	0.1	-24	-51	0.9	0.4	-1.0	9.6	-1.6	0.3	21	-0.2	1.0	0.8	0.1	0.2	-1.0	-0.8	-0.1
1y 1y3y5y	146.4/100/30.8	-8.4	-8.2	-0.6	-44	-57	-1.7	-0.6	2.8	-11.0	-1.8	-0.5	9	0.2	-2.8	-2.6	1.4	-0.2	2.8	2.6	-1.4
1y 1y3y7y	146.4/100/22.5	-3.9	-7.8	-0.2	-24	0	-1.5	-0.6	4.3	-12.0	0.6	-0.3	41	-3.8	-4.3	-8.1	0.9	3.8	4.3	8.1	-0.9
1y 2y3y5y	74.0/100/30.8	1.3	-1.7	0.4	-30	71	-0.1	-0.1	1.4	-3.1	-0.2	-0.4	73	-3.0	-1.4	-4.4	-0.4	3.0	1.4	4.4	0.4
1y 2y5y7y	120.2/100/36.6	-20.5	-11.0	-1.9	-53	-67	-4.9	-1.3	-1.6	-9.4	-6.9	-1.1	7	9.5	1.6	11.2	1.2	-9.5	-1.6	-11.2	-1.2
1y 2y5y10y	120.2/100/26.6	-19.6	-13.2	-2.2	-41	-44	-4.9	-1.8	0.4	-13.6	-5.1	-1.6	18	6.3	-0.4	5.9	0.9	-6.3	0.4	-5.9	-0.9
1y 2y10y30y	225.6/100/21.2	-12.5	7.8	-2.3	-39	-28	-16.9	-1.7	-2.3	10.1	-12.0	-1.3	29	20.2	2.3	22.6	2.3	-20.2	-2.3	-22.6	-2.3
1y 3y5y7y	81.3/100/36.6	-8.7	-4.4	-2.0	-15	-60	-2.2	-1.5	-0.8	-3.6	-2.2	-1.2	9	4.3	0.8	5.0	0.5	-4.3	-0.8	-5.0	-0.5
1y 3y5y10y	81.3/100/26.6	-7.7	-6.6	-2.0	30	65	-2.2	-2.0	1.3	-7.9	-0.3	-1.1	63	1.1	-1.3	-0.2	0.2	-1.1	1.3	0.2	-0.2
1y 3y7y10y	111.0/100/36.4	-16.7	-7.6	-1.7	-38	-67	-3.4	-1.1	-1.5	-6.0	-5.1	-0.9	9	9.2	1.5	10.7	1.3	-9.2	-1.5	-10.7	-1.3
1y 3y7y15y	111.0/100/25.9	-17.5	-8.8	-2.8	-20	-20	-5.8	-2.0	-0.2	-8.6	-3.9	-1.7	28	8.7	0.2	8.8	1.2	-8.7	-0.2	-8.8	-1.2
1y 3y10y30y	152.5/100/21.2	-0.6	14.3	-2.2	-23	-12	-13.9	-1.8	-1.5	15.8	-7.3	-1.4	45	15.0	1.5	16.5	1.6	-15.0	-1.5	-16.5	-1.6
1y 5y10y15y	93.9/100/35.6	-6.2	0.6	-1.5	-33	-52	-3.1	-1.3	-2.0	2.5	-3.0	-0.7	20	6.8	2.0	8.8	0.8	-6.8	-2.0	-8.8	-0.8
1y 5y10y30y	93.9/100/21.2	12.6	19.2	-1.6	-7	27	-10.9	-1.9	0.6	18.6	-2.7	-1.3	84	6.7	-0.6	6.0	0.5	-6.7	0.6	-6.0	-0.5
1y 10y20y30y	87.8/100/37.2	8.9	8.3	0.6	-20	-41	1.6	0.7	-1.7	10.0	-1.2	0.5	23	-0.5	1.7	1.2	0.0	0.5	-1.7	-1.2	0.0

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Note: Close 6/9/23. Conditional flies are structured by buying (selling) receivers in the belly vs selling (buying) receivers on the wings with 50/50 DV01 weights. The wings are struck ATM, while the strike on the belly is solved for zero cost. Spot, forward and costless strike rates flies are reported using -0.5/1/-0.5 weights. We compute correlations of the forward fly with the belly (using a 3m window in changes and 1y window in levels) and rich/cheap of the forward fly to the belly based on 1y regression in levels. We also compute historical % of the strike fly using data starting in 1990. 3m roll is computed assuming unchanged curve and vol cube and is expressed in bp of running yield. For each valuation metric, we mark three best **long belly receiver trades in red**, and three **best short belly receiver trades in green**. Note that the "trades" shown are backward looking, and suggest which actions would have been preferable at the pricing when the screen was run. These are illustrative only and not intended as open trade ideas.
Source: Citi Research

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Appendix A-1

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